Kingdom Come

A Quaternionic Geometric Approach to Number Theory and Physics

Extending Allen Hatcher's visual number theory
via quaternions and the Hopf fibration

Aaron Michael Kinder

Short name: QGA

Companion software: Kingdom Come portal
https://github.com/kinaar8340/kingdom_come

Book repository: https://github.com/kinaar8340/qga

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Epigraph

Make the integers visible.
— the geometric method of Hatcher's Topology of Numbers*,*
lifted here to S^3, the Hopf fibration, and flux lattices.

Contents

Preface

Kingdom Come: A Quaternionic Geometric Approach to Number Theory and Physics *(Short name: QGA)*

Mission. Lift Hatcher's visual, diagrammatic number theory from the Farey plane and binary quadratic forms to unit quaternions, the Hopf fibration, and gauged flux lattices—keeping pure geometry and arithmetic defensible, and isolating physical models and observational hypotheses so they never masquerade as theorems.

Why this book exists

Allen Hatcher's *Topology of Numbers* (free PDF; AMS paperback, 2022) teaches elementary number theory by **drawing**. Mediants, Farey edges, continued-fraction zigzags, and Conway topographs of binary quadratic forms turn dry Diophantine statements into spatial relationships you can almost walk along. Symmetries appear as linear fractional transformations; classification appears as finite lists of reduced forms; arithmetic depth appears as class groups of quadratic forms and fields. The method is as important as the theorems: **make the integers visible**.

Kingdom Come is a Hopf-fibration portal—code, figures, and a Gradio laboratory—built around a different but kindred intuition. Unit quaternions form the three-sphere S^3. The Hopf map $S^3 \to S^2$ fibers that sphere into linked circles. A **gauged Hopf lattice** discretizes the geometry; **flux flywheels** are topologically protected rotating configurations on that lattice. A map from atomic number Z to flywheel metrics produces a periodic-table proxy ("Magic Islands"). Across several observational domains a numerical signature $350/\pi \approx 111.41$ appears as a working topological clock W_g. The same "draw it, rotate it, see the links" ethos that animates Hatcher's diagrams drives the portal's visualizers.

This book, **Kingdom Come: A Quaternionic Geometric Approach to Number Theory and Physics**, is the **lift** of Hatcher's method:

Hatcher's plane	This book's stage
Farey diagram on the hyperbolic / rational plane	Discrete structure on S^3, projected by Hopf
Binary quadratic forms	Quaternion norms, quaternary forms, **flux topographs**
$SL(2,\mathbb{Z})$-type symmetries	Left/right unit-quaternion multiplications; lattice gauge actions
Representations by forms	Representations by norms **and** the $Z \mapsto$ flywheel map
Class groups of quadratic forms/fields	Ideal classes of quaternion orders and algebras

Parts I–II develop quaternions, the Hopf fibration, and the gauged lattice (Theorems plus labeled Models). Part III's topographs stay Model/OP; the $Z \mapsto$ map's chemistry-facing reading (former §§7.3–7.4) lives in **Part V**, not on the geometric spine. Part IV is arithmetic depth (class groups, quaternion algebras) with Models clearly marked. Part V states **models** and **hypotheses** about emergent physics and observational constants; those claims are labeled so they never masquerade as theorems.

What this book is not

It is **not** a modified reprint of Hatcher's PDF, nor a substitute for it. Hatcher remains the canonical geometric introduction to Farey geometry, topographs, and class groups of binary quadratic forms. We cite him freely, invite parallel reading, and—when a construction has a clear TN counterpart—say so explicitly (see `HATCHER_MAP.md` in the project root).

It is also **not** a claim that every numerical coincidence listed in Kingdom Come is a law of nature. Where the text says **Hypothesis**, treat it as a research prompt with an eventual validation or falsification protocol (Chapter 10).

Companion software and figures

Interactive experiments live in the Kingdom Come repository and Hugging Face Space. Pedagogical lattice/topograph/composition/algebra/validation helpers live in this book's `lib/` package. Static figures are under `book/figures/` (see attribution there).

Resource	Location
Book manuscript (this repo)	github.com/kinaar8340/qga
Kingdom Come source / portal	github.com/kinaar8340/kingdom_come
Shared Hopf / quaternion core	`flux_hopf_lib`
Figures	`book/figures/`
How to run labs	`book/HOW_TO_USE.md`

Module pointers at the end of each chapter name the Python entry points used in labs (`kingdom.core.*`, `lib.*`, and figure generators under `scripts/`).

How to read

- **Pure geometry and arithmetic.** Chapters 0–9 for the geometric/arithmetic development; treat Chapter 10 as optional (observations and validation). Chapter 7's chemistry-facing paragraphs and 7.3–7.4 are Model / Software fact, not theorem spine (Part V owns the empirical reading).
- **Model / portal narrative first.** Chapter 0 → Chapter 2 → Chapter 3 (lattice and flywheels) → Chapter 7 → Chapter 10.
- **Coming from Hatcher.** Keep TN open beside this book; use root `HATCHER_MAP.md` as a parallel syllabus.
- **Coming from the Gradio app.** The Preview chapter is the "Home + Hopf Visualizer + Flux Flywheel" tour in prose; see also `HOW_TO_USE.md`.
- **Open problems.** Living list in `notes/open_problems.md` (OP1–OP6).

Claim labels used throughout (see also `HOW_TO_USE.md`):

Label	Meaning
Theorem	Proved here or classical
Model	Consistent mathematical or physical construction, not claimed as nature's unique law
Hypothesis	Observational claim needing validation or falsification (Chapter 10 / Table T4)
Software fact	True of the current code or portal implementation

Acknowledgments

Allen Hatcher's freely available *Topology of Numbers* made geometric elementary number theory a shared public good; this project stands on that invitation to *see* number theory. The Kingdom Come codebase and observation program (Hopf visualizers, lattice simulations, flux flywheel element maps, Magic Island sweeps, and multi-domain $350/\pi$ investigations) supply the computational and visual backbone of the lift.

How to Use the Figures and Code

This short front-matter page is the practical companion to the Preface. It fixes the conventions used in every chapter so figures render correctly and labs stay reproducible. For navigation, start with root `TOC.md`; for Hatcher's book in parallel, use `HATCHER_MAP.md`.

Claim labels

Every major claim in the manuscript is one of four kinds:

Label	Meaning	Example
Theorem	Proved in this book or classical	Four-square theorem; Hopf linking; Hurwitz class number 1
Model	Consistent mathematical or physical construction; not claimed as nature's unique law	Gauged Hopf lattice adjacency; flux topographs; $Z \mapsto$ proxy
Hypothesis	Observational claim needing validation or falsification	H1a–H1e: $W_g = 350/\pi$ per domain (not one bundled clock)
Software fact	True of the current code or portal implementation	`map_z_to_flywheel` returns `stability_score`; there is **no** `magic_flag` key

Rule of thumb. Geometry and classical algebra → Theorem. Lattice / topograph design choices → Model. Portal numerics about the physical world → Hypothesis. "What this function returns today" → Software fact.

Validation protocols for Hypotheses live in Chapter 10 (Table **T4**) and `lib/validation.py`.

Repository layout

Path	Role
`book/*.md`	Manuscript (Preface, HOW_TO_USE, Chapters 0–10)
`book/figures/`	Static figures · `ATTRIBUTION.md`
`lib/`	Pedagogical helpers (not a replacement for Kingdom Come)
`scripts/generate_chN_figures.py`	Regenerate chapter figures
`TOC.md` · `HATCHER_MAP.md` · `notes/open_problems.md`	Navigation and open research edges
`https://github.com/kinaar8340/qga`	Canonical remote for this book

Live portal and domain code (separate repo):

Resource	Location
Kingdom Come	https://github.com/kinaar8340/kingdom_come · extra .[portal]
Shared Hopf / quaternion core	flux-hopf-lib (same extra)

Running the labs

One install path

From the clone root (any path; do not hard-code ~/Projects/...):

```
python3 -m pip install -e .
python3 -c "from lib.hopf_lattice import HURWITZ_UNITS, hopf_map; print(len(HURWITZ_UNITS))"
# -> 24
```

Labs that import kingdom or flux_hopf_lib need the documented extra. flux-hopf-lib is on PyPI. kingdom-come is not; pyproject.toml pins it to https://github.com/kinaar8340/kingdom_come until it is published:

```
python3 -m pip install -e ".[portal]"
```

Then:

```
from lib.hopf_lattice import HURWITZ_UNITS, hopf_map, sample_angle_lattice
from lib.flux_topograph import build_flux_topograph
from lib.composition import compose_flywheels
from lib.quaternion_algebra import HurwitzOrder, left_ideal_class_set
from lib.validation import table_t4_checklist, run_table_t4_demo, h2_ablation_table
```

Package	Chapters	Purpose
lib.hopf_lattice	3–4	Hurwitz units, angle lattice, adjacency (OP1), gauge sequences
lib.flux_topograph	5–7	Topographs, separators, classification, Magic Island scores
lib.composition	8	Flywheel composition, class-group analogue (OP6)
lib.quaternion_algebra	9	Quaternion algebras, Hurwitz order, toy class number
lib.validation	10	Table T4 checklist, Fisher combine, W_g diagnostics

Kingdom Come / portal modules

After pip install -e ".[portal]":

```
python3 -c "from kingdom.core.quaternion import Quaternion; print('ok')"
```

Do not set PYTHONPATH=~/Projects/kingdom/src:... as the official path.

Chapter	Primary modules
1	`kingdom.core.quaternion`
2	`kingdom.core.hopf`, `kingdom.viz.hopf_plotly`
3–4	`lib.hopf_lattice`, `kingdom.simulations.lattice.TwoGyroLattice`
5–6	`lib.flux_topograph`
7	`kingdom.core.flux_flywheel` · `lib.flux_topograph.stability_landscape_z`
8	`lib.composition`
9	`lib.quaternion_algebra`
10	`lib.validation`

API honesty (read once, save time)

These mismatch the sketch names that appear in early brainstorms; the manuscript and labs follow the **live** APIs:

Do use	Do not assume
`stability_score`, `stability_class`, `is_noble_gas`	`magic_flag`
`TwoGyroLattice.step_frame()`	`.step()`
`hopf_map(x1,x2,x3,x4)`, `hopf_map_quaternion`, `Quaternion.hopf_image()`	`hopf_map(q)` as a Quaternion object
`sample_fiber(...)` → **dict** of arrays	bare list of quaternions
`map_z_to_flywheel` / `map_z_to_flywheel_extended`	undocumented extra keys

Figures

1. **Paths.** In chapter markdown, figure paths are relative to `book/`: `figures/fig0_1_hopf_linked_fibers.png`.
1. **Index.** Full figure list: `book/FIGURES.md`.
2. **Attribution.** Sources and regenerate commands: `book/figures/ATTRIBUTION.md`.
3. **Regenerate.** From repo root run `python3 scripts/generate_ch1_figures.py`. Chapters 7 and 10 may need the portal extra: `python3 scripts/generate_ch7_figures.py`.
4. **Origins.** Chapter 0 partly vendors Kingdom Come assets; later chapters are mostly generated by the scripts above.

Naming pattern: `fig{N}_{k}_...` for main figures, `aux{N}_{k}_...` for auxiliary figures, `aux_z_map_*.png` for Z-map stills.

Notation (minimal sheet)

Symbol	Meaning
$\mathbb{H}$, S^3	Quaternions; unit quaternions
$h : S^3 \to S^2$	Hopf map
Λ_0	24 Hurwitz units on S^3
Λ_{ang}	Angle-sampled lattice (visualization density)
Λ_{dyn}	Dynamical two-gyro sites (portal)
W_g	$350/\pi$ (not 350π) topological clock (**Hypothesis** H1a–H1e; Ch. 10)
OP1–OP6	Open problems · `notes/open_problems.md`

Gradio portal ↔ book

Portal tab	Primary chapters
Book Mode	**0–10 map** (chapter picker → jump + mini demos)
Hopf Visualizer	0, 2
Lattice Simulator	3–4
Flux Flywheel	0, 5–7, 10
Observations	10
The Model / theory	Preface, 0, 8–9 (narrative; labs offline)

Book Mode (interactive companion)

In the Kingdom Come portal, open the **Book Mode** tab (between Help and Hopf Visualizer):

1. Choose a manuscript chapter from the dropdown — summary, claim focus, and linked live tab appear.
2. Press **Open linked live tab** to jump the main tab bar to Hopf / Lattice / Flux / Observations / Home.
3. Use the Part subtabs for **mini demos** (Classic Hopf snapshot, short lattice comparison, Flux Flywheel at a chosen Z) without leaving Book Mode.

Book Part	Live widgets
I Foundations (Ch. 1–2)	Hopf Visualizer
II Farey lift (Ch. 3–4)	Lattice Simulator
III Forms / Z-map (Ch. 5–7)	Flux Flywheel
IV Arithmetic (Ch. 8–9)	Home · The Model (+ `lib/composition`, `lib/quaternion_algebra`)
V Observations (Ch. 10)	Observations

Source of the map: `kingdom/app/pages/book_mode.py`. Portal: `https://github.com/kinaar8340/kingdom_come` · Space: `https://huggingface.co/spaces/kinaar111/kingdom`

Parallel reading with Hatcher

Keep *Topology of Numbers* open beside this book. Full map: root `HATCHER_MAP.md`.

Hatcher TN	This book
Ch. 0 Preview	Ch. 0
Ch. 1–3 Farey, CF, symmetries	Ch. 3–4 (after Ch. 1–2 foundations)
Ch. 4–6 Forms, classification, representations	Ch. 5–7
Ch. 7–8 Class group, quadratic fields	Ch. 8–9
(no TN counterpart)	Ch. 1–2 foundations; Ch. 10 observations

Appendices (expanded back matter)

Long listings and reference material live in appendices so the main chapters stay readable:

App	Content
A	Terminology and notation
B	Open Problems OP1–OP6 (full statements)
C	Laboratory code reference (full listings)
D	Table T4 validation protocols (full)
E	Figure atlas
F	Hatcher parallel dictionary

Chapters keep short lab "calls" and point here for copy-paste scripts.

Suggested first hour

1. Read the Preface and this page.
2. Open Chapter 0; confirm figures render under `book/figures/`.
3. Run `from lib.hopf_lattice import HURWITZ_UNITS` (expect 24).
4. Optionally open the Gradio Hopf Visualizer and match panels to Figures 0.1–0.3 / Chapter 2.
5. When ready for research edges, skim **Appendix B** (or `notes/open_problems.md`).

Part I

Foundations

Chapter 0

A Preview

This chapter is a guided tour, not a full development. By the end you should see a single chain of ideas:

$$\text{Pythagorean triples} \;\to\; \text{sums of two squares} \;\to\; \text{four squares / quaternions} \;\to\; S^3 \\ \to\; \text{Hopf fibration} \;\to\; \text{gauged lattice \& flux flywheels} \;\to\; Z \mapsto \text{flux map (model)}$$

and know which links are classical mathematics and which are working models or hypotheses from Kingdom Come.

Learning goals

1. Recall why geometry already sits inside elementary number theory.
2. Meet unit quaternions as S^3 and the Hopf map $S^3 \to S^2$.
3. Glimpse flux flywheels and the Z-to-flux idea without full machinery.
4. Spot the $350/\pi$ signature as a **Hypothesis**, not a theorem.
5. Own a visual roadmap for the rest of the book.

Figures in this chapter (all from Kingdom Come assets; paths relative to this file)

Tag	File	Role
Fig. 0.1	`figures/fig0_1_hopf_linked_fibers.png`	Linked Hopf fibers in stereographic view
Fig. 0.2	`figures/fig0_2_hopf_stereographic.png`	Stereographic preview of the fibration
Fig. 0.3	`figures/fig0_3_hopf_bundle.png`	Bundle picture: total space $\to$ base
Fig. 0.4	`figures/fig0_4_flux_flywheel_scales.jpg`	Flux flywheel scales / nested structure
Fig. 0.5	`figures/fig0_5_hopf_lattice_pi_cycle.jpg`	Gauged lattice meets $350/\pi$ cycle motif

Aux A0.1	`figures/aux_z_map_he.png`	Helium $Z = 2$ flux-flywheel still (closed-shell baseline)
Aux A0.2	`figures/aux_z_map_fe.png`	Iron $Z = 26$ flux-flywheel still (mid-table contrast)

0.1 From Pythagorean triples to sums of two squares

A **Pythagorean triple** is a triple of positive integers (a, b, c) with

$$a^2 + b^2 = c^2.$$

The triple $(3, 4, 5)$ is the smallest; infinitely many others exist. Already geometry and arithmetic talk: the equation is the integer-point condition on a circle of radius c, or the norm condition $N(a + bi) = c^2$ in the Gaussian integers $\mathbb{Z}[i]$.

Hatcher's *Topology of Numbers* opens by pushing this geometric instinct further. Rational slopes, mediants

$$\frac{a}{c} \oplus \frac{b}{d} \;=\; \frac{a+b}{c+d},$$

and the **Farey diagram** organize which fractions "touch" which others. Continued fractions become zigzag paths on that diagram. Binary quadratic forms

$$f(x, y) = ax^2 + bxy + cy^2$$

acquire **topographs**—value landscapes on the dual of the Farey triangulation—whose periodic separator lines encode Pell-type infinitude and classification.

We will not rebuild that plane geometry here (read Hatcher Chapters 0–4 for the full story). We only need the moral:

> Elementary number theory is already a theory of **pictures**: adjacency, paths, periods, and symmetries of integer data.

The present book asks what happens when those pictures are lifted from two variables and the rational/hyperbolic plane to **four real coordinates** organized by quaternion multiplication and a celebrated fiber bundle.

0.2 Four squares and the birth of quaternions

Lagrange's four-square theorem asserts that every natural number is a sum of four integer squares:

$$n = a^2 + b^2 + c^2 + d^2.$$

There is a classical algebraic home for this statement: the **quaternions**

$$\mathbb{H} = \{\, q = a + bi + cj + dk \mid a, b, c, d \in \mathbb{R} \,\},$$

with

$$i^2 = j^2 = k^2 = ijk = -1.$$

The multiplicative **norm**

$$N(q) = a^2 + b^2 + c^2 + d^2 = q\,\overline{q}$$

satisfies $N(q_1 q_2) = N(q_1)N(q_2)$. Integer quaternions (Lipschitz $\mathbb{Z}[i, j, k]$, better still the **Hurwitz order**) turn the four-square theorem into an arithmetic fact about norms—just as Gaussian integers turn two-square questions into unique factorization and primes congruent to 1 mod 4.

Claim type: four-square theorem and the norm identity are **Theorems** (classical). Their use as the *algebraic backbone* of Kingdom Come's geometry is a **Model** choice we make explicit.

Chapter 1 develops multiplication, conjugation, unit quaternions, and the Lipschitz/Hurwitz orders in full. For the preview, one geometric fact is enough:

$$\{q \in \mathbb{H} : N(q) = 1\} \;=\; S^3 \subset \mathbb{R}^4.$$

The unit sphere in four dimensions is not an ornament. It is the total space of the map we study next.

0.3 Circles and spheres as number-theoretic objects

In Hatcher's world, the circle S^1 already appears indirectly: slopes, rotations in the plane of a binary form, and periodic topograph orbits. The two-sphere S^2 is less prominent in elementary texts, but it is the natural **base** once we pass to unit quaternions.

Write a unit quaternion as a pair of complex numbers (up to the usual identification)

$$q \;\longleftrightarrow\; (z_1, z_2) \in \mathbb{C}^2, \qquad |z_1|^2 + |z_2|^2 = 1.$$

The classical **Hopf map** sends q to a point on S^2. The real-coordinate form used in this book (Chapter 2; `lib.hopf_lattice.hopf_map` and `flux_hopf_lib.hopf.fibration.hopf_map`) is

$$\begin{aligned} y_1 &= 2(x_1x_3 + x_2x_4),\\ y_2 &= 2(x_1x_4 - x_2x_3),\\ y_3 &= x_1^2 + x_2^2 - x_3^2 - x_4^2, \end{aligned}$$

for a unit 4-vector $(x_1, x_2, x_3, x_4) \in S^3$. On unit inputs this already lands on S^2 (no output-normalization). Equivalent complex forms identify the base point with a ratio $[z_1 : z_2] \in \mathbb{CP}^1 \cong S^2$. The older three-component formula $y_1 = x_1^2 - x_2^2$, $y_2 = 2x_1x_2$, $y_3 = 2(x_3x_4 + x_1x_2)$ is **not** a Hopf map; it is kept only as `legacy_portal_map`.

Fiber intuition. Over each base point sits a **circle's worth** of unit quaternions—the fiber—obtained by rotating a phase along that circle. Distinct fibers are **linked**: any two are linked once (Hopf invariant 1). Stereographic projection $S^3 \setminus \{\mathrm{pt}\} \to \mathbb{R}^3$ displays those fibers as linked Villarceau circles on nested tori.

Hatcher's Farey diagram arranges **rational slopes** so that adjacency and mediants are visible. Hopf arranges **circle fibers** over a sphere so that linking and phase are visible. The analogy is not identity of objects; it is identity of **method**: a picture that makes the right discrete or continuous relationships obvious.

Forward pointer. Calling Hopf a "Farey analogue" is a **Model** metaphor only. **Chapter 3 will turn this metaphor into precise discrete definitions**—adjacency, mediants, and gauge data on a lattice in S^3—so that the analogy becomes a construction rather than a slogan.

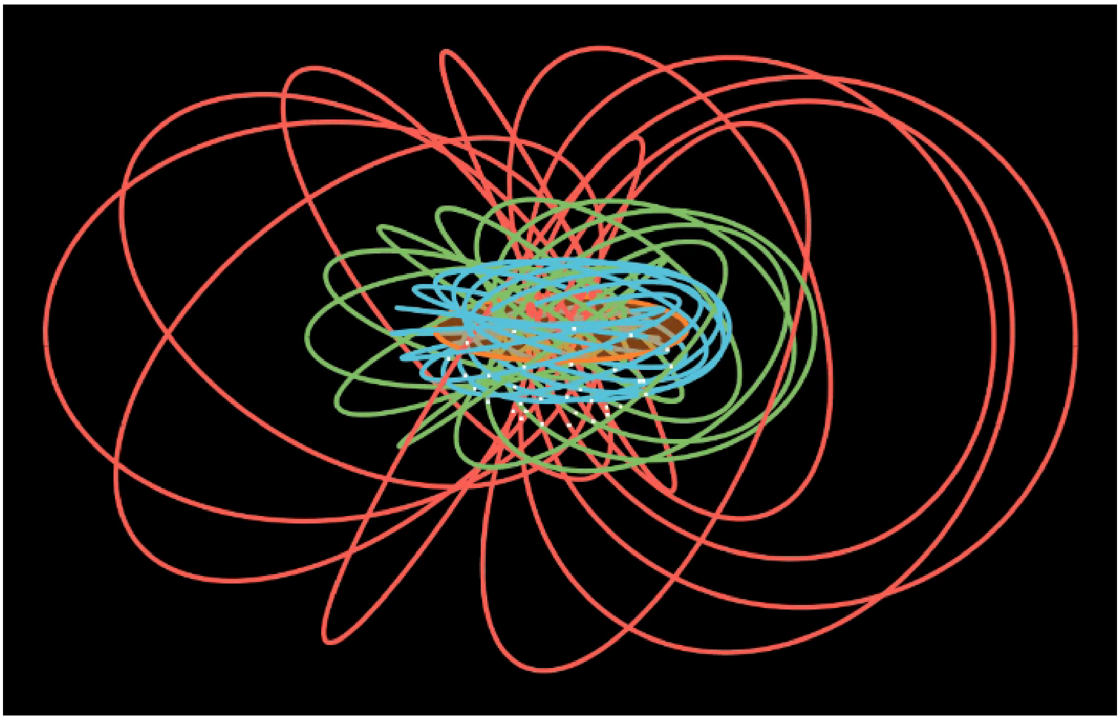


Figure 0.1: Two (or more) fibers of the Hopf fibration after stereographic projection to $\mathbb{R}^3$. Linking is the topological signature that will later underwrite **protection** of flux configurations: you cannot unlink fibers by continuous deformation without cutting.

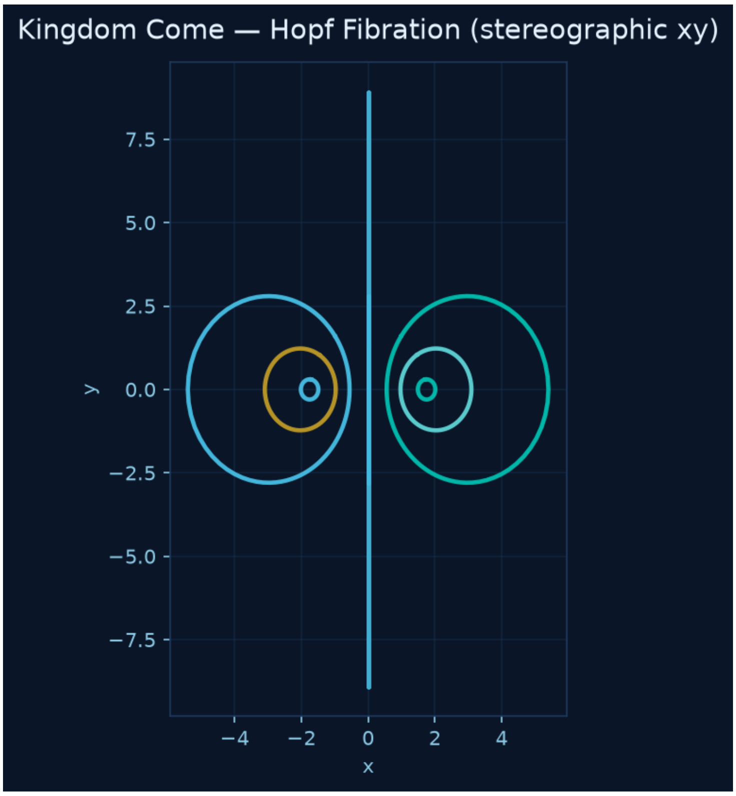


Figure 0.2: A portal-style stereographic preview of the fibration: the same total space seen as a single composed picture rather than isolated fibers. In the Gradio app this is the "Hopf Visualizer" family of views (2D projections are preferred on CPU-only hosts).

0.4 The Hopf fibration as a higher-dimensional Farey analogue

We state the comparison carefully so it does not pretend to be a theorem of uniqueness.

Farey diagram (Hatcher)	Hopf geometry (this book)
Vertices: rationals / cusps	Points of a discrete set in S^3 (or in $\mathbb{H}$)
Edges: adjacency of fractions	Fiber segments / lattice bonds projecting under Hopf
Mediant of two fractions	Quaternionic (or lattice) combination rule, **to be axiomatized in Ch. 3**
Paths: continued fractions	Phase winding along fibers; walks on the gauged lattice
Symmetries: linear fractional maps	Left/right multiplication by unit quaternions; gauge actions

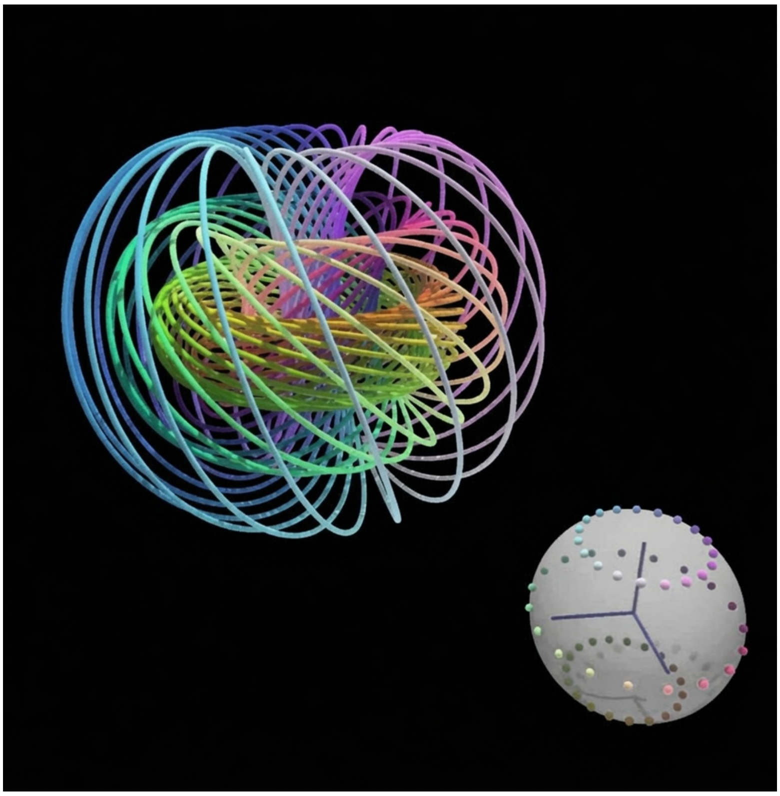


Figure 0.3: Conceptual bundle picture: total space S^3, base S^2, fibers circles. In Part II we **discretize** this picture—placing a lattice in the total space, projecting to the base, and defining adjacency—so that Farey-like combinatorial operations become available again.

Claim type: Hopf fibration, linking, and the double cover $\mathrm{Spin}(3) \to SO(3)$ via unit quaternions are **Theorems** of classical topology/geometry. Calling Hopf a "Farey analogue" remains a **Model** metaphor until Chapter 3 supplies the discrete axioms (see also Open Problem 1: *canonical quaternionic Farey structure*).

0.5 First sight of the gauged Hopf lattice and flux flywheels

Kingdom Come's physical **Model** (not yet a theorem of nature) can be stated in one paragraph:

> Physics emerges from **topologically protected flux structures** on a **gauged Hopf lattice** embedded in a **porous vacuum**. The Hopf fibration supplies the geometric backbone; quaternions supply the algebra; stable rotating configurations—**flux flywheels**—anchor emergent matter. Observer-linked phase holonomy between fibers damps toward synchrony with a coupling scale κ.

A **flux flywheel**, for preview purposes, is a lattice-supported circulating flux whose topology (linking, winding, charge) obstructs continuous decay to a trivial configuration. In Hatcher's topographs, **periodicity** of separator lines produces infinite families of representations (Pell). Here, **periodicity of a flywheel** is the dynamical counterpart: a closed orbit of gauge and phase that refuses to unwind.

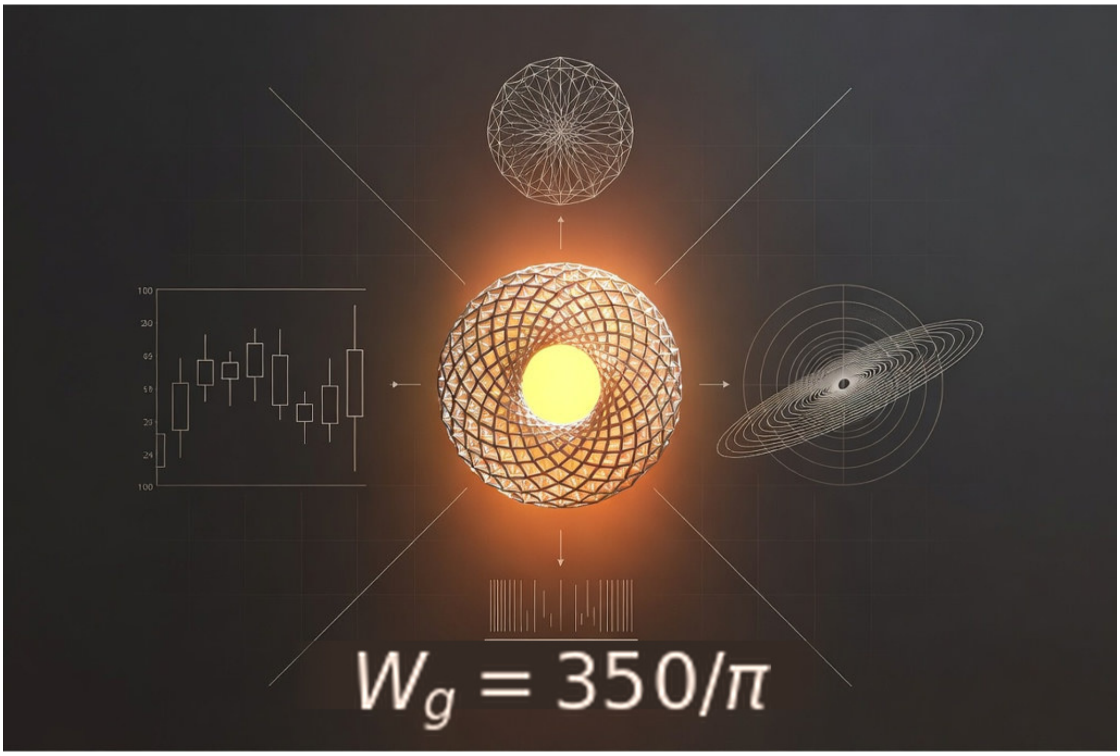


Figure 0.4: Nested / multi-scale flywheel imagery from the Kingdom Come observation assets. Read it as a **visual metaphor plus design sketch** for the lattice construction of Chapters 3–5: concentric scales, rotational identity, and coupling between levels. Formal definitions of lattice sites, gauge fields, and topological charge come later.

In code, lattice and flywheel experiments live under:

- `kingdom.core.lattice` — gauged lattice dynamics and stability comparisons
- `kingdom.core.flux_flywheel` — `map_z_to_flywheel`, `map_z_to_flywheel_extended`
- Gradio tabs: **Lattice Simulator**, **Flux Flywheel**

Chapter 5 will introduce **flux topographs**: value landscapes for quaternionic (or lattice flux) functionals, the direct conceptual child of Conway topographs.

0.6 A first Z-to-flux sketch and the $350/\pi$ signature

0.6.1 The Z-map (Model)

Kingdom Come implements a map from atomic number Z to a flywheel configuration and a bundle of stability / chemistry-facing metrics:

$$Z \longmapsto (\text{flywheel state, stability score, shell-like descriptors, } \ldots).$$

Primary entry points: `map_z_to_flywheel` and `map_z_to_flywheel_extended` in `kingdom.core.flux_flywheel`. Detuning parameters (frequency offset, gauge strength, depth/layers) select **stability islands** in parameter space—**Magic Islands**—including a documented ultra-stable lock used as a noble-gas-like template.

The portal's element tour renders each Z with shell clouds and a flux ring. Two static stills from that tour are kept beside the main figures:

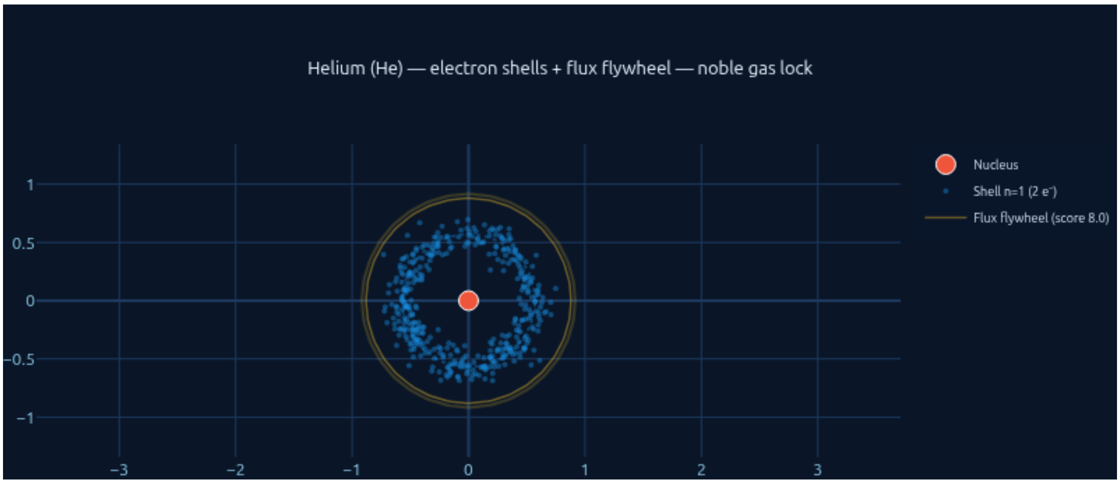


Figure 0.5: Helium ($Z = 2$). Flux-flywheel element card from the portal tour: noble-gas closed shell, high stability score in the working model, and a simple shell/flux-ring visualization. Use this as the "quiet baseline" when comparing later Z values. Source: `z_knowns/frame_0002.png`.

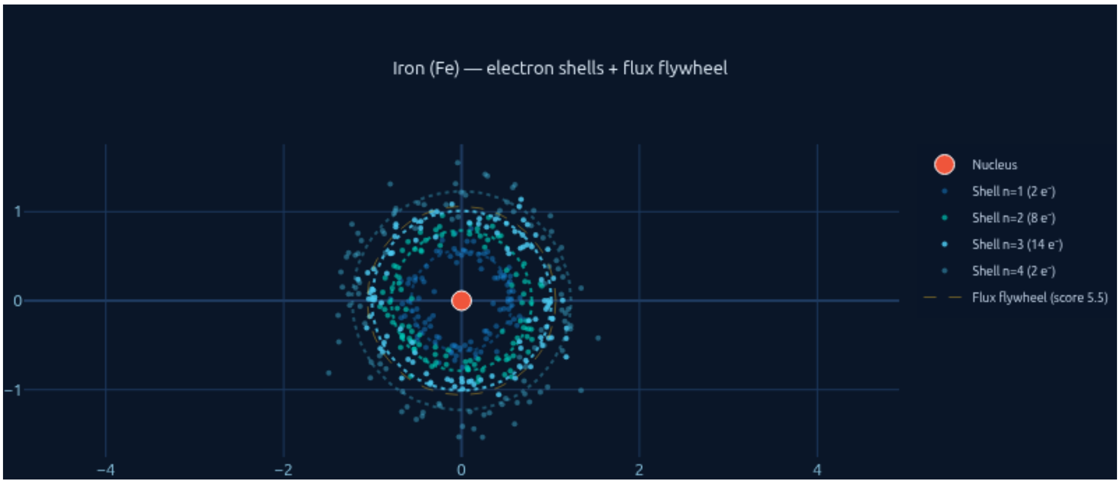


Figure 0.6: Iron ($Z = 26$). Same tour format at iron: a mid-table configuration where real-world nuclear and chemical specialness can later be checked against model metrics (stability, ionization-energy anchors, magnetic-moment proxies in Chapters 7 and 10). Source: `z_knowns/frame_0026.png`.

Claim type: the existence of a coded map $Z \mapsto$ configuration is a **Software fact**. Interpreting it as *the* emergence mechanism of the periodic table is a **Model**. Quantitative alignment with experiment is an empirical research program (Chapters 7 and 10), not a free theorem.

0.6.2 The $350/\pi$ signature (Hypothesis)

Kingdom Come's constants module records a topological clock scale

$$W_g = \frac{350}{\pi} \approx 111.408$$

(`WG_FROM_350_OVER_PI`, with a documented lock value $W_{g,\mathrm{lock}}$ in `flux_hopf_lib` / `kingdom.core.constants`). The same *numeral* appears in several **separate observation tracks** of the portal (pulsar-timing, Bitcoin Pi Cycle, TLS trees, cuprates, structural constants). Those tracks are **Hypothesis H1a–H1e**, one per domain — not one bundled clock. Chapter 10 / Appendix D split and pre-register them.

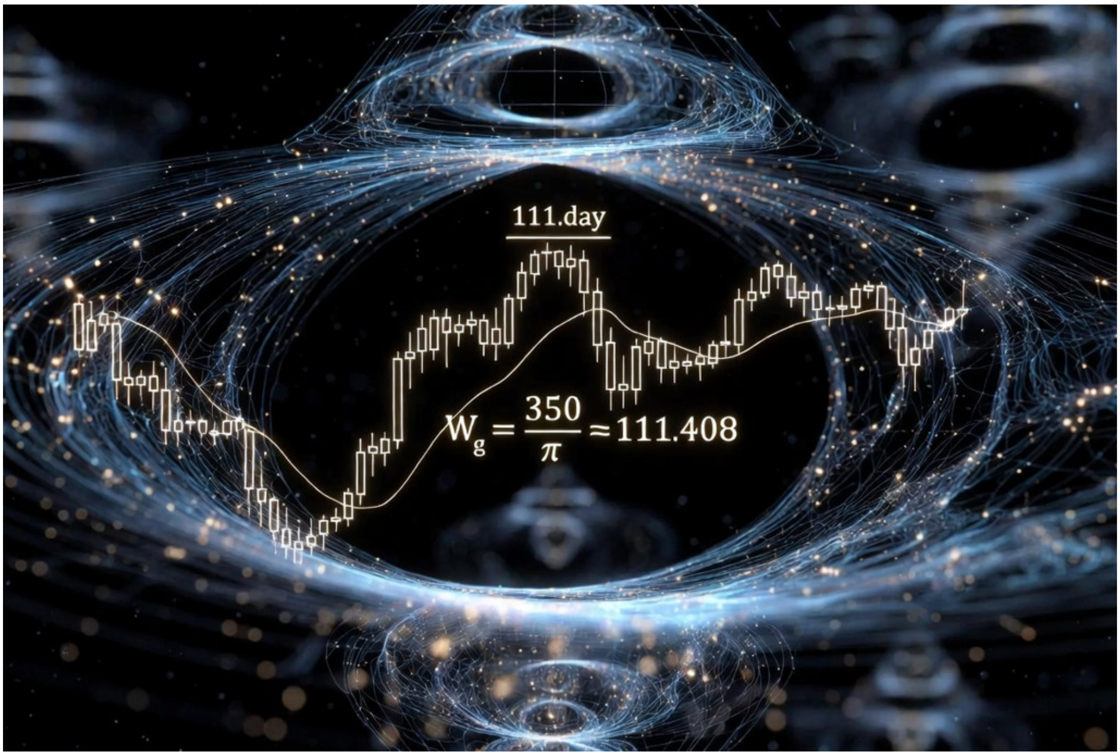


Figure 0.7: Asset bridging the **gauged Hopf lattice** picture with the π-cycle / $350/\pi$ observational storyline. In this book it marks the boundary between geometry (Parts I–IV) and the hypothesis layer (Part V).

Claim type: Hypothesis. Recurrence of $350/\pi$ across domains is **not** proved from first principles in this chapter. Chapter 10 will:

1. list each domain with source and method,
2. require a statistical validation checklist,
3. state what would **confirm, refine, or falsify** the hypothesis.

Until then, treat $W_g = 350/\pi$ as a **named constant of the working model**, not as a theorem of number theory.

0.7 Visual roadmap: Hatcher diagrams meet Hopf and Gradio

Think of the book as five floors of the same building:

```
Part V   Ch. 10     Observations & validation     <- Hypothesis & Model
Part IV  Ch. 8-9    Class groups & algebras       <- Hatcher Ch. 7-8 lift
Part III Ch. 5-7    Forms, topographs, Z-map      <- Hatcher Ch. 4-6 lift
Part II  Ch. 3-4    Gauged lattice (Farey lift)   <- Hatcher Ch. 1-3 lift
Part I   Ch. 1-2    Quaternions & Hopf            <- foundations (new floor)
         Ch. 0      This preview
```

How the five main figures will reappear

Figure	Returns in
0.1 Linked fibers	Ch. 2 (definition, Hopf invariant); Ch. 4 (symmetry orbits)
0.2 Stereographic preview	Ch. 1–2 (coordinates, charts); Gradio labs
0.3 Bundle diagram	Ch. 2–3 (discrete total space / base)
0.4 Flywheel scales	Ch. 3, 5, 7 (lattice, topographs, Z-map)
0.5 Lattice / π-cycle	Ch. 3 (lattice); Ch. 10 (observations)

Portal $\leftrightarrow$ chapter cheat sheet

Gradio / module	First serious chapter
Hopf Visualizer · `core.hopf` · `viz.hopf_plotly`	2
Quaternion helpers · `core.quaternion`	1
Lattice Simulator · `core.lattice`	3–4
Flux Flywheel slider · `core.flux_flywheel`	5–7
Magic Island · `viz.magic_island`	6
Observations tabs · `constants.WG_FROM_350_OVER_PI`	10

0.8 Reading plan for the rest of the book

1. **Part I (Ch. 1–2)** makes quaternions algebraic and geometric, then defines the Hopf fibration carefully and rebuilds Figures 0.1–0.3 from first principles.
2. **Part II (Ch. 3–4)** constructs the gauged Hopf lattice and its symmetries—the Farey lift (Open Problem 1).
3. **Part III (Ch. 5–7)** develops forms, flux topographs, Magic Islands, and the Z-map as representation theory (Open Problems 2–4).
4. **Part IV (Ch. 8–9)** treats composition, class groups, and quaternion algebras (Open Problem 6).
5. **Part V (Ch. 10)** collects multi-domain observations ($350/\pi$, $Z \mapsto$ correlations, Magic Islands), states validation protocols (Table T4), and summarizes Open Problems OP1–OP6 (Open Problem 5).

If you are impatient for physics, read $0 \to 2 \to 3$ (flywheel sections) $\to 7 \to 10$, then return to fill the arithmetic spine. If you are a number theorist first, read 0–9 with Hatcher open, and treat Chapter 10 as a speculative appendix until the checklists are green.

0.9 Exercises (preview level)

0.A. Verify $N(q_1 q_2) = N(q_1)N(q_2)$ for two random integer quaternions by hand or with `kingdom.core.quaternion`.

0.B. In the Gradio Hopf Visualizer, load **Classic Hopf** and identify which panel corresponds most closely to Figure 0.1 versus Figure 0.2.

0.C. Run `map_z_to_flywheel` for $Z \in \{2, 10, 26, 79\}$. Record stability scores; do **not** yet interpret them as chemical truth—only as model outputs.

0.D. Write one paragraph distinguishing **Theorem / Model / Hypothesis** for: (i) four-square theorem, (ii) Hopf linking, (iii) $W_g = 350/\pi$ as a multi-domain constant.

0.E. Open Hatcher TN Chapter 0 and list three pictures you expect to have quaternionic or Hopf analogues by the end of our Chapter 5.

0.10 Code and asset pointers

```
kingdom.core.quaternion      # Quaternion, from_hopf_coords, hopf_image
kingdom.core.hopf            # sample_fiber, sample_fiber_family
kingdom.core.lattice         # gauged lattice simulations
kingdom.core.flux_flywheel   # map_z_to_flywheel[_extended]
kingdom.core.constants       # WG_FROM_350_OVER_PI, W_G_LOCK, kappa defaults
kingdom.viz.hopf_plotly      # portal-style multi-panel figures
```

Figure sources (Kingdom Come): see `figures/ATTRIBUTION.md`.

Parallel reading: Hatcher, *Topology of Numbers*, Chapter 0; project file `HATCHER_MAP.md`.

Chapter 1

Quaternions: Algebra, Geometry, and Arithmetic

This chapter builds the algebraic and geometric language used everywhere later in the book. Chapter 2 fibers the unit sphere of quaternions; Chapters 3–4 act on it by left and right multiplication; later chapters repeatedly invoke norms, integer orders, and the four-square theorem.

Learning goals

1. Master the algebraic rules and geometric interpretation of quaternions.
2. See unit quaternions as the three-sphere S^3.
3. Distinguish Lipschitz and Hurwitz orders and understand why the latter is preferred for arithmetic.
4. Re-encounter the four-square theorem as a consequence of norm multiplicativity.
5. Meet the double-cover relationship to three-dimensional rotations.
6. Run first computational labs with `kingdom.core.quaternion`.

Figures in this chapter

Tag	File	Role
Fig. 1.1	`figures/fig1_1_ijk_multiplication.png`	i, j, k cycle and multiplication table
Fig. 1.2	`figures/fig1_2_s3_unit_quaternions.png`	Unit quaternions as $S^3 \subset \mathbb{R}^4$
Fig. 1.3	`figures/fig1_3_lipschitz_hurwitz.png`	Lipschitz vs Hurwitz lattice (schematic)
Fig. 1.4	`figures/fig1_4_norm_multiplicativity.png`	Norm multiplicativity $\rightarrow$ four squares
Aux A1.1	`figures/aux1_1_quaternion_rotation.png`	Rotation via $v \mapsto qvq^{-1}$ ($\rightarrow$ Ch. 2)

Claim discipline

Claim	Type
Algebra of $\mathbb{H}$; $N(q_1q_2) = N(q_1)N(q_2)$; four-square theorem; $\mathrm{Spin}(3) \rightarrow SO(3)$ double cover	**Theorem** (classical)
Preferring the Hurwitz order as the primary integer lattice for later flux/lattice work	**Model** choice
Kingdom Come / `flux_hopf_lib` APIs and demos	**Software fact**

1.1 Definition and basic operations

1.1.1 The algebra $\mathbb{H}$

A **quaternion** is an expression

$$q = a + bi + cj + dk,$$

where $a, b, c, d \in \mathbb{R}$ and the symbols i, j, k satisfy Hamilton's rules

$$i^2 = j^2 = k^2 = ijk = -1.$$

These force the cyclic products

$$ij = k, \quad jk = i, \quad ki = j$$

and the anticyclic products

$$ji = -k, \quad kj = -i, \quad ik = -j.$$

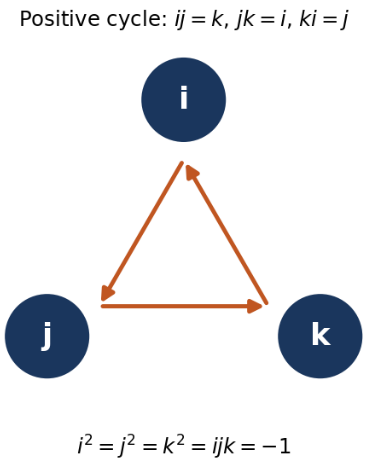
In particular, multiplication is **associative** and **distributive** over addition, but **not commutative**: $ij \neq ji$.

As a real vector space,

$$\mathbb{H} \cong \mathbb{R}^4,$$

with basis $\{1, i, j, k\}$. Addition and scalar multiplication are componentwise. The interesting structure is the bilinear product determined by the table below.

Figure 1.1 — Quaternion imaginary units and multiplication



Multiplication table (row x column)

	1	**i**	**j**	**k**
1	1	i	j	k
i	i	-1	k	-j
j	j	-k	-1	i
k	k	j	-i	-1

Figure 1.1: Left: the positive cycle $i \to j \to k \to i$. Right: the 4×4 multiplication table for $\{1, i, j, k\}$. Red-tinted cells are negative results; green-tinted cells are the pure imaginary basis elements.

Notation. We write $q = (a, b, c, d)$ or $q = a + \mathbf{v}$ with vector part $\mathbf{v} = (b, c, d) \in \mathbb{R}^3$. Kingdom Come stores components as `(w, x, y, z)` with $w = a$ the real part—the same convention as `flux_hopf_lib.quaternion.core.Quaternion`.

1.1.2 Explicit product formula

If $q_1 = a_1 + b_1 i + c_1 j + d_1 k$ and $q_2 = a_2 + b_2 i + c_2 j + d_2 k$, then

$$\begin{aligned} q_1 q_2 = {} & (a_1 a_2 - b_1 b_2 - c_1 c_2 - d_1 d_2) \\ & + (a_1 b_2 + b_1 a_2 + c_1 d_2 - d_1 c_2)\, i \\ & + (a_1 c_2 - b_1 d_2 + c_1 a_2 + d_1 b_2)\, j \\ & + (a_1 d_2 + b_1 c_2 - c_1 b_2 + d_1 a_2)\, k. \end{aligned}$$

This is exactly the formula implemented by `Quaternion.multiply` in both `flux_hopf_lib` and `kingdom.core.quaternion`.

1.1.3 Complex pairs

Identifying $\mathbb{C}$ with $\operatorname{span}_{\mathbb{R}}\{1, i\}$ inside $\mathbb{H}$, every quaternion can be written uniquely as

$$q = z_1 + z_2\, j, \qquad z_1, z_2 \in \mathbb{C},$$

or as a pair $(z_1, z_2) \in \mathbb{C}^2$. This view is the natural bridge to the Hopf map in Chapter 2.

1.2 Conjugation and the multiplicative norm

1.2.1 Conjugate

The **conjugate** of $q = a + bi + cj + dk$ is

$$\overline{q} = a - bi - cj - dk.$$

Properties (all **Theorems**, elementary verifications):

$$\overline{\overline{q}} = q, \qquad \overline{q_1 + q_2} = \overline{q_1} + \overline{q_2}, \qquad \overline{q_1 q_2} = \overline{q_2}\,\overline{q_1}.$$

Note the **order reversal** under conjugation of products—the noncommutative analogue of the complex rule $\overline{zw} = \overline{z}\,\overline{w}$.

1.2.2 Norm

The **(Euclidean) norm** is

$$N(q) \ := \ q\,\overline{q} \ = \ a^2 + b^2 + c^2 + d^2 \ = \ |q|^2$$

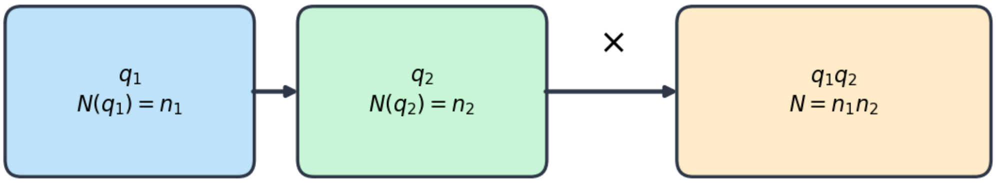
if one writes $|q|$ for the Euclidean length in $\mathbb{R}^4$. In code, `Quaternion.norm()` returns $\sqrt{N(q)} = |q|$; the multiplicative quantity used in number theory is $N(q) = |q|^2$. We will say **multiplicative norm** for $N(q)$ and **length** for $|q|$ when both appear.

Theorem (multiplicativity of the norm). For all $q_1, q_2 \in \mathbb{H}$,

$$N(q_1 q_2) = N(q_1)\, N(q_2).$$

Sketch. $N(q_1 q_2) = (q_1 q_2)\overline{(q_1 q_2)} = q_1 q_2 \overline{q_2}\,\overline{q_1} = q_1 N(q_2) \overline{q_1} = N(q_1) N(q_2)$, using that $N(q_2)$ is real (hence central).

Figure 1.4 — Multiplicativity of the quaternion norm



Theorem: $N(q_1 q_2) = N(q_1)N(q_2) \Rightarrow$ products of sums of four squares

Example: $(1^2 + 1^2 + 1^2 + 1^2)(1^2 + 0 + 0 + 0) = 4 \cdot 1 = 4$

Figure 1.2: The product of two quaternions multiplies their norms. Every identity among sums of four squares is, at bottom, this theorem in arithmetic clothing.

1.2.3 Inverse

If $q \neq 0$, then

$$q^{-1} = \frac{\overline{q}}{N(q)},$$

so every nonzero quaternion is a unit in the division ring $\mathbb{H}$. In floating-point code, `inverse()` divides by $|q|^2$ and raises if $|q|$ is near zero.

1.3 Unit quaternions and the identification $S^3 \subset \mathbb{H} \cong \mathbb{R}^4$

1.3.1 The three-sphere

The set of **unit quaternions** is

$$\mathrm{Sp}(1) \;:=\; \{q \in \mathbb{H} : N(q) = 1\} \;=\; S^3 \subset \mathbb{R}^4.$$

It is a compact Lie group under quaternion multiplication (associative, identity 1, inverses $\overline{q}$). Geometrically:

- As a manifold, S^3 is the unit sphere in four-dimensional Euclidean space.
- As a group, left (or right) multiplication by a fixed unit quaternion is an isometry of S^3.
- As a complex pair, unit length means $|z_1|^2 + |z_2|^2 = 1$.

1.3.2 Stereographic projection (preview)

Removing a point (classically -1, or any fixed unit quaternion) and projecting to a tangent $\mathbb{R}^3$ yields a global chart on $S^3 \setminus \{\mathrm{pt}\}$. Kingdom Come's Hopf visualizer uses stereographic (or orthographic 2D) projections heavily; here we only need that S^3 **is drawable** once a chart is chosen. The full Hopf fibration story—fibers, linking, base S^2—is Chapter 2.

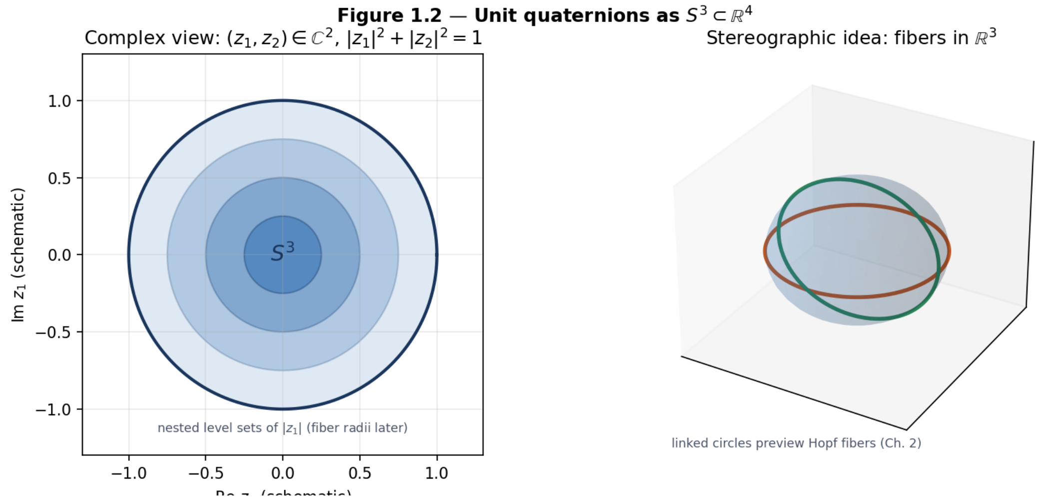


Figure 1.3: Left: schematic of the constraint $|z_1|^2 + |z_2|^2 = 1$ via nested level sets of $|z_1|$. Right: after stereographic projection $S^3 \setminus \{\mathrm{pt}\} \to \mathbb{R}^3$, great-circle fibers will appear as linked circles (Chapter 2; compare Figs. 0.1–0.2).

1.3.3 Normalization in software

`Quaternion.normalize()` returns $q/|q|$ (or the identity if q is numerically zero). Portal methods that construct new quaternions return `type(self)` so subclass helpers such as `hopf_image()` survive `normalize()` and `multiply()`.

1.4 Integer quaternions: Lipschitz order vs Hurwitz order

1.4.1 Lipschitz order

The **Lipschitz order** is the set of quaternions with integer coordinates:

$$L \;:=\; \mathbb{Z}[i,j,k] \;=\; \{a + bi + cj + dk : a,b,c,d \in \mathbb{Z}\}.$$

It is a free $\mathbb{Z}$-module of rank 4, closed under multiplication, and contains the eight **Lipschitz units**

$$\{\pm 1, \pm i, \pm j, \pm k\}.$$

Norms of Lipschitz quaternions are nonnegative integers. Multiplicativity still holds, so products of sums of four integer squares are again such sums.

1.4.2 Hurwitz order

The **Hurwitz order** $\mathcal{H}$ consists of all quaternions whose coordinates are **either all integers or all half-integers**:

$$\mathcal{H} = L \,\cup\, \Big(\tfrac{1}{2} + \tfrac{1}{2}i + \tfrac{1}{2}j + \tfrac{1}{2}k + L\Big).$$

Equivalently: $a,b,c,d \in \mathbb{Z}$ or $a,b,c,d \in \mathbb{Z} + \frac{1}{2}$. This is still a ring (in fact a maximal order in the rational quaternion algebra $\mathbb{H}_{\mathbb{Q}}$), and it is **larger** than L.

The unit group of $\mathcal{H}$ has **24** elements—the binary tetrahedral group—including

$$\pm 1,\ \pm i,\ \pm j,\ \pm k, \quad \tfrac{1}{2}(\pm 1 \pm i \pm j \pm k)$$

(all sign combinations). The extra units and lattice points make $\mathcal{H}$ **Euclidean** with respect to the norm: for any $q, d \in \mathcal{H}$ with $d \neq 0$ there exist $m, r \in \mathcal{H}$ with $q = md + r$ and $N(r) < N(d)$. That Euclidean property underwrites unique factorization phenomena used in classical proofs of the four-square theorem and in ideal theory (Chapter 9).

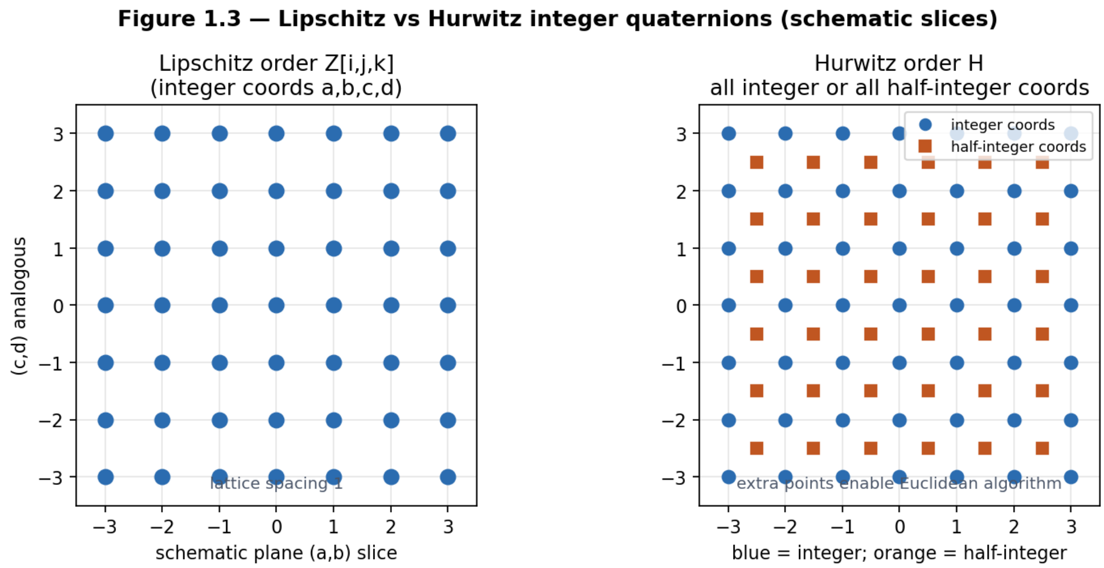


Figure 1.4: Left: integer lattice (Lipschitz), shown in a 2D schematic slice. Right: Hurwitz adds half-integer points (orange squares) when all four coordinates share the same half-integer type. The picture is only a slice—true lattice geometry lives in $\mathbb{R}^4$.

1.4.3 Why prefer Hurwitz later? (Model)

Feature	Lipschitz L	Hurwitz $\mathcal{H}$
Definition	Integer coords	Integer or half-integer coords
Units	8	24
Euclidean algorithm	fails in general	holds
Unique factorization flavor	weaker	stronger (classical)
Relation to four squares	works, clumsier proofs	clean norm arguments

Claim type: Model. For the gauged Hopf lattice and flywheel arithmetic of Parts II–IV, this book **defaults to Hurwitz** (or to lattices containing $\mathcal{H}$) unless a section explicitly restricts to Lipschitz. That is a design choice aligned with classical quaternion number theory, not a theorem that Lipschitz is "wrong."

Open thread. Discrete adjacency rules built on $\mathcal{H}$ (or on unit spheres over $\mathcal{H}$) feed Open Problem 1 in Chapter 3.

1.5 The four-square theorem via norms

Theorem (Lagrange). Every natural number n is a sum of four integer squares:

$$n = a^2 + b^2 + c^2 + d^2 \quad \text{for some } a, b, c, d \in \mathbb{Z}.$$

Quaternionic reading. The statement is exactly: for every $n \in \mathbb{N}$ there exists $q \in L$ (Lipschitz) with $N(q) = n$.

Why multiplicativity matters. Euler's four-square identity is $N(q_1 q_2) = N(q_1)N(q_2)$ written in coordinates. Therefore it suffices to prove the theorem for primes (or for n square-free) and multiply representations. Classical proofs then show every prime is a norm from $\mathcal{H}$ (or handle 2 and primes 1 mod 4 / 3 mod 4 by standard casework). Full proofs appear in many number-theory texts; Appendix D will collect a sketch. For this chapter, the essential message is:

> Four squares are not an isolated curiosity—they are the **norm theory of integer quaternions**.

Comparison with Hatcher. Hatcher's early chapters revolve around **two** squares and binary quadratic forms (Gaussian integers, topographs, class groups). Quaternions are the four-dimensional upgrade: the same geometric instinct (norms, units, lattices), one dimension higher. Chapter 5 will lift topographs themselves.

Claim type: Theorem (classical). No new proof is claimed here.

1.6 Quaternions as rotations: the double cover $\mathrm{Spin}(3) \to SO(3)$

1.6.1 Conjugation action on pure imaginaries

Identify $\mathbb{R}^3$ with the pure imaginary quaternions

$$\operatorname{Im}\mathbb{H} = \{bi + cj + dk : b, c, d \in \mathbb{R}\}.$$

For a **unit** quaternion $q \in S^3$ and $v \in \operatorname{Im}\mathbb{H}$, set

$$\rho_q(v) \;:=\; q\,v\,q^{-1} = q\,v\,\overline{q}.$$

Then $\rho_q(v)$ is again pure imaginary, and ρ_q is an orthogonal linear map of $\mathbb{R}^3$ with determinant $+1$. Thus we obtain a homomorphism

$$\rho : S^3 \longrightarrow SO(3), \qquad q \mapsto \rho_q.$$

1.6.2 Axis–angle formula

If $q = \cos(\theta/2) + \sin(\theta/2)\,u$ with unit pure imaginary u (i.e. $u^2 = -1$), then ρ_q is rotation by angle θ about the axis u. This is the content of Rodrigues' formula; Kingdom Come exposes both `Quaternion.from_axis_angle(axis, theta)` and `rodrigues_rotation`.

1.6.3 The kernel is $\{\pm 1\}$

Clearly $\rho_{-q} = \rho_q$, since $(-q)v(-q)^{-1} = qvq^{-1}$. In fact

$$\ker\rho = \{\pm 1\},$$

so ρ is a **2-to-1** covering homomorphism

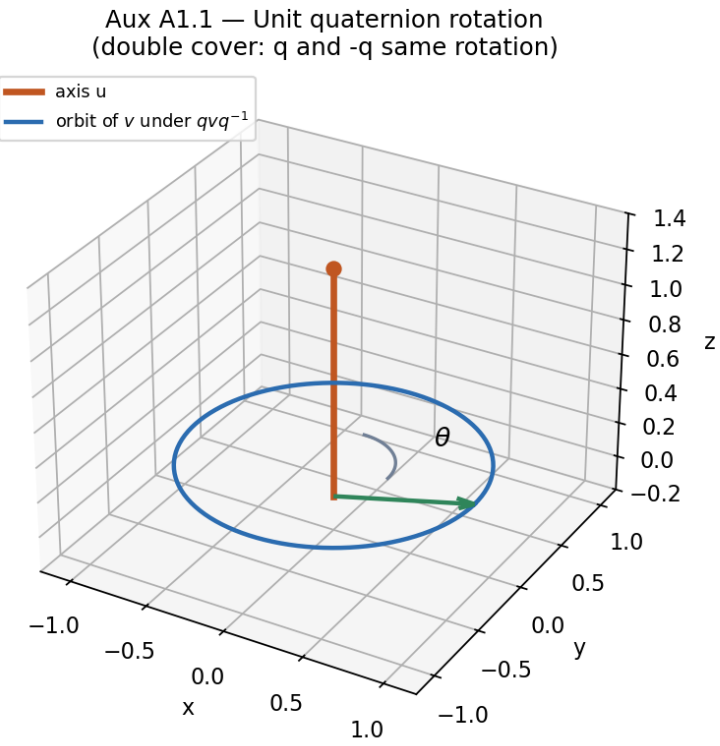


Figure 1.5: A vector v swept around an axis u under the conjugation action of a one-parameter family of unit quaternions. The same rotation arises from q and from $-q$.

$$S^3 \cong \mathrm{Spin}(3) \longrightarrow SO(3).$$

Topologically this is the unique nontrivial double cover of $SO(3)$. Paths in $SO(3)$ that rotate by 2π lift to paths in S^3 from q to $-q$; only a 4π rotation closes in the cover—the origin of spinorial sign changes in physics.

Claim type: Theorem (classical Lie theory / geometry).

Forward link. The same unit quaternions that rotate $\mathbb{R}^3$ will, in Chapter 2, be points of the **total space** of the Hopf fibration. Left and right multiplications that preserve the fibration become the symmetry language of Chapters 3–4.

1.7 First computational examples and labs

Core API: `kingdom.core.quaternion` (see also Appendix C §C.1).

Labs (short form). Full listings: **Appendix C**.

- **1.A** Verify $ij = k$, $ji = -k$, $i^2 = -1$ with `Quaternion`.
- **1.B** Check $N(q_1q_2) = N(q_1)N(q_2)$ numerically (residual $\sim 10^{-14}$).
- **1.C** Enumerate small Lipschitz norms.
- **1.D** Confirm q and $-q$ induce the same conjugation action on a pure vector.
- **1.E** Optional: `from_hopf_coords` / `hopf_image` (preview of Ch. 2).

```
from kingdom.core.quaternion import Quaternion
i, j = Quaternion(0,1,0,0), Quaternion(0,0,1,0)
print(i.multiply(j))  # k
```

1.8 Exercises

1.A (hand). Expand $(a + bi + cj + dk)(e + fi + gj + hk)$ and match the formula in §1.1.

1.B (hand). Prove $\overline{q_1 q_2} = \overline{q_2}\,\overline{q_1}$ from the multiplication table (or from the explicit formula).

1.C (hand). Show that $N(q) = q\overline{q}$ is real and nonnegative, and that $N(q) = 0 \Leftrightarrow q = 0$.

1.D (hand). Verify that the 24 Hurwitz units listed in §1.4 all have norm 1. How many Lipschitz units are among them?

1.E (hand / reading). Using multiplicativity, show that if $n = N(q_1)$ and $m = N(q_2)$ then $nm = N(q_1 q_2)$. Explain in one sentence why this reduces Lagrange's theorem to representing primes.

1.F (hand). Let $q = \cos(\theta/2) + \sin(\theta/2)\,i$. Compute $q\,j\,\overline{q}$ and identify the rotation in the jk-plane.

1.G (code). Complete Labs 1.A–1.D. Submit residuals for Lab 1.B and a printed set of norms for Lab 1.C with $N_{\max} = 3$.

1.H (code). Using `chordal_distance` (upstream) or $\min(|q - r|, |q + r|)$, verify numerically that q and $-q$ are close in the rotation sense even when far in $\mathbb{R}^4$.

1.I (Hatcher bridge). In Hatcher, sums of two squares and Gaussian integers organize binary quadratic forms. Write three sentences mapping: (Gaussian integers $\leftrightarrow$ Lipschitz/Hurwitz), (complex modulus $\leftrightarrow$ quaternion norm), (unit circle S^1 $\leftrightarrow$ unit quaternions S^3).

1.J (forward). Without reading Chapter 2, guess: if unit quaternions are points of S^3, what might a natural map $S^3 \to S^2$ do to a whole circle of phases? (You will check your guess against the Hopf map next.)

1.9 Code and asset pointers

```
kingdom.core.quaternion.Quaternion
kingdom.core.quaternion.rodrigues_rotation
flux_hopf_lib.quaternion.core.Quaternion   # upstream implementation
flux_hopf_lib.quaternion.core.q_mult, q_conj, q_normalize
```

Figures: generated for this chapter under `book/figures/fig1_*` and `aux1_1_*` (see `FIGURES.md`). **Related portal:** Gradio rotation intuition is secondary here; Hopf Visualizer becomes primary in Chapter 2.

1.10 Looking ahead

With the algebraic and geometric language of quaternions in hand—product, conjugate, norm, S^3, Hurwitz arithmetic, and the double cover of $SO(3)$—we are ready to **fiber** S^3 over S^2 in Chapter 2. The same unit quaternions will reappear as total-space points; their left and right multiplications will become the symmetries of the gauged Hopf lattice in Part II.

Chapter 2

The Hopf Fibration

This chapter defines the Hopf fibration from first principles and rebuilds the visual intuition of Figures 0.1–0.3. The same unit quaternions introduced in Chapter 1 become the total space S^3. Left and right multiplications on this total space will become the symmetry language of the gauged Hopf lattice in Chapters 3–4.

Learning goals

1. Define the Hopf fibration in both complex-pair and real four-vector coordinates.
2. Understand the fibers as circles and see their linking (Hopf invariant 1).
3. Work with stereographic projections and the visualizations used throughout the book.
4. Recognize the bundle structure and basic charts.
5. Connect the fibration back to unit quaternions and forward to the discrete lattice.
6. Run labs with `kingdom.core.hopf` and the Gradio Hopf Visualizer.

Figures in this chapter

Tag	File	Role
Fig. 2.1	`figures/fig2_1_hopf_definition.png`	Hopf map diagram (total space → base)
Fig. 2.2	`figures/fig2_2_linked_fibers.png`	Linked Hopf fibers (stereographic) — rebuild of Fig. 0.1
Fig. 2.3	`figures/fig2_3_stereographic_preview.png`	Multi-panel stereographic preview — rebuild of Fig. 0.2
Fig. 2.4	`figures/fig2_4_bundle_structure.png`	Total space → base with fibers — rebuild of Fig. 0.3
Aux A2.1	`figures/aux2_1_fiber_phase_sweep.png`	Single fiber phase sweep
Aux A2.2	`figures/aux2_2_hopf_charts.png`	Local charts on base S^2

Claim discipline

Claim	Type
Hopf fibration definition, fiber structure, linking (Hopf invariant = 1); double cover Spin(3) → $SO(3)$ from Ch. 1	**Theorem** (classical differential topology / geometry)
Interpreting the Hopf fibration as a higher-dimensional "Farey analogue" for the gauged lattice	**Model**
Kingdom Come visualizations and `kingdom.core.hopf` / `flux_hopf_lib` implementations	**Software fact**

2.1 Definition of the Hopf fibration

2.1.1 Complex-pair form (standard in the literature)

Identify unit quaternions with pairs of complex numbers (Chapter 1):

$$q = z_1 + z_2\, j \quad \longleftrightarrow \quad (z_1, z_2) \in \mathbb{C}^2, \qquad |z_1|^2 + |z_2|^2 = 1.$$

One classical form of the **Hopf map** $h : S^3 \to S^2$ is

$$h(z_1, z_2) = \big(2\,\mathrm{Re}(\overline{z_1} z_2),\ 2\,\mathrm{Im}(\overline{z_1} z_2),\ |z_1|^2 - |z_2|^2\big) \in S^2 \subset \mathbb{R}^3,$$

or, equivalently, the projectivized ratio

$$[z_1 : z_2] \in \mathbb{CP}^1 \cong S^2.$$

The second description makes the **structure-group circle** obvious: multiplying (z_1, z_2) by a common phase $e^{i\phi}$ does not change the projective point. That circle is the fiber of h.

2.1.2 Real four-vector form (the same map)

Identify $z_1 = x_1 + ix_2$, $z_2 = x_3 + ix_4$, and $(w, x, y, z) \equiv (x_1, x_2, x_3, x_4)$. The complex formula is exactly

$$\begin{aligned} y_1 &= 2(x_1x_3 + x_2x_4), \\ y_2 &= 2(x_1x_4 - x_2x_3), \\ y_3 &= x_1^2 + x_2^2 - x_3^2 - x_4^2. \end{aligned}$$

On unit 4-vectors this already lands on S^2; there is no output-normalization step. This is the **one** library map: `lib.hopf_lattice.hopf_map` (alias `hopf_map_classical`). The same formula is `flux_hopf_lib.hopf.fibration.hopf_map`, re-exported as `kingdom.core.hopf`. Do not fork. A non-unit input is a numerical guard (homogeneous degree 2: divide by $\|q\|^2$), not a feature of the map.

Not a Hopf map. The old three-component formula $y_1 = x_1^2 - x_2^2$, $y_2 = 2x_1x_2$, $y_3 = 2(x_3x_4 + x_1x_2)$ (then divide by $\|y\|$) ignores (x_3, x_4) in y_1, y_2, vanishes at $(0, 0, 1, 0)$, and is not constant on Hopf fibers. It is kept only as `legacy_portal_map` and is **not** called Hopf.

Angle coordinates vs the structure-group fiber. Hopf angles (η, ξ_1, ξ_2) via `hopf_coordinates` are

$$x_1 = \cos\eta\, \cos\xi_1, \qquad x_2 = \cos\eta\, \sin\xi_1,$$
$$x_3 = \sin\eta\, \cos\xi_2, \qquad x_4 = \sin\eta\, \sin\xi_2,$$

with $\eta \in [0, \pi/2]$ and $\xi_1, \xi_2 \in [0, 2\pi)$. Fixing (η, ξ_1) and sweeping ξ_2 traces the **angle-chart ξ_2-circle**. That is **not** the structure-group fiber. The Hopf fiber through (z_1, z_2) is the diagonal $U(1)$ action

$$(z_1, z_2) \ \longmapsto\ (e^{i\phi} z_1,\ e^{i\phi} z_2),$$

which shifts ξ_1 and ξ_2 **together**. In this book's identification $q = z_1 + z_2 j$, that action is left multiplication by $e^{i\phi} = \cos\phi + i \sin\phi \in \mathrm{span}\{1, i\}$ (`common_phase`, `structure_group_unit`). General left or right multiplication by an arbitrary unit of S^3 **moves fibers**.

API note (Software fact). Call signatures in code are component-wise, not "pass a Quaternion object" to `hopf_map`:

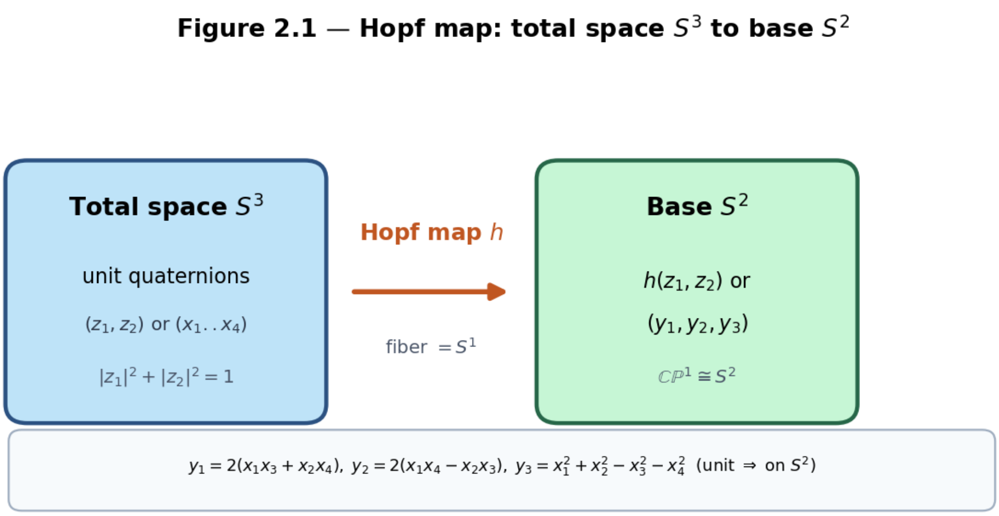


Figure 2.1: From total space S^3 (unit quaternions) to base S^2. The map is smooth and surjective; every base point has a circle's worth of preimages.

```
hopf_map(x1, x2, x3, x4)          -> (y1, y2, y3)
hopf_map_quaternion(w, x, y, z)   -> (y1, y2, y3)
Quaternion(...).hopf_image()      -> (y1, y2, y3)
```

2.2 Fibers and linking

2.2.1 Fibers are circles

Over each point $p \in S^2$ the preimage $h^{-1}(p)$ is diffeomorphic to a circle S^1. That circle is the structure-group $U(1)$ action

$$(z_1, z_2) \longmapsto (e^{i\phi}z_1,\ e^{i\phi}z_2),$$

sampled in code by `sample_structure_group_fiber`. The angle-chart ξ_2-circle at fixed (η, ξ_1) is a different loop in S^3; do not call it the Hopf fiber.

2.2.2 Linking and the Hopf invariant

Any two distinct fibers are **linked exactly once** in S^3. After stereographic projection to $\mathbb{R}^3$, this appears as linked Villarceau circles on nested tori. The classical **Hopf invariant** of the map $h : S^3 \to S^2$ equals 1.

Linking is the topological feature that later underwrites **topological protection** of flux configurations on the gauged lattice: continuous deformations that preserve the fiber structure cannot unlink configurations without a cut or a singular transition.

Claim type: Theorem. Fiber structure and Hopf invariant $= 1$ are classical results in differential topology.

Software diagnostics. The library exposes linking helpers (`linking_number_pair`, `fiber_linking_number`, `fiber_pair_diagnostics`) for numerical checks on sampled curves—useful in labs, not substitutes for the classical theorem.

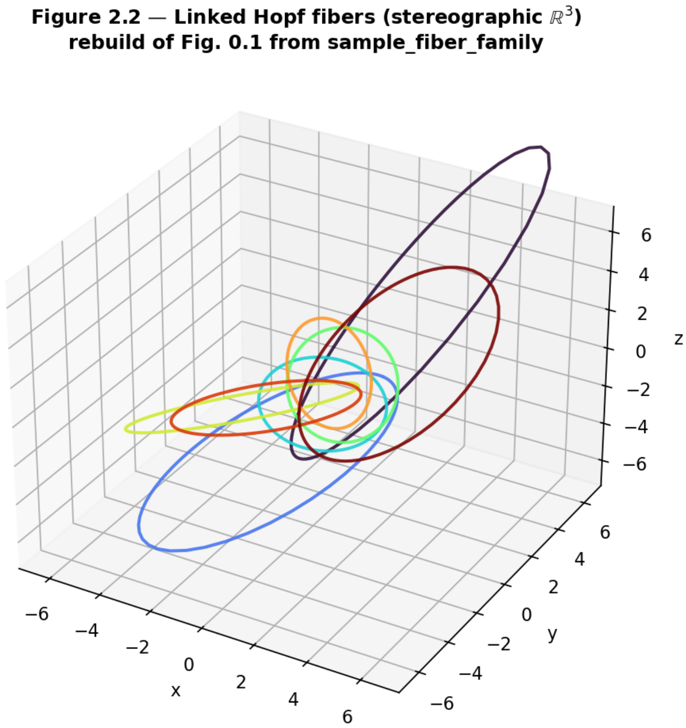


Figure 2.2: Rebuild of Figure 0.1, generated from `sample_fiber_family`. Several fibers after stereographic projection $S^3 \setminus \{\mathrm{pt}\} \to \mathbb{R}^3$. Linking cannot be undone by continuous deformation without cutting a fiber.

2.3 Stereographic projections and visualizations

2.3.1 Projection formula

Kingdom Come / `flux_hopf_lib` use a stereographic chart with pole related to the fourth coordinate (TOE-compatible convention):

$$(x_1, x_2, x_3, x_4) \longmapsto \left(\frac{s\,x_2}{1 - x_4},\ \frac{s\,x_3}{1 - x_4},\ \frac{s\,x_1}{1 - x_4}\right),$$

with default scale $s = 2$ (`stereographic_project`). Fibers map to closed curves in $\mathbb{R}^3$.

2.3.2 Portal layout

The Gradio **Hopf Visualizer** defaults to multi-panel **2D** views (xy, xz, phase map, S^2 base markers) for CPU-only hosts, with optional 3D WebGL modes locally. That multi-panel layout is the live counterpart of Figure 2.3.

Sampling API (Software fact). `sample_fiber(eta, xi1, n_points=...)` returns a **dictionary**, not a bare list of quaternions:

{`eta`, `xi1`, `xi2`, `x1..x4`, `y1..y3`, `px,py,pz`, `base_y1..base_y3`, ...}.

`sample_fiber_family(...)` returns a list of such dictionaries over a spread of base points.

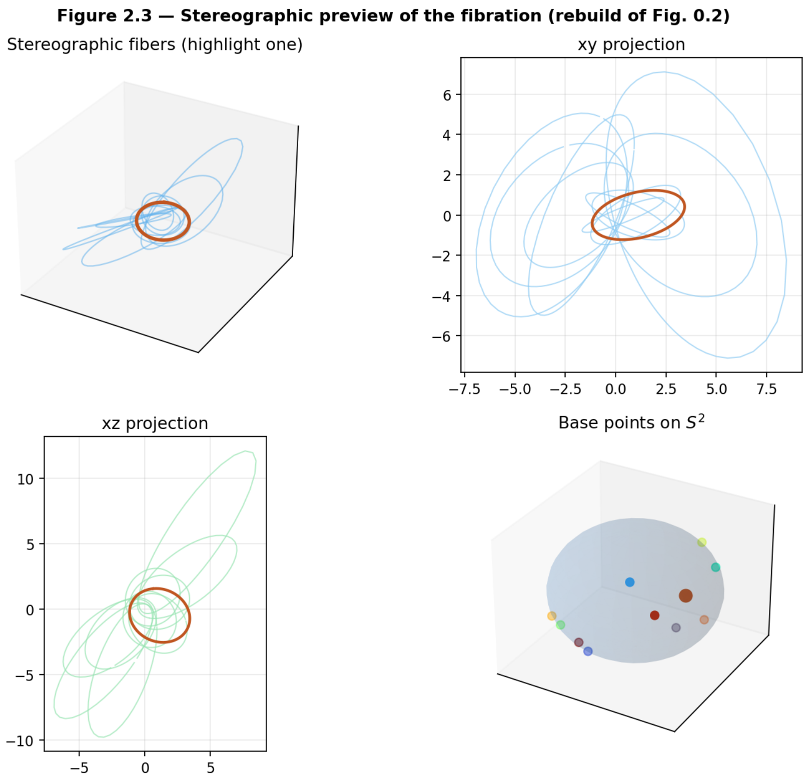


Figure 2.3: Rebuild of Figure 0.2. Composed stereographic fibers, planar projections, and base points on S^2, with one fiber highlighted—the same visual language as the portal's Classic Hopf preset.

2.4 Bundle structure and charts

2.4.1 Fiber bundle

The Hopf fibration is a fiber bundle

$$S^1 \hookrightarrow S^3 \twoheadrightarrow S^2.$$

It is the canonical example of a nontrivial principal $U(1)$-bundle (equivalently, an S^1-bundle). Locally it is a product $U \times S^1$; globally the twisting produces the linking of fibers.

2.4.2 Local charts on the base

The base S^2 is covered by the usual stereographic charts (north and south). Over each chart $U \subset S^2$ there is a local trivialization

$$h^{-1}(U) \cong U \times S^1,$$

so a fiber coordinate (phase) can be chosen continuously on U. Crossing from one chart to another changes the phase by a transition function valued in $U(1)$—the seed of **gauge** language for Part II.

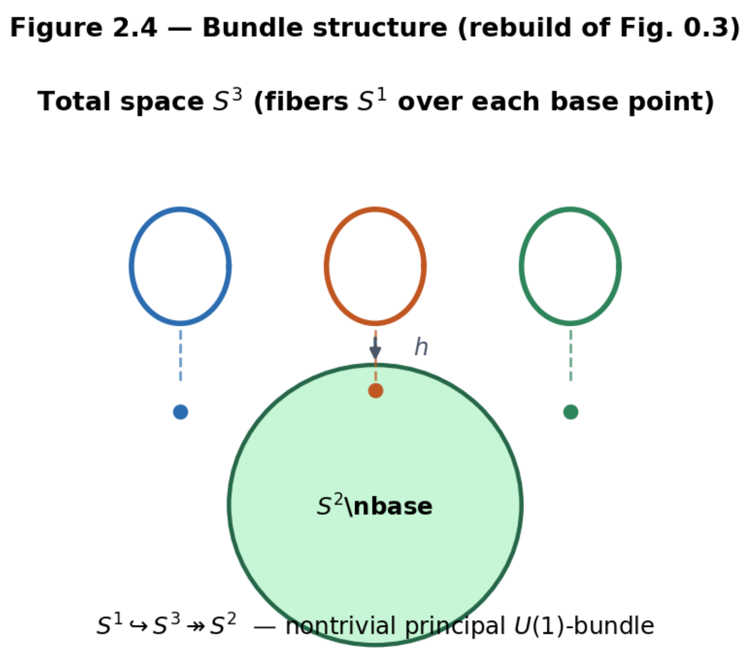


Figure 2.4: Rebuild of Figure 0.3. Conceptual diagram of the bundle structure that Chapter 3 will **discretize**: lattice points in the total space, Hopf projection to the base, adjacency and gauge along fibers.

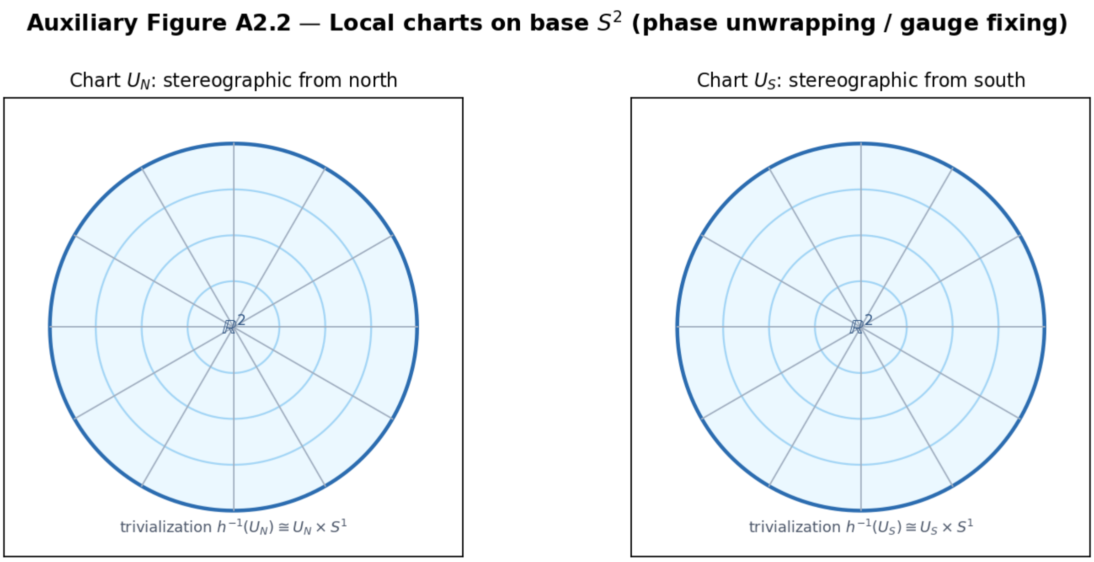


Figure 2.5: Schematic north/south charts on the base. Phase unwrapping and gauge fixing on the discrete lattice will live over such charts.

2.4.3 Phase sweep along a fiber

Holding the base point fixed and varying the fiber phase is the continuous analogue of "walking in the fiber direction." In Hatcher's Farey world, continued-fraction paths walk along edges of a planar diagram; here the primitive closed walk is a fiber circle.

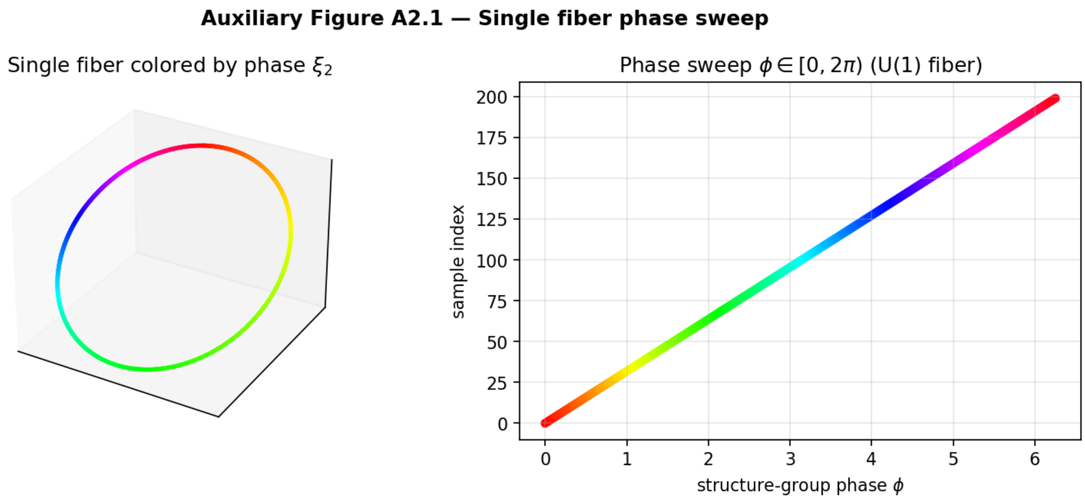


Figure 2.6: One **structure-group** fiber colored by phase $\phi \in [0, 2\pi)$. The path closes: a full common-phase sweep returns to the same unit quaternion (up to sampling).

Model reminder. Calling Hopf a higher-dimensional "Farey analogue" remains a **Model** metaphor until Chapter 3 axiomatizes discrete adjacency and mediants on the gauged lattice (Open Problem 1).

2.5 Connection to quaternions and rotations

2.5.1 Every unit quaternion lies on exactly one fiber

The Hopf map partitions S^3 into fibers. Chapter 1's unit group $\mathrm{Sp}(1) = S^3$ is therefore a circle bundle over S^2.

2.5.2 Left and right multiplications

If $u \in S^3$ is fixed:

- **Left multiplication** $q \mapsto uq$ is an isometry of S^3. It maps fibers to fibers and induces a rotation of the base S^2.
- **Right multiplication** $q \mapsto qu$ is also an isometry. It is **fiberwise** $U(1)$ only for the structure-group circle $\{e^{i\phi}\}$ in the identification where that circle is the fiber (and for Hurwitz units that normalize it). Under $q = z_1 + z_2 j$ as in §2.1, the structure-group circle is left multiplication by $e^{i\phi} \in \mathrm{span}\{1, i\}$. General right multiplication by an arbitrary unit of S^3 **moves fibers**.

Together, left and right actions generate the symmetry repertoire we will **gauge** and discretize in Chapters 3–4. The double cover $\mathrm{Spin}(3) \to SO(3)$ from §1.6 sits naturally here: directions in $\mathbb{R}^3$ relate to the base, while the fiber phase is the extra spinorial degree of freedom.

Claim type: Theorem for the continuous group actions and bundle structure; **Model** for their later role as lattice gauge symmetries.

2.6 First computational labs

Core API: `kingdom.core.hopf` · **Appendix C §C.1**.

- **2.A** `lib.hopf_lattice.hopf_map` — unit image on S^2 **without** output-normalization; $h(0,0,1,0) = (0,0,-1)$.
- **2.B** `sample_structure_group_fiber` — common-phase circle; check unit length on S^3 and constant base point.
- **2.C** Check fiber constancy: $h(e^{i\phi}z_1, e^{i\phi}z_2) = h(z_1, z_2)$. Separately, contrast the ξ_2-circle.
- **2.D–E** Phase sweep of the **structure-group** fiber; optional portal visualizer.

```
from lib.hopf_lattice import hopf_map, common_phase
import numpy as np
q = np.array([0.5, 0.5, 0.5, 0.5]); q = q / np.linalg.norm(q)
print(hopf_map(q), hopf_map(np.array([0.0, 0.0, 1.0, 0.0])))
```

2.7 Exercises

2.A (hand). On a few unit test points, check that the complex-pair formula $h(z_1, z_2) = (2\mathrm{Re}(\overline{z_1}z_2), 2\mathrm{Im}(\overline{z_1}z_2), |z_1|^2 - |z_2|^2)$ agrees componentwise with the real form in §2.1. Confirm $\|h(q)\| = 1$ with **no** extra normalization, and that $h(0,0,1,0) = (0,0,-1)$.

2.B (hand). Argue that every point of S^2 has a circle's worth of preimages under the Hopf map, using either the $\mathbb{CP}^1$ description or the (η, ξ_1, ξ_2) coordinates.

2.C (hand). Prove that simultaneous phase rotation $(z_1, z_2) \mapsto (e^{i\phi}z_1, e^{i\phi}z_2)$ leaves the formula of §2.1 invariant. Interpret this as motion along the **structure-group** fiber. Then explain in one sentence why sweeping ξ_2 at fixed (η, ξ_1) is a different circle.

2.D (code). Complete Labs 2.A–2.B. For three random unit quaternions, print `hopf_image()` and confirm each image has Euclidean norm ≈ 1.

2.E (code). Sample one structure-group fiber (`sample_structure_group_fiber`, $n \geq 64$). Confirm unit length, constant Hopf image, and a closed stereographic loop. Optionally contrast an angle-chart ξ_2-circle.

2.F (visual). In the Gradio Hopf Visualizer, switch between **Classic Hopf** and a custom phase sweep. In one paragraph, describe how the linked-fiber view (Fig. 2.2) emerges from the multi-panel layout (Fig. 2.3).

2.G (Hatcher bridge). In Hatcher, continued-fraction convergents trace zigzag paths on the Farey diagram. Guess (and later verify in Ch. 3) what the discrete analogue of "walking along a fiber" might be on the gauged Hopf lattice.

2.H (forward). Why might left and right multiplications by unit quaternions be natural candidates for the gauge symmetries of a lattice built on S^3? (Answered in Chapter 3.)

2.I (connection to Ch. 1). Using `from_axis_angle` and `hopf_image`, pick a one-parameter family of unit quaternions rotating about a fixed axis. Plot or tabulate how the Hopf image moves on S^2. Relate to left multiplication acting on the base.

2.8 Code and asset pointers

```
lib.hopf_lattice.hopf_map              # the book's map (this chapter / labs)
lib.hopf_lattice.common_phase, sample_structure_group_fiber
flux_hopf_lib.hopf.fibration.hopf_map  # same formula
kingdom.core.hopf                      # re-export
lib.hopf_lattice.legacy_portal_map     # NOT Hopf; old 3-component formula
```

Figures: generated under `book/figures/fig2_*` and `aux2_*` via `scripts/generate_ch2_figures.py` (samples real fibers from `flux_hopf_lib` when available). **Related portal:** Hopf Visualizer tab — live counterpart of Figures 2.2–2.4 and Aux A2.1. **Chapter 0 assets:** Figs. 0.1–0.3 remain the narrative first look; Figs. 2.2–2.4 are the first-principles rebuilds.

2.9 Looking ahead

The Hopf fibration supplies the geometric backbone: S^3 as total space, S^2 as base, and linked circle fibers as the primitive topological objects. In **Chapter 3** we discretize this picture by placing a lattice (built on the Hurwitz preference of Chapter 1) inside S^3, projecting via the Hopf map, and defining adjacency and gauge actions. The same linking that makes the fibers topologically nontrivial will become the protection mechanism for **flux flywheels**.

With the algebraic language of quaternions and the bundle geometry of the Hopf fibration now in place, we are ready to construct the **gauged Hopf lattice**—the direct higher-dimensional lift of Hatcher's Farey diagram.

Part II

The Gauged Hopf Lattice (The Farey Lift)

Chapter 3

Construction of the Gauged Hopf Lattice

This chapter discretizes the Hopf fibration. We place a lattice inside the total space S^3 (built on the Hurwitz preference of Chapter 1), project it via the Hopf map of Chapter 2, and define adjacency and gauge actions. The resulting **gauged Hopf lattice** is the direct higher-dimensional analogue of Hatcher's Farey diagram—**as a Model construction**, pending a uniqueness theorem (Open Problem 1). Left and right multiplications become gauge symmetries; linked fibers become the carriers of topological protection for flux configurations.

Learning goals

1. Construct discrete point sets inside S^3 using Hurwitz units and denser samplings.
2. Define the gauged Hopf lattice via Hopf projection and adjacency rules.
3. Introduce mediant-like combination rules and discrete fiber walks.
4. Define gauge fields and the first notion of flux on the lattice.
5. Meet flux flywheels as topologically protected rotating configurations.
6. State Open Problem 1 (canonical quaternionic Farey structure) with a clear research target.

Figures in this chapter

Tag	File	Role
Fig. 3.1	`figures/fig3_1_hurwitz_lattice_in_s3.png`	24 Hurwitz units on S^3 (stereographic)
Fig. 3.2	`figures/fig3_2_hopf_projected_lattice.png`	Angle-sampled lattice $\to$ base S^2
Fig. 3.3	`figures/fig3_3_adjacency_and_fibers.png`	Candidate along-/inter-fiber adjacency
Fig. 3.4	`figures/fig3_4_gauge_action.png`	Left/right multiplication as gauge actions
Fig. 3.5	`figures/fig3_5_flux_flywheel_schematic.png`	Flux flywheel schematic
Aux A3.1	`figures/aux3_1_lattice_simulator_still.png`	Two-gyro Lattice Simulator still

Claim discipline

Claim	Type
Existence of the Hurwitz order, 24 units, Euclidean property (Ch. 1)	**Theorem**
Hopf fibration structure and linking (Ch. 2)	**Theorem**
Specific discrete adjacency / mediant rules; "the" gauged Hopf lattice as Farey lift	**Model** (Open Problem 1)
Flux flywheels as protected circulating configurations; dynamical two-gyro demos	**Model** + **Software fact**
`kingdom.core.lattice.LatticeConfig`, `kingdom.simulations.lattice.TwoGyroLattice`, book helper `qga/lib/hopf_lattice.py`	**Software fact**

3.1 The Hurwitz lattice inside S^3

Chapter 1 established the **Hurwitz order** $\mathcal{H}$ as the preferred integer lattice in $\mathbb{H}$ (**Model** preference for later arithmetic; classical Euclidean properties are **Theorems**). Its 24 units form a finite group—the binary tetrahedral group—lying on the unit sphere:

$$\Lambda_0 \;:=\; \{q \in \mathcal{H} : N(q) = 1\} \;=\; \mathcal{H} \cap S^3.$$

Explicitly, the units are

$$\{\pm 1, \pm i, \pm j, \pm k\} \;\cup\; \left\{\tfrac{1}{2}(\pm 1 \pm i \pm j \pm k)\right\}$$

(all sign combinations in the second family).

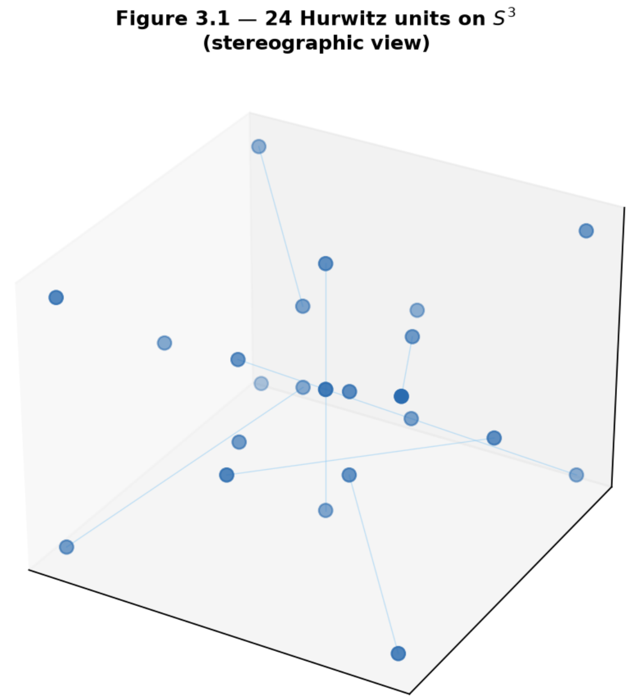


Figure 3.1: The 24 Hurwitz units after stereographic projection to $\mathbb{R}^3$. This is the smallest nontrivial discrete substrate for total-space geometry compatible with Chapter 1.

3.1.1 Density beyond 24 points

A theory with only 24 sites is too sparse for Farey-style adjacency, continued-fraction walks, or continuum-like flux. We therefore work with **nested levels** of discreteness:

Level	Object	Role
Λ_0	Hurwitz units ($N = 1$)	Exact arithmetic skeleton; discrete gauge group
Λ_n	$\{q/\lvert q\rvert : q \in \mathcal{H},\ N(q) = n\}$ (optional)	Norm shells; arithmetic depth (Ch. 9)
Λ_{ang}	Samples of Hopf angles (η, ξ_1, ξ_2)	Dense visualization / numerical OP1 tests
Λ_{dyn}	Site quaternions in `TwoGyroLattice`	Dynamical Model used in the portal

Software fact. The book helper `qga/lib/hopf_lattice.py` implements Λ_0 and Λ_{ang} for labs and figures. Kingdom Come's live portal currently emphasizes Λ_{dyn} (see §3.6).

3.2 The gauged Hopf lattice

3.2.1 Projection via the Hopf map

Apply the Hopf map $h : S^3 \to S^2$ (Chapter 2) to a discrete set $\Lambda \subset S^3$. The image $h(\Lambda)$ is a discrete set of base points. Each fiber over those base points may contain multiple samples from Λ (phase discretization).

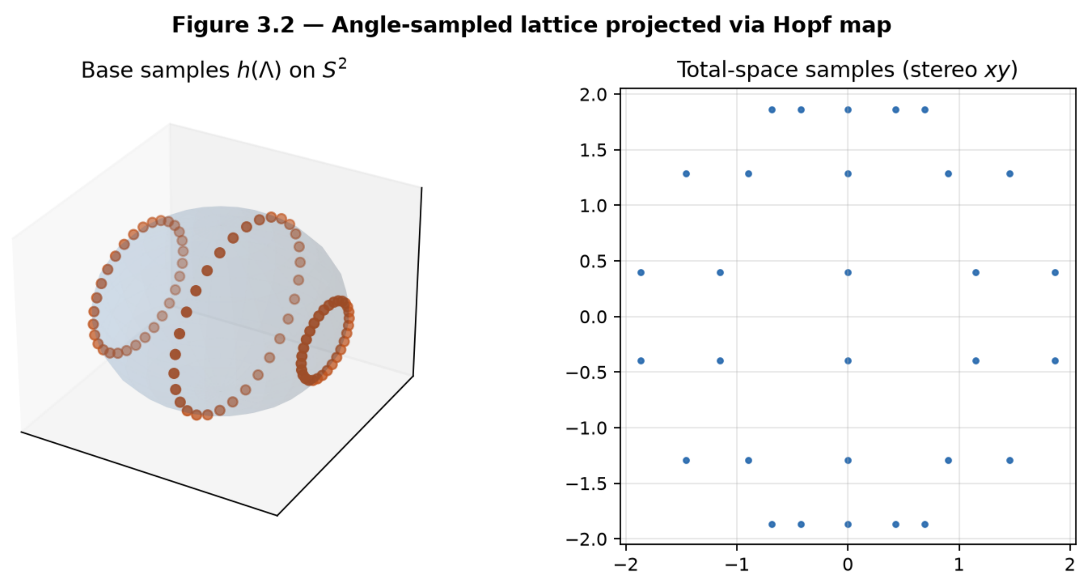


Figure 3.2: Left: base samples on S^2. Right: total-space samples in a stereographic xy view. Generated from `sample_angle_lattice` + `hopf_project_points` in the book helper.

3.2.2 Definition (working)

We define a **gauged Hopf lattice** as a tuple

$$(\Lambda,\ h(\Lambda),\ E_{\|},\ E_{\perp},\ \mathcal{G})$$

where:

- $\Lambda \subset S^3$ is a discrete point set (one of the levels above),
- $h(\Lambda)$ is its Hopf image on the base,
- $E_{\|}$ is the set of **along-fiber** edges (phase neighbors),
- $E_{\perp}$ is the set of **inter-fiber** edges (base-neighbor lifts),
- $\mathcal{G}$ is a group of **gauge transformations** acting on Λ (left/right multiplications by units, and continuous one-parameter subgroups in the dynamical Model).

3.2.3 Adjacency rules (Model / OP1)

Two lattice points $q_1, q_2 \in \Lambda$ may be declared adjacent in either of two ways:

1. **Along-fiber ($E_{\|}$).** They lie on (approximately) the same Hopf fiber and are neighbors in discrete phase ξ_2.
2. **Inter-fiber ($E_{\perp}$).** Their base points $h(q_1), h(q_2)$ are neighbors under a discrete rule on S^2, and a lift to S^3 is chosen (gauge condition).

A **mediant-like rule** would supply, for adjacent base points, a canonical "sum" or interpolant in the total space—exactly the quaternionic lift of Hatcher's

$$\frac{a}{c} \oplus \frac{b}{d} = \frac{a+b}{c+d}.$$

No single such rule is yet proved canonical.

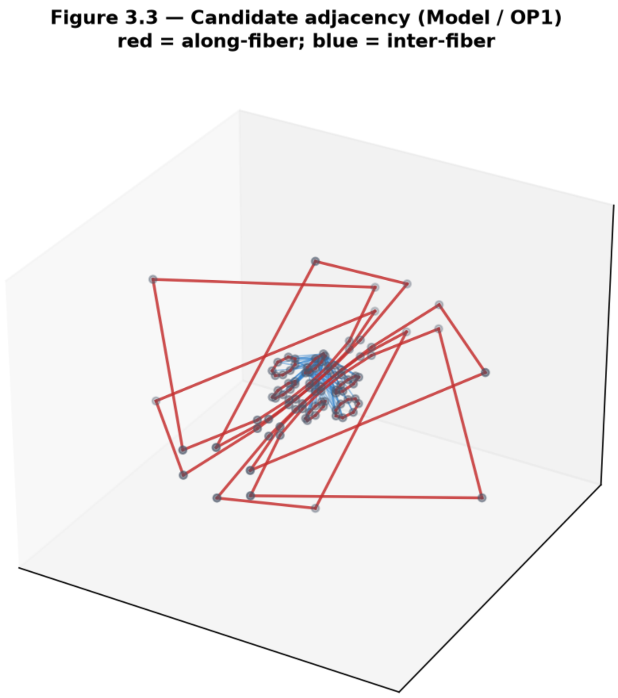


Figure 3.3: Red: along-fiber edges. Blue: inter-fiber edges from a **candidate** rule (angular nearest neighbors on the base after Hopf projection, plus phase neighbors). This is an experimental combinatorial object, not a theorem.

Model note. The adjacency relation is a design choice that *should* reduce to classical Farey adjacency under a suitable projection or restriction. That reduction (and uniqueness among reasonable choices) is **Open Problem 1**.

3.2.4 Candidate rule used in figures and labs

The book helper implements one explicit candidate (`candidate_adjacency`):

- Recover rough (η, ξ_1, ξ_2) from coordinates.
- Angle-chart ξ_2-circle (not the Hopf $U(1)$ fiber): close in (η, ξ_1), neighboring in ξ_2.
- Inter-chart: angular distance on S^2 below a threshold; exclude ξ_2-circle pairs.

Until Open Problem 1 is resolved, do not treat this adjacency as if the base were the Hopf fibration of Chapter 2.

Other candidates (spherical Delaunay on the base, Hurwitz-norm determinants, geodesic mediants, ...) are welcome; record them against OP1.

3.3 Gauge actions: left and right multiplications

From Chapters 1–2:

- **Left multiplication** $q \mapsto uq$ (unit u) is an isometry of S^3, maps fibers to fibers, and induces a rotation of the base S^2.
- **Right multiplication** $q \mapsto qu$ is an isometry. Fiberwise $U(1)$ holds only for the structure-group circle $\{e^{i\phi}\}$ (and Hurwitz units that normalize it); under this book's $q = z_1 + z_2 j$ that circle is left multiplication by $e^{i\phi}$. General right S^3 **moves fibers**.

On a discrete set Λ we therefore equip two families of **gauge transformations**:

Action	Continuous	Discrete (Hurwitz)
Left	$u \in S^3$	$u \in \Lambda_0$ (24 units)
Right	$u \in S^3$ (fiberwise only for $\{e^{i\phi}\}$)	$u \in \Lambda_0$

When $\Lambda = \Lambda_0$, left/right multiplications by units **permute** the lattice exactly. On denser samples they move points within S^3; one may re-snap to nearest lattice points (another OP1 design choice).

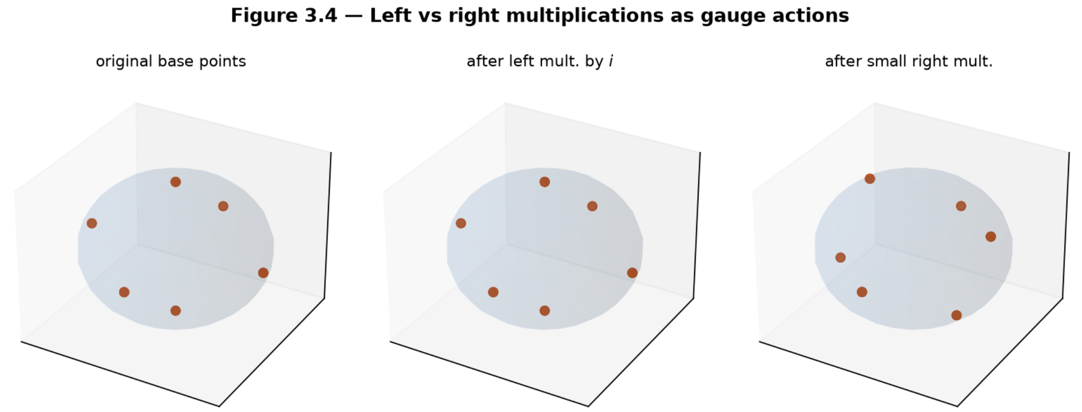


Figure 3.4: Hopf images of the 24 Hurwitz units: original; after left multiplication by i; after a small right multiplication (phase-like). Left multiplication by a Hurwitz unit **permutes** Λ_0, so the multiset of base points is invariant even though individual assignments change (see Lab 3.C note and Ch. 4 Lab 4.A).

3.3.1 Dynamical gauge in the portal (Model + software)

Kingdom Come's `TwoGyroLattice` does not store a static Farey graph. It evolves n site quaternions with left/right rotors and a global **gauge rotation** driven by mean twist imbalance:

$$q_i \;\leftarrow\; \delta_L\, q_i\, \overline{\delta_R}, \quad \text{then} \quad q_i \;\leftarrow\; q_i\, g(\alpha),$$

with gauge angle α proportional to average twist. **Identity preservation** tracks how well a parallel-transported identity field stays aligned—stable vs chaotic modes differ mainly in `gauge_strength` and detuning. This is a **dynamical Model** of gauged flux, not yet a combinatorial Farey theory.

3.4 Flux configurations and flywheels (first introduction)

3.4.1 Discrete flux

A **flux configuration** on the gauged Hopf lattice is an assignment of oriented edge weights

$$\Phi : E_{\parallel} \cup E_{\perp} \to \mathbb{Z}$$

(or to $\mathbb{R}$, in continuum limits) that transforms naturally under gauge actions and satisfies a discrete conservation law at vertices (coboundary / Kirchhoff condition)—to be refined in Chapter 5 with topographs.

3.4.2 Flux flywheel

A **flux flywheel** is a closed, topologically protected circulating flux configuration: a periodic orbit of gauge and phase that cannot unwind continuously without cutting a linked fiber or violating gauge constraints. In the continuous / dynamical limit these become the rotating flux structures of the Kingdom Come Model (Ch. 0, Fig. 0.4; portal Flux Flywheel tab).

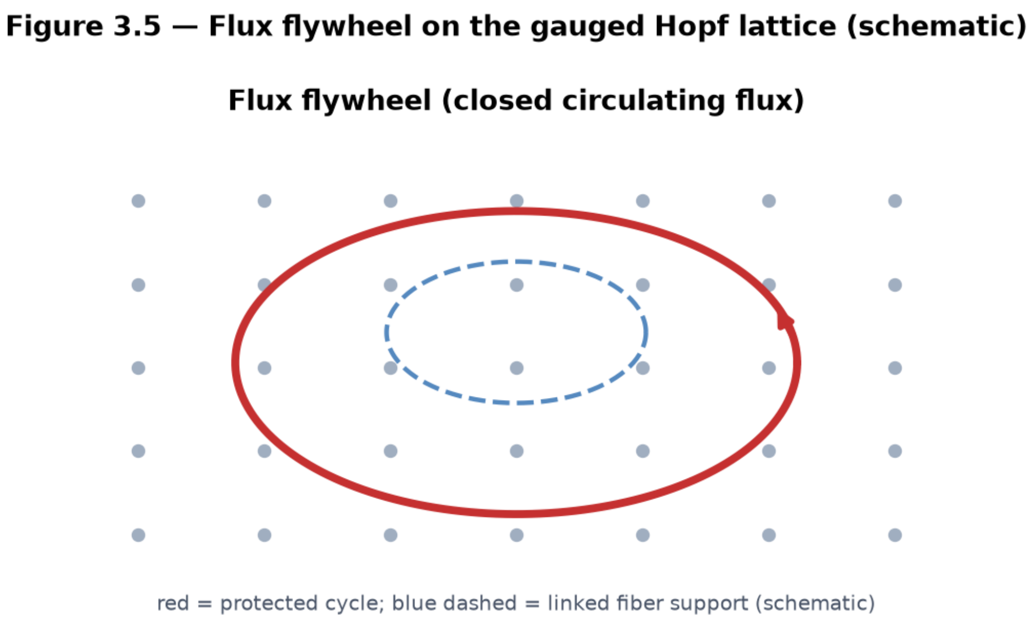


Figure 3.5: Closed circulating flux (red) supported on lattice edges and fibers; a linked fiber (blue dashed) hints at the topological obstruction to trivialization.

Claim type. Existence and stability of flywheels as *physical* objects: **Model**. Linking of Hopf fibers as a classical topological fact: **Theorem** (Ch. 2). Mapping flywheels to atomic number Z: later chapters (**Model** / **Hypothesis**).

3.5 Open Problem 1 — Canonical quaternionic Farey structure

Open Problem 1 (core of this chapter and of Part II). Prove uniqueness (or classify all reasonable choices) of discrete adjacency / mediant rules on the gauged Hopf lattice such that:

1. The structure **reduces to the classical Farey diagram** under a suitable projection or restriction (e.g. a fixed complex line, a parabolic subgroup, or a 2D slice of base coordinates).
2. Adjacency is **compatible** with left and right gauge actions (equivariance).
3. The resulting graph admits a natural notion of **continued-fraction paths** (discrete walks that refine approximants), including walks that advance primarily **along fibers**.
4. (Optional but desirable.) A Euclidean or greedy algorithm on the lattice recovers shortest paths analogous to Farey mediants.

Current status (draft).

Layer	Status
Hurwitz units Λ_0	Classical, settled
Hopf projection of discrete sets	Implemented (portal + book helper)
Candidate adjacency (`candidate_adjacency`)	Numerical experiments only — **not** proved canonical
Reduction to Farey $\lvert ad - bc\rvert = 1$	**Open**
Equivariant mediant	**Open**
Dynamical two-gyro lattice	Portal demo; parallel track to combinatorial theory

Research target. Any reader who solves or substantially advances Open Problem 1 produces the central combinatorial object of the entire book.

See also `notes/open_problems.md` (OP1 status tracking).

3.6 Software map and first computational labs

Component	Location
`LatticeConfig`	`kingdom.core.lattice`
`TwoGyroLattice`	`kingdom.simulations.lattice`
Hurwitz / adjacency / gauge	`lib/hopf_lattice.py`

Honest gap. No production `build_hurwitz_lattice_sample` in Kingdom Come yet — pedagogy lives in `lib/`.

Labs (short form). Full code: **Appendix C §C.2**.

- **3.A** 24 Hurwitz units; Hopf project; count base points.
- **3.B** `sample_angle_lattice` + `candidate_adjacency` (OP1 sandbox).
- **3.C** Left *imesi*: base **set** invariant, total-space points move (see Lab 4.A).
- **3.D** `run_lattice_comparison` stable vs chaotic.
- **3.E** Toy `discrete_flux_cycle`.

```python
from lib.hopf_lattice import HURWITZ_UNITS, hopf_project_points
print(len(HURWITZ_UNITS), hopf_project_points(HURWITZ_UNITS).shape)
```

3.7 Exercises

3.A (hand). List the 24 Hurwitz units and verify $N(q) = 1$ for each family ($\pm 1, \pm i, \pm j, \pm k$ and half-integer units).

3.B (hand). In two sentences, describe how left multiplication by a fixed unit quaternion acts on the base S^2 via the Hopf map.

3.C (code). Complete Labs 3.A–3.C. Report the number of approximate distinct base points for Λ_0 and for one angle lattice. For Lab 3.C, confirm that the sorted base multiset is invariant under left multiplication by i while some index-wise base coordinates still move.

3.D (code). Apply a sequence of small right multiplications to Λ_0 (or one fiber of Λ_{ang}) and check approximate return after total angle 2π.

3.E (visual / dynamical). Run Lab 3.D (simulator) or the Gradio Lattice Simulator. Describe what "identity preservation" looks like for stable vs chaotic modes.

3.F (Hatcher bridge). In Hatcher, two fractions a/c and b/d are Farey-adjacent iff $|ad - bc| = 1$. Propose a quaternionic analogue (e.g. $|N(q_1\overline{q_2} - q_2\overline{q_1})|$ or a 2×2 minor from complex pairs) and test it numerically on Λ_0 or Λ_{ang}. Record whether it recovers a subgraph of `candidate_adjacency`.

3.G (forward). Why is topological linking of fibers a natural candidate for the protection mechanism of flux flywheels? (Developed in Chapters 5 and 10.)

3.H (Open Problem 1 teaser). Run the current `candidate_adjacency` implementation. Does it reduce to classical Farey adjacency under any obvious projection or 2D slice? Document one negative or partial observation—progress often starts as a failed reduction.

3.I (software honesty). In one paragraph, distinguish: (i) Hurwitz arithmetic lattice, (ii) angle-sampled visualization lattice, (iii) two-gyro dynamical lattice. Which claim types apply to each?

3.8 Code and asset pointers

```
# Book pedagogy (this repo)
qga/lib/hopf_lattice.py
  HURWITZ_UNITS, sample_angle_lattice, hopf_project_points,
  candidate_adjacency, left_multiply, right_multiply, discrete_flux_cycle

# Kingdom Come
kingdom.core.lattice.LatticeConfig
kingdom.simulations.lattice.TwoGyroLattice
kingdom.simulations.lattice.run_lattice_comparison
kingdom.simulations.lattice.build_lattice_figure
kingdom.core.hopf                    # projection / fibers
```

Figures: `scripts/generate_ch3_figures.py` **Related portal:** Lattice Simulator tab — dynamical counterpart of Aux A3.1. **Open problems:** `notes/open_problems.md` · OP1.

3.9 Looking ahead

We now possess the discrete geometric object at the heart of the book: the **gauged Hopf lattice**, still with a **Model**-level adjacency rule pending OP1. In **Chapter 4** we study its symmetries in detail—the full gauge group generated by left and right actions, periodicity, and comparison with Hatcher's linear fractional transformations. Chapters 5–7 develop flux topographs, classification, and the $Z \mapsto$ flywheel map on this lattice.

Forward pointer (invariant discipline). The lattice of this chapter is already a place where discrete combinatorial structure meets continuous S^3 geometry. Later chapters will assign **labels**, **scores**, and **classes** to configurations. When that happens, keep the discipline developed fully in **Chapter 9, section 9.5**: algebraic cycle structure, sequential label progression, and true angle–label lock are *different* invariants. Visual order is not statistical locking unless tested against a null. Full treatment, figure, and controls: Chapter 9 section 9.5 · `notes/RESEARCH_NOTE_moduli.md` · supporting stack `vortex_math`.

With the gauged Hopf lattice constructed, we are ready to examine its symmetries—the higher-dimensional lift of the linear fractional transformations that organize Hatcher's Farey diagram.

Chapter 4

Symmetries of the Gauged Hopf Lattice

This chapter examines the symmetries generated by left and right multiplications on the gauged Hopf lattice of Chapter 3. These actions play the role that linear fractional transformations play for Hatcher's Farey diagram. We classify periodic structures, glide-like motions, and rotational symmetries, and see how the gauge group acts on flux configurations and flywheels.

Learning goals

1. Treat left and right multiplications as the fundamental gauge symmetries of the lattice.
2. Compare the quaternionic gauge actions with Hatcher's linear fractional transformations.
3. Identify periodic orbits, glide-like actions, and rotational symmetries.
4. Understand how symmetries act on flux configurations and preserve (or map) flywheels.
5. Prepare the symmetry language required for flux topographs (Chapter 5).

Figures in this chapter

Tag	File	Role
Fig. 4.1	`figures/fig4_1_left_right_symmetries.png`	Left vs right gauge actions on base and fibers
Fig. 4.2	`figures/fig4_2_periodic_orbit.png`	Orbit under combined gauge sequence
Fig. 4.3	`figures/fig4_3_glide_reflection_analogue.png`	Glide-like helical symmetry
Fig. 4.4	`figures/fig4_4_symmetry_on_flywheel.png`	Gauge symmetry preserving a flux flywheel
Aux A4.1	`figures/aux4_1_gauge_group_elements.png`	Discrete gauge elements (24 Hurwitz units)

Claim discipline

Claim	Type
Left and right multiplications are isometries of S^3; classical Hopf bundle structure group	**Theorem**
Discrete gauge actions as "the modular group of the lattice"; equivariance of candidate adjacency; glide analogues	**Model** (pending OP1)
Action of gauge on flux/flywheels in the combinatorial and dynamical Models	**Model**
`qga/lib/hopf_lattice.py` symmetry helpers; `TwoGyroLattice` gauge evolution	**Software fact**

4.1 Left and right multiplications as gauge symmetries

Chapter 3 equipped the gauged Hopf lattice with two families of gauge transformations coming directly from quaternion multiplication:

- **Left multiplication** $q \mapsto uq$ (fixed unit u) is an isometry of S^3. It maps fibers to fibers and induces a rotation of the base S^2 via the Hopf map.
- **Right multiplication** $q \mapsto qu$ is an isometry. It is fiberwise $U(1)$ **only** for the structure-group circle $\{e^{i\phi}\}$ (and Hurwitz units that normalize it). Under $q = z_1 + z_2 j$, that circle is left multiplication by $e^{i\phi} \in \mathrm{span}\{1, i\}$. General right multiplication by an arbitrary unit of S^3 **moves fibers**.

On a discrete set Λ these become the operational gauge symmetries.

Domain	Left action	Right action
Λ_0 (24 Hurwitz units)	Permutes Λ_0 exactly	Permutes Λ_0 exactly
Λ_{ang} (angle sample)	Moves samples in S^3	Moves samples (phase-like)
Λ_{dyn} (`TwoGyroLattice`)	Continuous rotors δ_L	Continuous rotors δ_R + global gauge $g(\alpha)$

Re-snapping a moved dense sample to a nearest lattice point is a design choice tied to **Open Problem 1** (canonical adjacency must decide how continuous symmetries interact with discrete graphs).

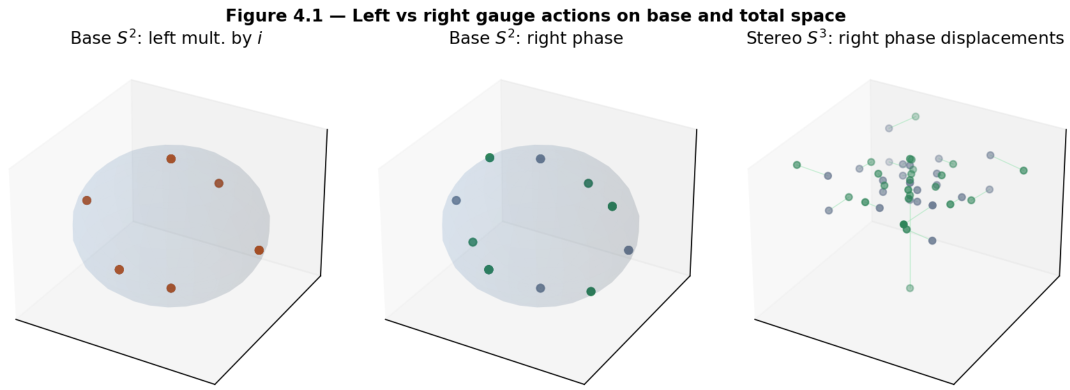


Figure 4.1: Left: base points of the Hurwitz units under left multiplication by i. Middle: base under a small right phase. Right: stereographic displacements under the same right phase. Because left multiplication by a Hurwitz unit **permutes** Λ_0, the multiset of projected base points is invariant—individual assignments change, but the sorted base set does not.

Theorem. For any unit $u \in S^3$, the maps $q \mapsto uq$ and $q \mapsto qu$ are isometries of S^3 with respect to the round metric (they preserve the Euclidean norm on $\mathbb{R}^4$ and fix the sphere).

4.2 Comparison with Hatcher's linear fractional transformations

Hatcher organizes the Farey diagram with the action of linear fractional transformations

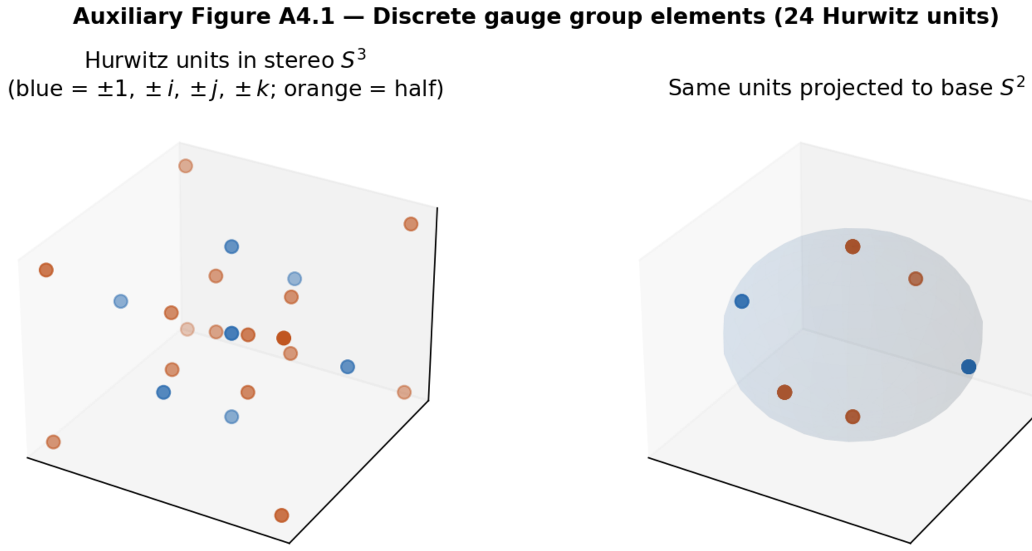


Figure 4.2: The 24 Hurwitz units as the discrete gauge group generators/elements: stereographic S^3 and Hopf projection to S^2.

$$T(z) = \frac{az+b}{cz+d}, \qquad ad - bc = \pm 1 \quad \text{(this is PGL}(2,\mathbb{Z})\text{, not PSL}(2,\mathbb{Z})\text{)}.$$

These preserve adjacency, generate translations, glide reflections, and rotations, and produce the periodic structures of continued fractions and topographs.

In the gauged Hopf lattice the analogous organizing actions are **left and right quaternion multiplications**. The structural parallel is strong but remains a **Model** until a canonical adjacency rule is established (Open Problem 1) and equivariance is proved.

Feature in Hatcher	Quaternionic analogue
Linear fractional transformations (PGL$(2,\mathbb{Z})$; PSL plus orientation-reversing glides)	Left + right multiplications by units
Preservation of Farey adjacency	Preservation of candidate adjacency (**Model** / OP1 test)
Translations along periodic strips	Repeated right phase advance along fibers
Glide reflections	Combined left+right actions producing helical orbits
Rotations about vertices or triangle centers	Left actions that fix or cycle base points
Periodic continued-fraction expansions	Periodic orbits under combined gauge sequences

The quaternionic version is richer—two interacting actions rather than a single matrix group—and lives on the three-sphere rather than the hyperbolic plane.

Claim type. The parallel is a **Model** metaphor until OP1 is resolved and an equivariant reduction to the classical Farey case is proved.

4.3 Periodic orbits and glide-like symmetries

4.3.1 Periodic orbits

A finite sequence of left and right multiplications often produces a **periodic orbit** on the lattice: after k applications of a composite map g, a point returns (exactly, for Λ_0; approximately, for dense samples). These orbits are discrete analogues of the periodic separator lines that appear in Hatcher's topographs and of the closed continued-fraction periods that generate Pell solutions.

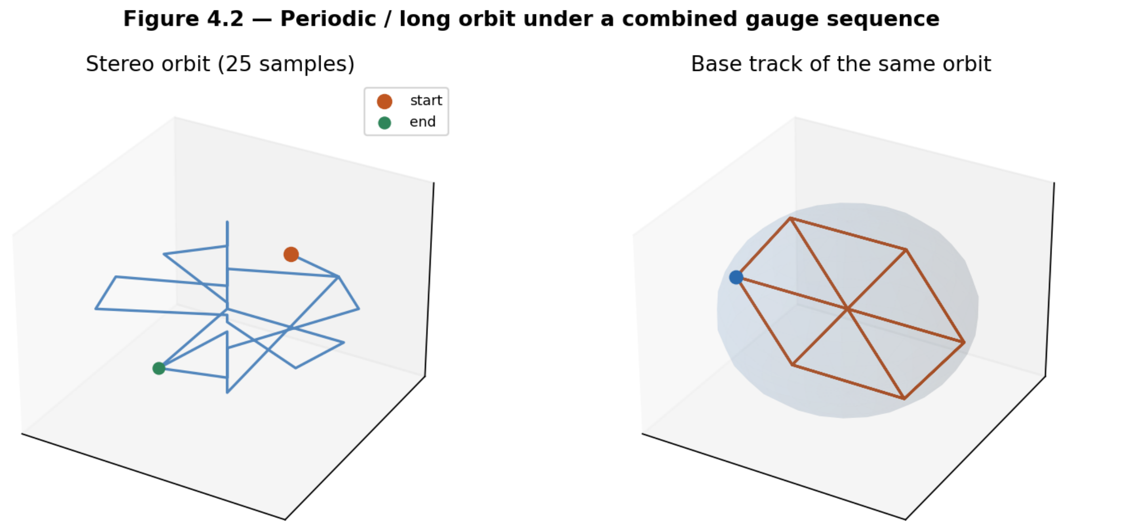


Figure 4.3: Total-space (stereographic) and base tracks of a point under iterated left and right steps. Exact period depends on the sequence; on Λ_0 periods divide the order of the induced permutation.

4.3.2 Glide-like symmetries

Glide-like symmetries arise when a right multiplication (phase advance along a fiber) is paired with a left multiplication that shifts the base point. The resulting motion is helical—a higher-dimensional lift of a classical glide reflection along a Farey strip.

4.3.3 Book API for sequences

```
phase_unit(theta)                  # right-phase unit quaternion
apply_gauge_step(points, 'L'|'R', unit)
apply_gauge_sequence(points, [('L', u), ('R', v), ...])
orbit_of_point(q, sequence, max_periods=..., tol=...)
permutes_hurwitz_units(unit, side='L'|'R')
adjacency_equivariance_score(points, unit, side=...)  # OP1 diagnostic
```

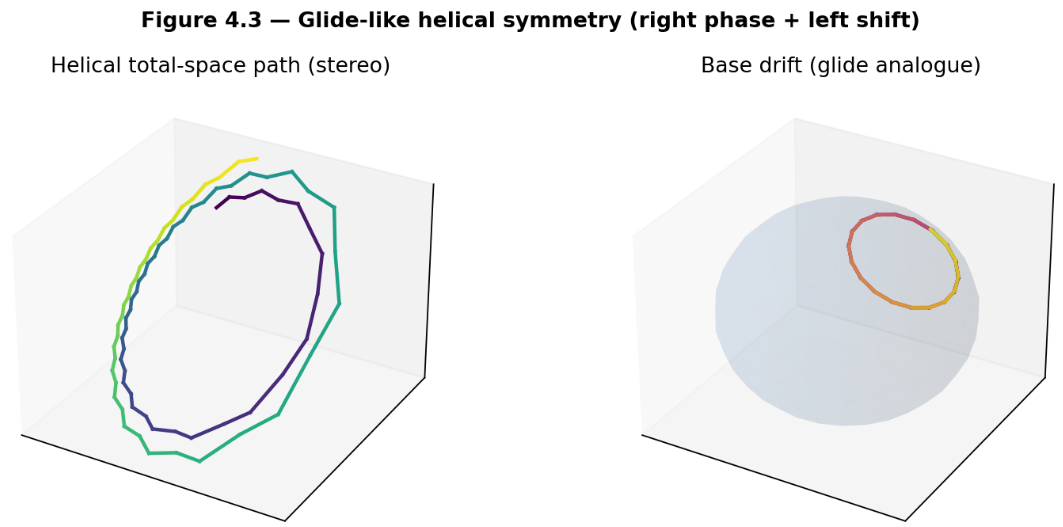


Figure 4.4: Combined right phase + left shift iterated: a helical path in the total space and a drifting track on the base. **Model** analogue of Hatcher's glide reflections, not a theorem of uniqueness.

4.4 Symmetries acting on flux configurations and flywheels

4.4.1 Push-forward of discrete flux

If $\Phi : E \to \mathbb{Z}$ assigns oriented flux to edges of the lattice graph, a gauge symmetry g that maps vertices (and therefore edges) induces

$$(g \cdot \Phi)(g(e)) = \Phi(e).$$

Conservation at vertices (discrete $\mathrm{d}^*\Phi = 0$) is preserved whenever g is a graph automorphism. When g only approximately preserves candidate adjacency (OP1 unfinished), flux push-forward is well-defined on the abstract edge set of Λ with indices, but may leave the "along-fiber / inter-fiber" typing.

Helpers: `discrete_flux_cycle`, `transform_flux`, `nearest_index_map` in `qga/lib/hopf_lattice.py`.

4.4.2 Flywheels as equivariant cycles

Topologically protected **flux flywheels** (Ch. 3) are closed circulating configurations whose supporting cycles are mapped to equivalent cycles by the gauge group. The linking of Hopf fibers (Ch. 2, **Theorem**) supplies a topological obstruction to trivialization that is preserved by all continuous isometries of the fibration—and therefore by left/right multiplications.

4.4.3 Dynamical gauge (software / Model)

In `TwoGyroLattice`, each frame applies left/right rotors and a global right-type gauge rotation driven by mean twist. **Identity preservation** is a numerical proxy for "how well the gauged configuration remembers its initial orientation." Stable vs chaotic modes (Ch. 3 Aux A3.1) are dynamical illustrations of ordered vs disordered gauge orbits—not combinatorial periods on Λ_0.

Figure 4.4 — Gauge symmetry mapping a flux flywheel to an equivalent cycle

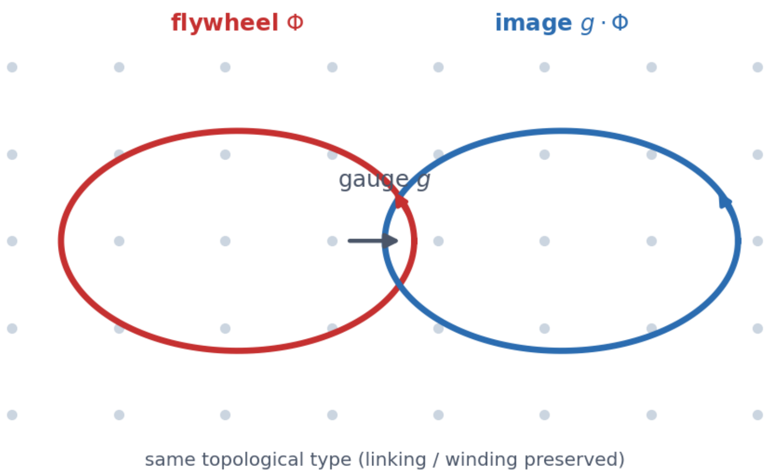


Figure 4.5: A gauge transformation maps a closed circulating flux cycle to another cycle of the same topological type. In the **Model**, flywheels are equivariant under the symmetry group developed here—exactly the structure Chapter 5 will demand of flux topographs.

4.5 First computational labs

Helpers: `lib/hopf_lattice.py` · dynamics: `TwoGyroLattice` · **Appendix C §C.2**.

- **4.A** `permutes_hurwitz_units`; base-set invariance vs total-space motion.
- **4.B** `orbit_of_point` under a left/right sequence.
- **4.C** `step_frame()` identity preservation (stable vs chaotic).
- **4.D** Flux push-forward + `adjacency_equivariance_score` (OP1).

```python
from lib.hopf_lattice import HURWITZ_UNITS, permutes_hurwitz_units
import numpy as np
print(permutes_hurwitz_units(np.array([0.,1.,0.,0.]), side="L"))
```

4.6 Exercises

4.A (hand). Prove that left multiplication by a unit quaternion is an isometry of S^3 (preserve $N(uq) = N(u)N(q) = N(q)$).

4.B (hand). In one paragraph, identify the structure-group circle $\{e^{i\phi}\}$ for the Hopf map of Chapter 2, and explain why **general** right multiplication by an arbitrary unit of S^3 moves fibers rather than acting fiberwise.

4.C (code). Complete Lab 4.A. Confirm `permutes_hurwitz_units` for several Hurwitz units on both left and right. For each unit, report: (i) sorted base multiset invariant? (ii) total-space index-wise motion? (iii) base index-wise motion?

4.D (code). Complete Lab 4.B. Find a short non-trivial sequence that returns (approximately) to the start; report period length.

4.E (visual / dynamical). Run Lab 4.C or the Lattice Simulator. Identify visibly ordered gauge-pointer behavior and contrast it with chaotic regimes.

4.F (Hatcher bridge). In Hatcher, the translation $T_n(z) = z + n$ generates periodic strips. Construct a quaternionic analogue (repeated right phase advance + compensating left shift) on a small angle lattice and describe the base track (cf. Fig. 4.3).

4.G (forward). Why must any flux topograph defined in Chapter 5 be equivariant under the gauge actions studied here?

4.H (Open Problem 1 connection). Using `adjacency_equivariance_score`, test whether `candidate_adjacency` is preserved by left and right multiplications by a few Hurwitz units on Λ_{ang}. Record failures—they constrain what a canonical rule can be.

4.I (software honesty). In two sentences, distinguish combinatorial periods on Λ_0 from dynamical "identity preservation" in `TwoGyroLattice`. Which claim type applies to each?

4.7 Code and asset pointers

```
qga/lib/hopf_lattice.py
kingdom.core.lattice.LatticeConfig
kingdom.simulations.lattice.TwoGyroLattice
kingdom.simulations.lattice.run_lattice_comparison
kingdom.core.hopf
```

Figures: `scripts/generate_ch4_figures.py` **Related portal:** Lattice Simulator tab — dynamical illustration of gauge evolution and identity preservation. **Open problems:** OP1 equivariance tests (Exercise 4.H); see `notes/open_problems.md`.

4.8 Looking ahead

The gauge symmetries of the gauged Hopf lattice are now operational: left and right multiplications generate a rich group whose action mirrors (in the **Model** sense) the modular group on Hatcher's Farey diagram. In **Chapter 5** we lift Conway's topographs to this setting, producing **flux topographs** whose value landscapes, separator structures, and periodicity will be required to be equivariant under the symmetries developed here. Classification, Magic Islands, and the $Z \mapsto$ flywheel map will follow from the reduced configurations and periodic orbits these symmetries make visible.

With the lattice and its symmetries in place, we are ready to develop the visual and arithmetic theory of flux topographs—the direct quaternionic descendant of Hatcher's topographs of binary quadratic forms.

Part III

Forms, Topographs, and Representations

Chapter 5

Quaternionic Forms and Flux Topographs

This chapter lifts Conway's topographs of binary quadratic forms (Hatcher Chapters 4–5) to the gauged Hopf lattice. We define **flux topographs**—value landscapes on the discrete lattice whose periodic separator structures encode stability, periodicity, and classification. The resulting theory is the direct higher-dimensional, quaternionic analogue of Hatcher's visual number theory, with Magic Islands as a natural generalization of class-number phenomena and reduced forms—stated carefully as a **Model**.

Learning goals

1. Generalize binary quadratic forms and their topographs to flux functionals on the gauged Hopf lattice.
2. Define flux topographs, value landscapes, and separator structures.
3. Understand periodicity, reduced configurations, and Magic Islands.
4. See how gauge symmetries act on topographs (equivariance).
5. Prepare the visual and arithmetic foundation for classification (Chapter 6) and the $Z \mapsto$ flywheel map (Chapter 7).

Figures in this chapter

Tag	File	Role
Fig. 5.1	`figures/fig5_1_flux_topograph_schematic.png`	Flux topograph value landscape
Fig. 5.2	`figures/fig5_2_separator_structure.png`	Periodic / sign-crossing separators
Fig. 5.3	`figures/fig5_3_magic_island.png`	Magic Island schematic + model Z-sweep
Fig. 5.4	`figures/fig5_4_symmetry_on_topograph.png`	Gauge action on a topograph
Aux A5.1	`figures/aux5_1_topograph_evolution.png`	Evolution under repeated right phase

Claim discipline

Claim	Type
Classical topograph properties (periodicity, arithmetic progression rule, reduced forms) from Hatcher	**Theorem** (classical; cited)
Flux topographs, separator structures, Magic Islands, gauge equivariance of candidate constructions	**Model** (core of OP2)
`qga/lib/flux_topograph.py`, `kingdom.core.flux_flywheel` APIs	**Software fact**

5.1 From binary quadratic forms to flux functionals

In Hatcher, a binary quadratic form

$$f(x, y) = ax^2 + bxy + cy^2$$

acquires a **topograph**—a value landscape drawn on the dual of the Farey triangulation. Adjacent regions differ by an arithmetic progression rule, and periodic separator lines appear for indefinite forms (positive non-square discriminant).

We lift this picture to the gauged Hopf lattice of Chapters 3–4:

A **flux functional** on the lattice assigns a real (or integer) value to vertices—and, in richer versions, to edges or small cycles—such that the assignment transforms naturally under gauge actions. A **flux topograph** is the visual landscape of these values together with their level sets and separator structures.

Natural candidate functionals (all **Model** choices for OP2 experiments):

Name (book helper)	Geometric meaning
`norm`	$x^2 + y^2 + z^2$ on pure-imaginary part (quadratic form on $\operatorname{Im} \mathbb{H}$)
`hopf_y1` / `hopf_height`	Classical Hopf base coordinates (Ch. 2)
`phase`	Fiber phase proxy $\operatorname{atan2}(x_4, x_3)$
portal stability	`map_z_to_flywheel` scores (Ch. 7 preview)

Model note. The precise choice of functional is part of the definition of flux topographs and is tied to **Open Problem 2**. No uniqueness claim is made in this chapter.

5.2 Flux topographs and separator structures

5.2.1 Definition (working)

A **flux topograph** is a tuple

$$(\Lambda,\ E,\ V,\ \mathcal{S})$$

where Λ is a discrete point set in S^3, E is an edge set (along-fiber and/or inter-fiber from Ch. 3), $V : \Lambda \to \mathbb{R}$ is a flux functional, and $\mathcal{S}$ is the separator structure extracted from V (sign crossings, critical levels).

In code:

```
FluxTopograph(points, values, edges, functional)
build_flux_topograph(points, edges=..., functional=...)
detect_separators(topo, threshold=..., mode='sign'|'level')
```

5.2.2 Separator structures

Separator structures are the loci where the functional changes sign or crosses a critical threshold. On a graph, they appear as edges (i, j) with $V(i)$ and $V(j)$ on opposite sides of a threshold. Connected components of those edges are discrete analogues of Hatcher's separator curves.

In the indefinite / oscillatory case, separators can become **periodic** under the gauge group—exactly analogous to Hatcher's periodic separator lines for indefinite binary forms, which encode infinitude (Pell-type families).

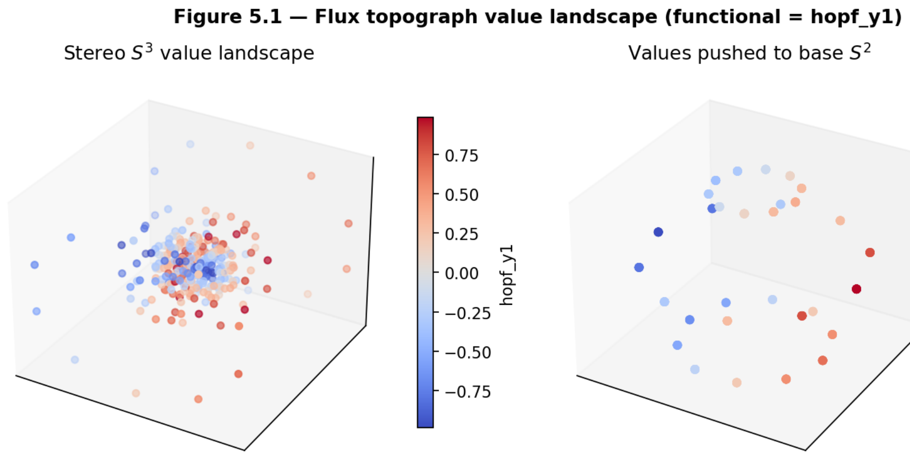


Figure 5.1: Value landscape for the functional `hopf_y1` on an angle-sampled lattice, shown in stereographic S^3 and pushed to the base S^2.

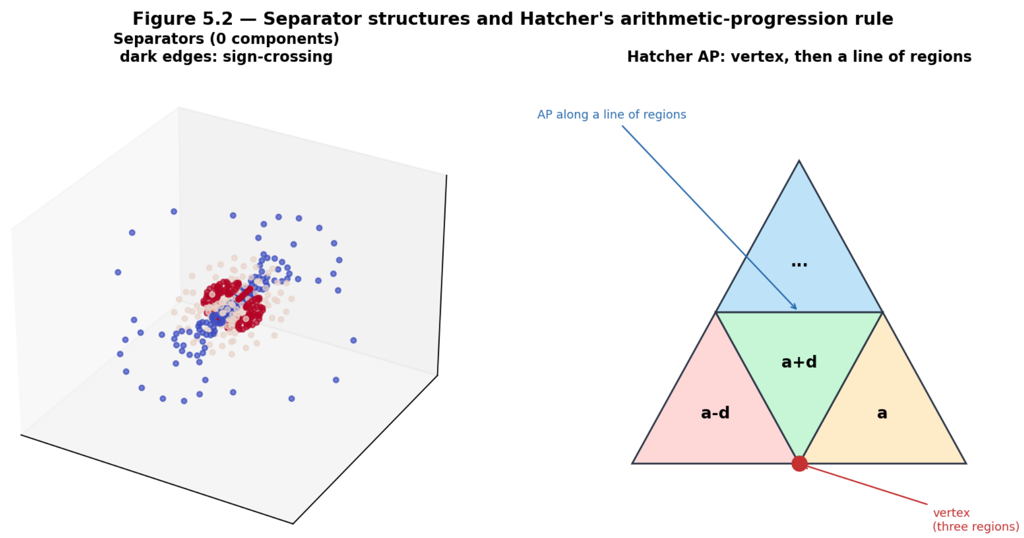


Figure 5.2: Dark edges: sign-crossing separators of `hopf_height` on the candidate adjacency graph. Component count and edge sets are experimental OP2 diagnostics, not theorems.

5.2.3 Arithmetic progression rule (lift)

Classically (Hatcher TN Ch. 4), values on **three regions meeting at a vertex** of the topograph form an arithmetic progression; that progression continues **along a line of regions**. (It is not "three regions meeting at an edge.") Figure 5.2's schematic panel records this vertex/line picture. On the gauged Hopf lattice we do **not** lift the misstated edge rule into OP2: we only measure jumps $V(j) - V(i)$ on graph edges (`arithmetic_progression_residuals`) as a diagnostic. A genuine equivariant AP rule, if it exists, must match Hatcher at vertices and along lines of regions.

5.3 Periodicity, reduced configurations, and Magic Islands

5.3.1 Periodicity under the gauge group

When a flux topograph is (approximately) periodic under a composite gauge sequence g from Chapter 4—i.e. $g^k \cdot \Phi \approx \Phi$ for some k—the orbit corresponds to a **reduced configuration**. The book helper `periodicity_score` reports nearest-neighbor total-space return and multiset distance of values.

5.3.2 Magic Islands

Magic Islands are pockets of enhanced stability in parameter space (detuning $\delta\omega$, gauge strength, layers, or atomic number Z in the portal Model). Inside an island the topograph (or the derived stability score) shows strong order: high `stability_score`, noble-gas locks, low variation. Outside, scores drop and dynamical simulations become more chaotic (Ch. 3 Aux A3.1).

They are the narrative generalization of finite class number and reduced forms in Hatcher's theory. **Arithmetic characterization** (class-number-like invariants predicting islands) is deferred to Open Problem 3 / Chapter 6.

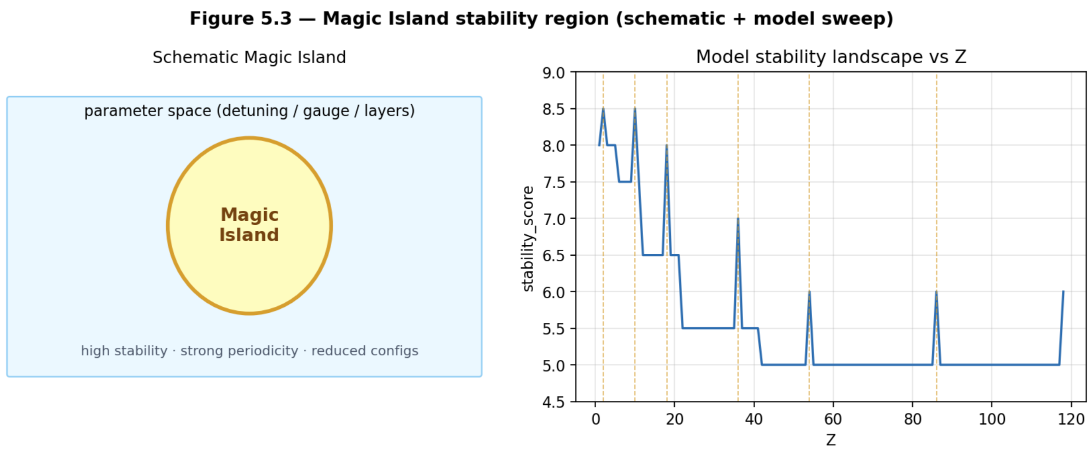


Figure 5.3: Left: schematic island in parameter space. Right: Kingdom Come Model `stability_score` vs Z (gold lines mark noble-gas Z). **Software fact** for the right panel; physical interpretation remains **Model**.

Claim type. Visual and software properties of Magic Islands in the portal: **Model + Software fact**. Classical reduced forms / class number: **Theorem** (Hatcher). Correspondence between the two: **Open** (OP3).

5.4 Gauge equivariance of flux topographs

Any well-defined flux topograph must be **equivariant** under the gauge group of Chapter 4: applying a left or right multiplication must map the topograph to another topograph of the same type (possibly after transporting values or recomputing a geometric functional).

This is the direct analogue of Hatcher's requirement that topographs transform properly under linear fractional transformations.

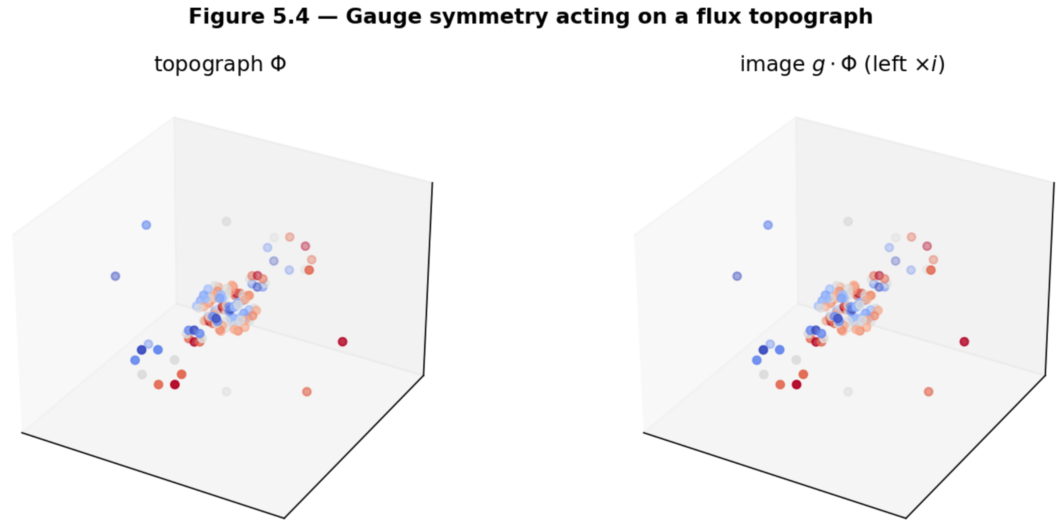


Figure 5.4: Left multiplication by i maps a `hopf_y1` topograph to its image. For geometric functionals recomputed after the move, values track the geometry exactly.

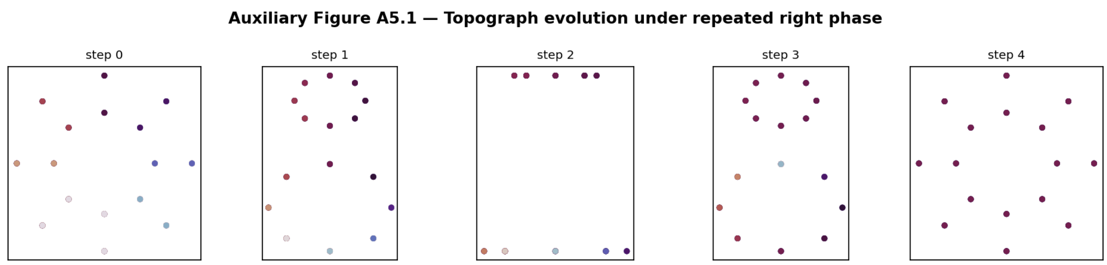


Figure 5.5: Repeated right phase advances: base scatter colored by `phase` functional. Separator patterns should transform predictably if OP2 axioms hold.

Open Problem 2 (core of this chapter). Formalize the axioms of a flux topograph so that:

1. The structure **reduces** to Conway/Hatcher topographs under a suitable projection or restriction.
2. Separator periodicity and an arithmetic progression rule lift naturally.
3. **Gauge equivariance** holds for a fixed adjacency rule from Chapter 3 (OP1).
4. Magic Islands emerge as finite or periodic reduced regions with measurable stability.

Sandbox: `qga/lib/flux_topograph.py` — `build_flux_topograph`, `detect_separators`, `periodicity_score`, `separator_equivariance_score`.

5.5 First computational labs

Helpers: `lib/flux_topograph.py` · portal: `map_z_to_flywheel` · **Appendix C §C.3**.

- **5.A** `build_flux_topograph` + `detect_separators`.
- **5.B** `periodicity_score` under a gauge sequence.
- **5.C** Z-stability via `map_z_to_flywheel` (no `magic_flag` — use `stability_score` / `is_noble_gas`).
- **5.D** `separator_equivariance_score` (OP2 diagnostic).

```
from lib.flux_topograph import build_flux_topograph, detect_separators
from lib.hopf_lattice import sample_angle_lattice, candidate_adjacency
pts = sample_angle_lattice(3, 8, 8)
along, inter = candidate_adjacency(pts)
topo = build_flux_topograph(pts, edges=along+inter, functional="hopf_height")
print(len(detect_separators(topo)))
```

5.6 Exercises

5.A (hand). State the classical arithmetic progression rule for Hatcher topographs (or look it up in TN Ch. 4) and propose a quaternionic lift for flux values on adjacent lattice elements.

5.B (hand). In two sentences, explain why gauge equivariance is a necessary condition for a flux topograph to be well-defined.

5.C (code). Complete Labs 5.A–5.B. Report separator component count and `periodicity_score` for your functional.

5.D (code). Run Lab 5.C for $Z \in \{2, 10, 26, 79, 118\}$. Which values show high `stability_score` or `is_noble_gas=True`? Record the pattern.

5.E (visual). Generate a flux topograph and apply one gauge transformation (Fig. 5.4). Describe how separator curves / value colors transform.

5.F (Hatcher bridge). In Hatcher, reduced forms and class number classify quadratic forms up to equivalence. Sketch how Magic Islands might play an analogous role for flux configurations on the gauged Hopf lattice.

5.G (Open Problem 2 teaser). Using `build_flux_topograph` and `detect_separators`, test whether separator counts survive under `candidate_adjacency` + left/right multiplications (`separator_equivariance_score`). Document failures—they inform OP2 axioms.

5.H (forward). Why will the classification theory of Chapter 6 depend on the periodic separator structures developed here?

5.I (software honesty). In one paragraph, distinguish: (i) pedagogical flux functionals in `qga/lib/flux_topograph.py`, (ii) portal `stability_score` from `map_z_to_flywheel`, (iii) classical Conway topographs. Assign claim types.

5.7 Code and asset pointers

```
qga/lib/flux_topograph.py
  build_flux_topograph, detect_separators, periodicity_score,
  apply_gauge_to_topograph, separator_equivariance_score,
  arithmetic_progression_residuals, stability_landscape_z

qga/lib/hopf_lattice.py
kingdom.core.flux_flywheel.map_z_to_flywheel[_extended]
kingdom.viz.magic_island.build_magic_island_heatmap
```

Figures: `scripts/generate_ch5_figures.py` **Related portal:** Flux Flywheel tab; Magic Island heatmap; Lattice Simulator (dynamical backdrop). **Open problems:** OP2 (`notes/open_problems.md`); depends on OP1 adjacency.

5.8 Looking ahead

Flux topographs are the visual and arithmetic heart of Part III—the direct quaternionic lift of Hatcher's topographs, still with **Model**-level axioms pending OP2. In **Chapter 6** we classify these topographs, enumerate reduced configurations, and formalize Magic Islands as higher-dimensional analogues of class numbers and reduced forms. **Chapter 7** connects classified objects to the $Z \mapsto$ flywheel map, completing the representation-theoretic bridge from number theory to the periodic-table proxy in the Kingdom Come Model.

With flux topographs in hand, we are ready for the classification theory that turns visual landscapes into arithmetic invariants.

Chapter 6

Classification of Flux Topographs and Magic Islands

This chapter classifies flux topographs on the gauged Hopf lattice, enumerates reduced configurations, and formalizes **Magic Islands** as higher-dimensional analogues of class numbers and reduced forms. The theory mirrors Hatcher Chapters 5–6 while remaining a **Model** construction pending the axioms of Open Problems 2–3.

Learning goals

1. Classify flux topographs by invariants analogous to discriminant and type (elliptic / hyperbolic / ...).
2. Define equivalence of topographs under gauge actions.
3. Enumerate reduced configurations and compute class-number-like invariants.
4. Formalize Magic Islands as pockets of enhanced periodicity and stability.
5. Prepare the arithmetic foundation for the $Z \mapsto$ flywheel map (Chapter 7) and the class-group lift (Chapter 8).

Figures in this chapter

Tag	File	Role
Fig. 6.1	`figures/fig6_1_four_types.png`	Four types of flux topographs (schematic)
Fig. 6.2	`figures/fig6_2_reduced_config.png`	Reduced configuration (minimal-variance rep)
Fig. 6.3	`figures/fig6_3_class_number_analogue.png`	Class-number-like count of reduced islands
Fig. 6.4	`figures/fig6_4_equivalence.png`	Gauge equivalence of two topographs
Aux A6.1	`figures/aux6_1_island_sweep.png`	Z-sweep revealing Magic-Island-like peaks

Claim discipline

Claim	Type
Classical classification of quadratic forms and topographs (Hatcher Ch. 5)	**Theorem** (cited)
Flux topograph types, equivalence, reduced configurations, Magic Islands	**Model** (core of OP3; depends on OP2)
`qga/lib/flux_topograph.py` classification helpers; Kingdom Come `stability_score`	**Software fact**

6.1 The four types of flux topographs

In Hatcher, binary quadratic forms (and their topographs) are classified by the sign of the discriminant $\Delta = b^2 - 4ac$:

Discriminant	Classical type (Conway–Hatcher)	Topograph character
$\Delta < 0$	elliptic	**no river**; finite reduced forms
$\Delta > 0$ nonsquare	hyperbolic	**periodic river**; finite reduced cycles
$\Delta > 0$ square	0-hyperbolic	degenerate river
$\Delta = 0$	parabolic	boundary / degenerate

Hyperbolic forms have infinitely many automorphisms, but the **reduced cycle is finite** (the period of the river). Do not write "infinite families of reduced forms" for the hyperbolic case.

We lift this classification to flux topographs by examining the flux functional V and separator / river-like structures under the gauge group of Chapter 4—**not** by computing a classical Δ (that remains OP2–3 territory).

Type	Conway–Hatcher character	Working Model lift
Elliptic	No river; finite reduced forms	Few / bounded separators; finite reduced set
Hyperbolic	Periodic river; finite reduced cycles	Periodic separators under gauge; finite reduced cycle
0-hyperbolic	Degenerate river	Flat or near-constant regions; degenerate
Parabolic	Boundary / degenerate	Transitional / default bin

Model / Hypothesis box (not a type). Noble-gas locks and portal `stability_score` peaks are **not** elliptic/hyperbolic/0-hyperbolic types. They belong to Hypothesis H2 / H3 and the $Z \mapsto$ map in Part V. Do not mix them into the type table.

6.1.1 Heuristic classifier (software / Model)

```
classify_topograph_type(topo) ->
  type, reason, n_separator_edges, value_variance,
  best_period_found, signature, ...
```

The decision tree is aligned with the Conway–Hatcher table as a *proxy* (still not a discriminant Δ):

1. near-constant values $\to$ `0-hyperbolic`
2. no sign separators (no river) $\to$ `elliptic`
3. periodic under a small gauge dictionary + separators $\to$ `hyperbolic`
4. separators present but no period found $\to$ `0-hyperbolic`
5. otherwise transitional `parabolic`

Software fact. The previous tree (`few separator components -> elliptic`, and `no separators + low variance -> parabolic`) could relabel a definite (no-river) form as the wrong type. The tree above matches the type table; finite samples can still mislabel. This is **not** a theorem of classification—only a lab tool for OP3 experiments.

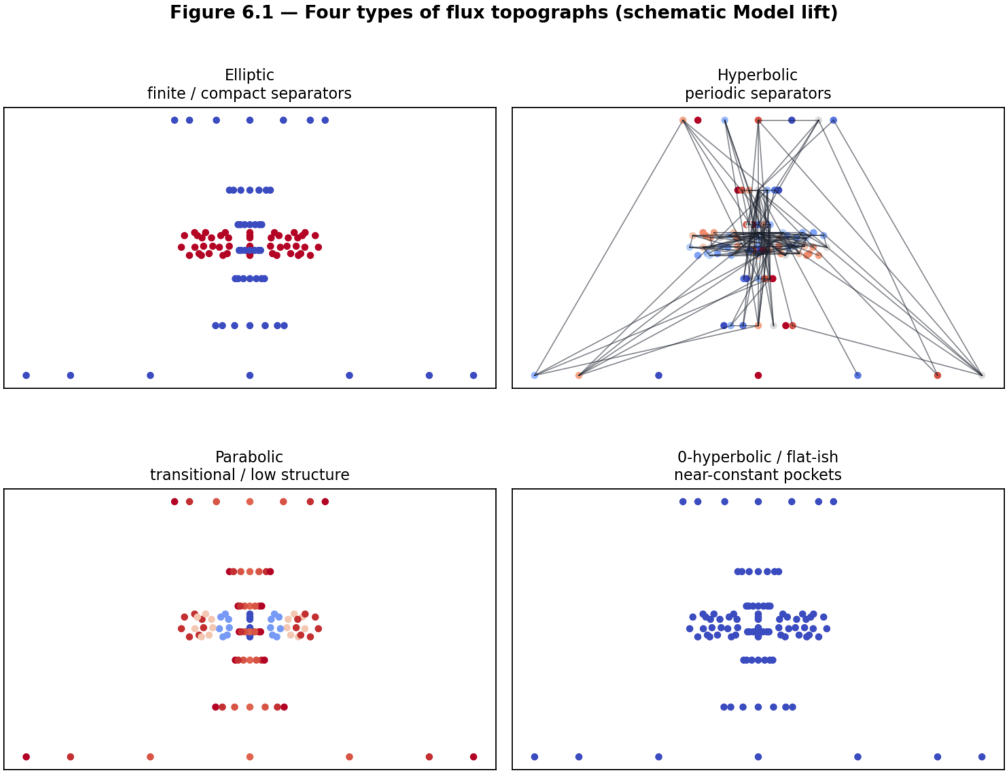


Figure 6.1: Schematic value landscapes and separator patterns for the four lifted types. Generated with pedagogical functionals; labels are Model categories.

6.2 Equivalence of flux topographs

Two flux topographs are **equivalent** (in the working Model) if there exists a gauge transformation—or finite sequence of gauge actions—that maps one to the other, with values recomputed from a geometric functional or transported by index.

This is the direct analogue of Hatcher's equivalence of quadratic forms under $SL(2, \mathbb{Z})$ / linear fractional transformations.

6.2.1 Distance and reduced representatives

```
equivalence_distance(topo_a, topo_b)
reduced_representative(topo)           # minimize value variance in a gauge orbit
enumerate_reduced(topo, dedup_tol=...)  # inequivalent reduced orbit reps
```

Reduced configurations are representatives chosen from each approximate equivalence class by a normalization rule (here: minimal value variance among a bounded gauge orbit). The number of distinct reduced configurations is a **class-number analogue**.

6.3 Magic Islands as a working class-number analogue (Hypothesis H3, OP3)

Magic Islands are regions in parameter space (or in the space of flux functionals) that contain a high density of reduced configurations with strong periodicity and high stability scores.

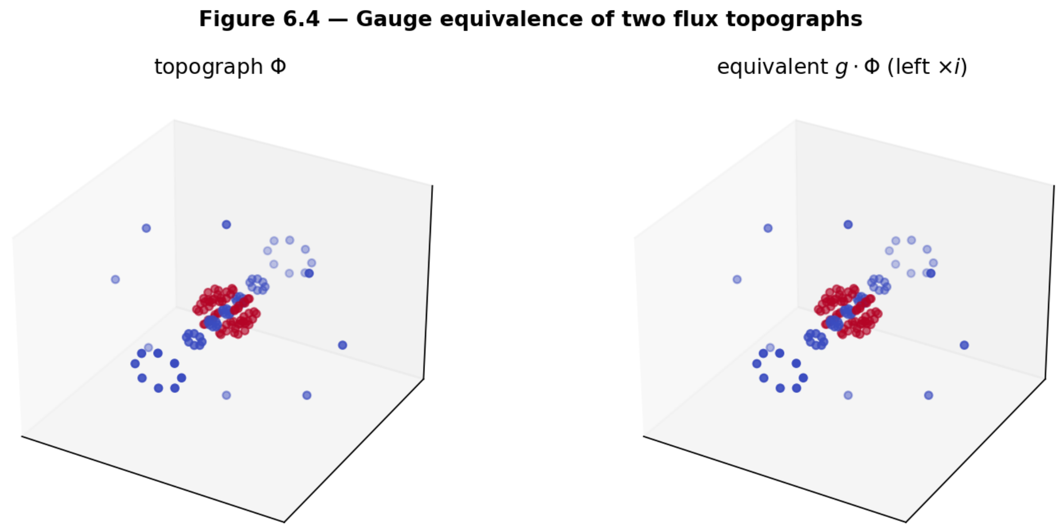


Figure 6.2: A topograph and its image under left multiplication by i. Value multisets and sorted point clouds match for geometric functionals (`equivalence_distance` near zero).

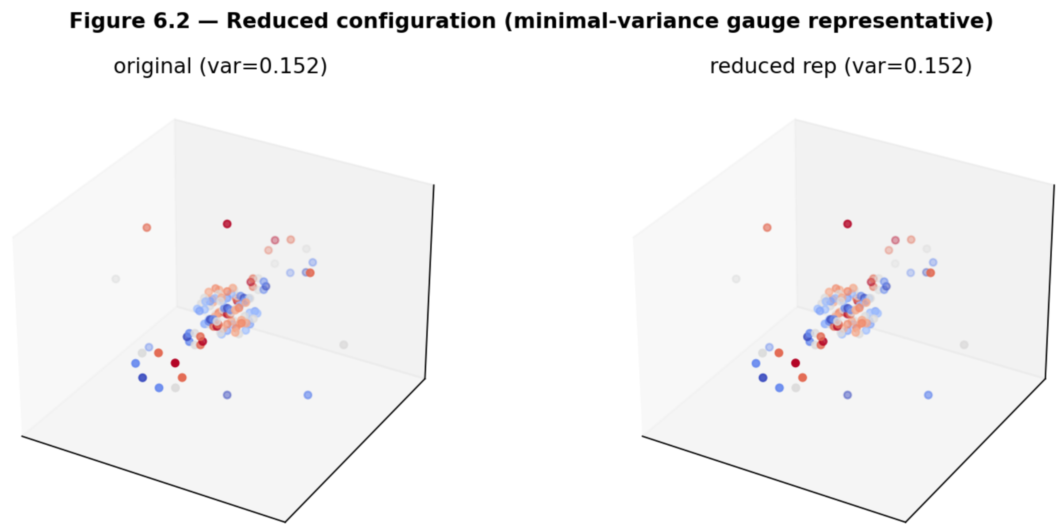


Figure 6.3: Original topograph vs reduced representative selected by minimal variance under the standard gauge dictionary.

They generalize:

- the finite set of reduced forms for negative discriminant (elliptic case);
- the periodic reduced cycles for positive nonsquare discriminant (hyperbolic case).

In the Kingdom Come Model, Magic Islands appear as pockets where `stability_score` is high, `stability_class` is noble-like, and dynamical simulations show ordered long-term behavior (Ch. 3–5).

Invariant discipline (forward pointer). When classifying topograph types and Magic Islands, keep separate: (i) *algebraic* structure of discrete labels or class-like counts, (ii) *sequential / Model flow* of scores along parameter or Z sweeps, and (iii) *statistical geometric lock* of a pattern to a continuous base beyond chance. The distinction among these three layers—developed with controls in **Chapter 9 section 9.5**—applies directly here: high `stability_score` clusters and visual island order are not automatically the same as an algebraic class number, nor as a proved lock to a continuous geometric coordinate. See also Chapter 10's checklist and `notes/RESEARCH_NOTE_moduli.md`.

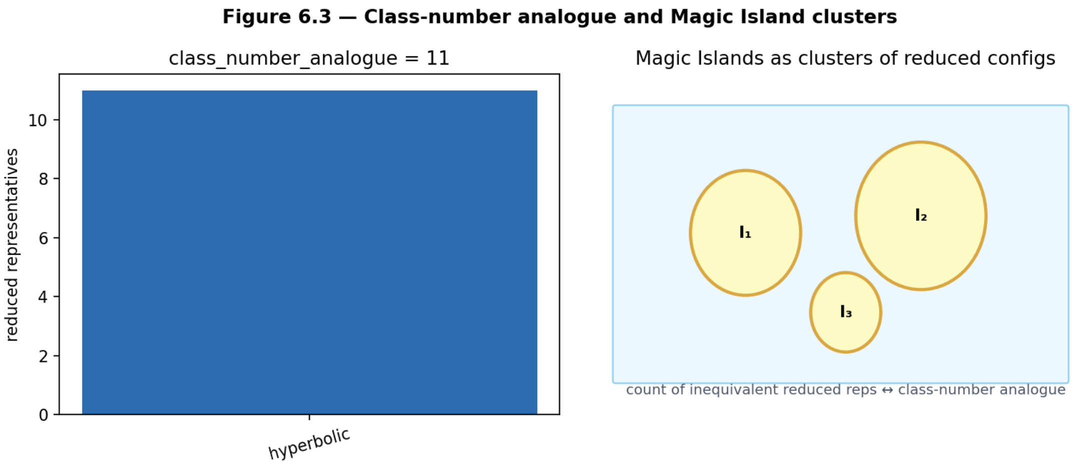


Figure 6.4: Left: count of inequivalent reduced representatives by type (`class_number_analogue`). Right: schematic islands as clusters of reduced configs.

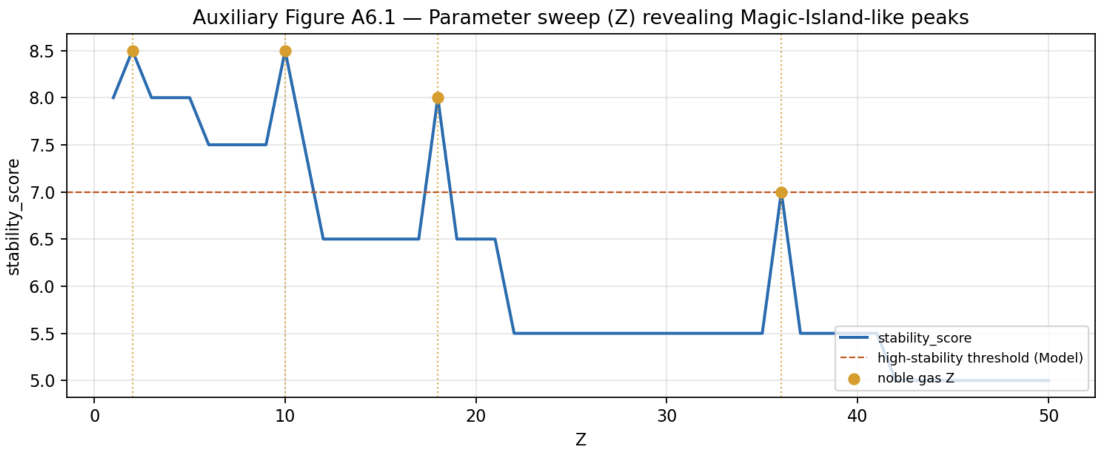


Figure 6.5: Portal Model `stability_score` vs Z (gold: noble-gas Z; dashed: high-stability threshold). **Software fact** for the curve; island interpretation remains **Model**.

6.3.1 Heuristic island score

```
magic_island_score(topo) -> score, period_term, variance_term, separator_term, type
```

```
class_number_analogue(topo) -> class_number_analogue, by_type, reduced
```

Higher magic_island_score means more "island-like" under the pedagogical formula (periodicity + controlled variance + ordered separators). It is **not** a classical invariant.

Open Problem 3 (core of this chapter). Develop invariants (discriminant analogues, class-number-like counts) that predict the location and size of Magic Islands from lattice data and flux functionals **without** exhaustive parameter sweeps—and that reduce to classical class numbers under a Hatcher-style restriction.

See notes/open_problems.md. Depends on OP1 (adjacency) and OP2 (topograph axioms).

6.4 First computational labs

Helpers: lib/flux_topograph.py · **Appendix C §C.3**.

- **6.A** classify_topograph_type.
- **6.B** enumerate_reduced / class_number_analogue (use modest lattices).
- **6.C** Portal stability_score vs Z (no magic_flag).
- **6.D** equivalence_distance after left multiplication.

```python
from lib.flux_topograph import classify_topograph_type, class_number_analogue
# build topo as in Appendix C SC.3
# print(classify_topograph_type(topo)["type"])
```

6.5 Exercises

6.A (hand). State the four classical types of quadratic forms/topographs in Hatcher and map each to the lifted types in this chapter.

6.B (hand). Explain why reduced configurations are useful for classification (one paragraph).

6.C (code). Complete Labs 6.A–6.B. Report the type and number of reduced representatives for your topograph.

6.D (code). Run Lab 6.C for $Z = 1$ to 50. Tabulate stability_score and mark noble-gas Z. Identify any visible Magic-Island-like clusters.

6.E (visual). Generate two equivalent topographs (via gauge action) and confirm equivalence_distance is small; compare separator counts.

6.F (Hatcher bridge). In Hatcher, the class number counts inequivalent reduced forms for a given discriminant. Sketch how a "Magic Island count" / class_number_analogue could play an analogous role here.

6.G (Open Problem 3 teaser). Using enumerate_reduced and magic_island_score, test whether the current candidate adjacency produces consistent island scores across small changes in functional (hopf_height vs hopf_y1). Document inconsistencies—they constrain OP3 invariants.

6.H (forward). Why will the ideal-class-group lift in Chapter 8 depend on the classification developed here?

6.I (software honesty). In one paragraph, distinguish: (i) heuristic classify_topograph_type, (ii) portal stability_score vs Z, (iii) classical class numbers. Assign claim types.

6.6 Code and asset pointers

```
qga/lib/flux_topograph.py   # Ch. 5-6 helpers
qga/lib/hopf_lattice.py
kingdom.core.flux_flywheel
kingdom.viz.magic_island
```

Figures: `scripts/generate_ch6_figures.py` **Related portal:** Flux Flywheel tab; Magic Island heatmap. **Open problems:** OP3 (this chapter); OP2 (topograph axioms); OP1 (adjacency).

6.7 Looking ahead

We now have a working **Model** classification of flux topographs, reduced configurations, and Magic Islands—the quaternionic narrative lift of Hatcher's classification theory. In **Chapter 7** we connect these classified objects to the $Z \mapsto$ flywheel map, turning arithmetic-looking invariants into a periodic-table proxy. **Chapter 8** will then lift the class group itself to the quaternionic / Hopf setting. **Chapter 9 section 9.5** will sharpen the arithmetic–geometry boundary with a controlled three-layer analysis (algebra / sequential flow / angle lock) that you can already apply when reading island plots in this chapter.

With classification in hand, we are ready to bridge number theory and the emergent physics of the Kingdom Come Model.

Chapter 7

The $Z \mapsto$ Flux Map and Representation Theory

This chapter connects the classified flux topographs and Magic Islands of Chapters 5–6 to the atomic number Z. We define the $Z \mapsto$ **flux map**, interpret it as a representation-theoretic bridge (which "numbers" are represented by stable flux configurations), and examine how Magic Islands align with chemically and physically special elements. The construction remains a **Model** with explicit validation pathways (Chapter 10).

Learning goals

1. Define the $Z \mapsto$ flywheel / flux configuration map formally.
2. Interpret the map as representation theory on the gauged Hopf lattice.
3. Connect Magic Islands to the periodic table and noble-gas stability.
4. Examine chemical and physical proxies (stability scores, shell descriptors, extended fields).
5. Prepare the bridge to class-group ideas (Chapter 8) and observational validation (Chapter 10).

Figures in this chapter

Tag	File	Role
Fig. 7.1	`figures/fig7_1_z_to_flywheel.png`	Schematic of the $Z \mapsto$ flux map
Fig. 7.2	`figures/fig7_2_magic_island_periodic_table.png`	Magic Island scores on a schematic table
Fig. 7.3	`figures/fig7_3_stability_vs_z.png`	`stability_score` vs Z (portal data)
Fig. 7.4	`figures/fig7_4_electron_cloud_flux.png`	Electron cloud as flux distribution
Aux A7.1	`figures/aux7_1_he_fe_au_stills.png`	He, Fe, Au flywheel stills

Claim discipline

Claim	Type
Four-square theorem and classical representation by norms (Ch. 1)	**Theorem**
Existence and implementation of the coded $Z \mapsto$ map	**Software fact**
Map as representation theory; Magic Islands as periodic-table proxy; physical emergence	**Model**
Any claim that topology *derives* real chemistry as a law of nature	**Hypothesis** (Ch. 10 validation)
`map_z_to_flywheel[_extended]` metrics	**Software fact**

7.1 The $Z \mapsto$ flux map

The Kingdom Come portal implements a function that takes an atomic number Z and returns a flywheel configuration together with stability and chemistry-facing metrics:

$$Z \longmapsto (\text{flywheel / detuning state}, \\ \texttt{stability\_score}, \texttt{stability\_class}, \\ \texttt{is\_noble\_gas}, \ldots).$$

Primary entry points

Function	Role
`map_z_to_flywheel(z)`	Core map: detuning, gauge params, `stability_score`, class labels
`map_z_to_flywheel_extended(z)`	Adds alignment, IE proxies, magnetic-moment validation fields

There is still **no** `magic_flag` key. Treat high `stability_score`, noble-gas class strings, and `is_noble_gas=True` as the Model's island indicators.

The map is built on the gauged Hopf lattice and flux-topograph language of previous chapters. Different detuning parameters select different stability islands; a documented ultra-stable lock (Magic Island Sweep reference in portal notes) serves as a noble-gas-like template.

Figure 7.1 — Schematic of the $Z \mapsto$ flux / flywheel map

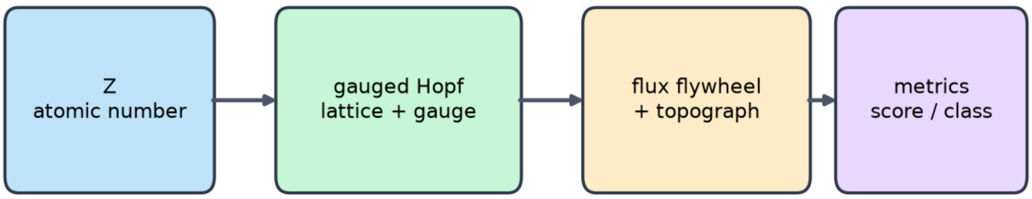


Figure 7.1: Pipeline: atomic number → lattice / gauge data → flux flywheel + topograph → scored metrics.

Claim type. Reproducible coded map: **Software fact**. Interpreting outputs as *the* emergence mechanism of the periodic table: **Model**.

7.1.1 Book helper

```
stability_landscape_z(z_values=...)          # list of dict rows
stability_landscape_z(z_range=(1, 50))       # inclusive range
stability_landscape_z(z_range=(1, 30), extended=True)
```

7.2 Representation theory on the gauged Hopf lattice

In classical number theory, representation by quadratic forms asks which integers n satisfy $f(x, y) = n$ for a given form (Hatcher Chapter 6). Quaternion norms answer which n are sums of four squares (Chapter 1, **Theorem**).

Here we ask an analogous **Model** question:

> Which atomic numbers Z are "represented" by stable flux configurations on the gauged Hopf lattice?

Stable flywheels correspond to reduced configurations with high `magic_island_score` (Chapter 6) and high portal `stability_score`. The map therefore selects, for each Z, a configuration (or orbit) intended to match observed chemical/nuclear specialness.

Classical	Quaternionic / Hopf Model
Form f	Lattice + gauge group + flux functional
Represented integer n	Atomic number Z (and related quantum numbers)
Class / genus of forms	Equivalence class of topographs / flywheels
Class number	`class_number_analogue` / island counts (Ch. 6)

This is a **representation-theoretic reading**, not a theorem that every chemical property is a lattice invariant.

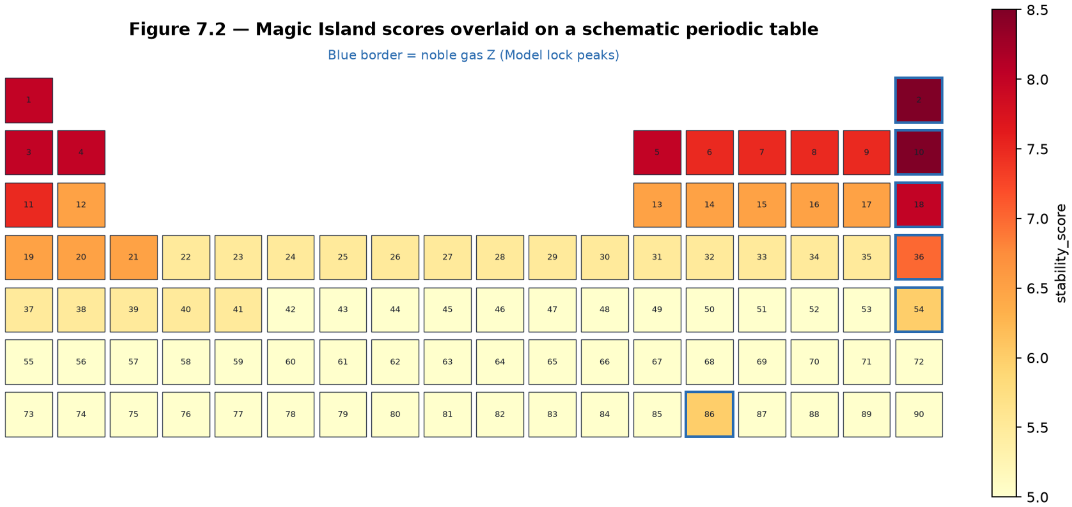


Figure 7.2: Cells colored by portal `stability_score`; blue borders mark noble-gas Z. Layout is a simplified table for visualization (f-block truncated)—software + Model.

7.3 Chemistry-facing scores (Hypothesis — developed in Part V)

Magic Islands as *chemical* stability, noble-gas locks, and ionization-energy alignment are **not** theorems of Part III. They are Hypothesis H2 / H3, with Table T4 protocols and an H2 ablation (noble-gas bonuses off) in **Chapter 10** and Appendix D. This section keeps only a pointer so the $Z \mapsto$ map can be used as a **Model** representation without promoting chemistry.

Figure 7.3 is a portal software curve, not a finding.

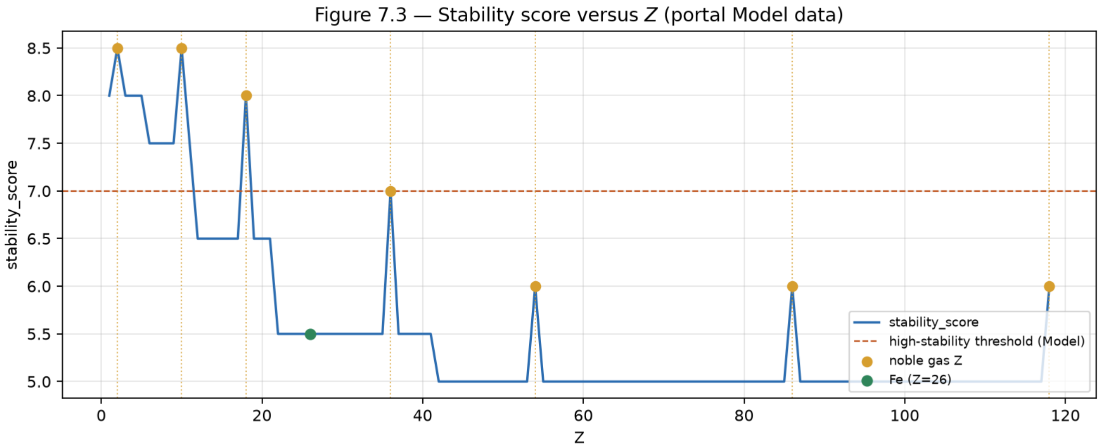


Figure 7.3: Portal `stability_score` vs atomic number. Gold: noble-gas Z (explicit bonuses live in the portal map). Interpretation: **Hypothesis**, Part V. Lab 7 must ablate those bonuses before any alignment claim.

Open Problem 4 (shared with Ch. 10). Up to gauge, is the map $Z \mapsto$ flywheel configuration unique under stated axioms? See `notes/open_problems.md`.

7.4 Electron-cloud visuals (Software fact — Part V)

Portal electron-cloud pictures (`kingdom.viz.electron_cloud`) are **Software facts** about the renderer, not Schrödinger solutions and not a Part III theorem. Chemistry-facing reading: **Part V**.

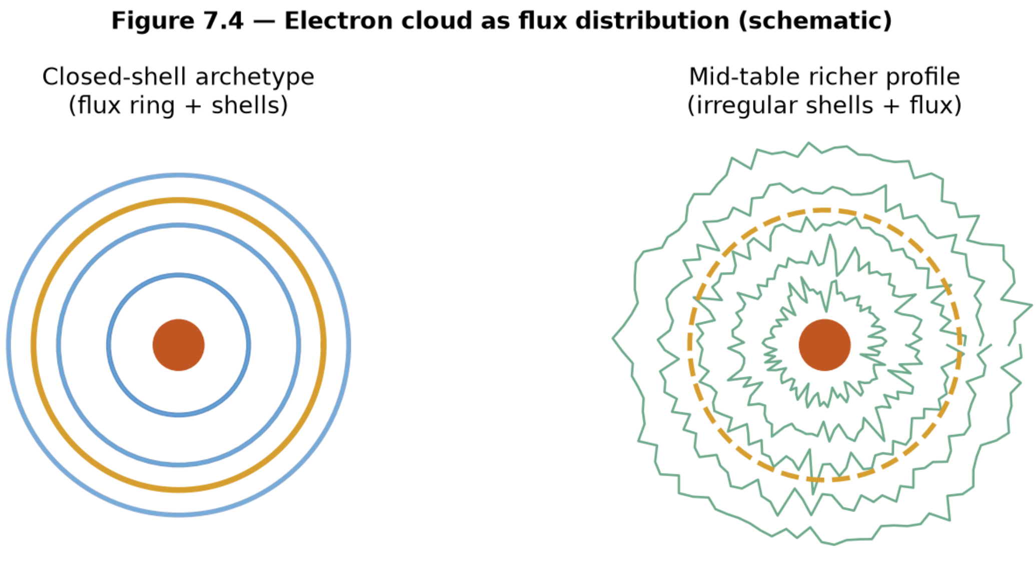


Figure 7.4: Schematic closed-shell vs mid-table flux-derived profiles (portal style).

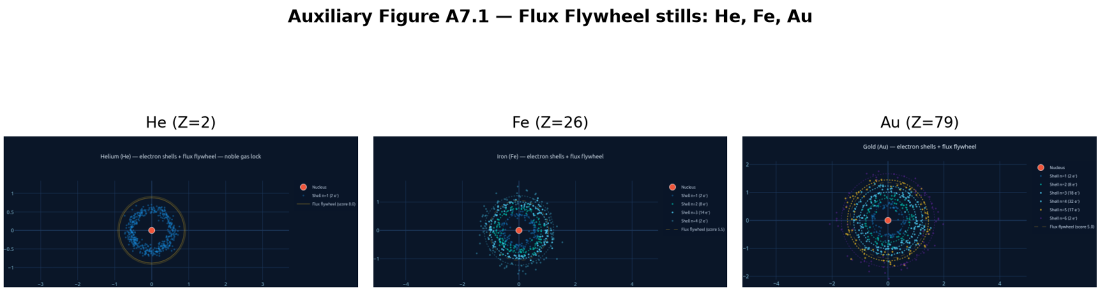


Figure 7.5: Portal Flux Flywheel stills: He ($Z = 2$), Fe ($Z = 26$), Au ($Z = 79$). Sources: `kingdom/z_knowns/frame_0002.png`, `frame_0026.png`, `frame_0079.png`.

7.5 First computational labs

Portal: `map_z_to_flywheel[_extended]` · `stability_landscape_z` · **Appendix C §C.3**.

- **7.A** Scores for $Z \in \{2, 10, 26, 79, 118\}$.
- **7.B** `stability_landscape_z(z_range=(1,50))` high islands.
- **7.C** Noble vs neighbors ($Z = 10, 11, 12$).
- **7.D** Gradio Flux Flywheel / Aux A7.1.
- **7.E** Extended fields He vs Fe.

```
from kingdom.core.flux_flywheel import map_z_to_flywheel
print(map_z_to_flywheel(2)["stability_score"], map_z_to_flywheel(2).get("is_noble_gas"))
```

7.6 Exercises

7.A (hand). In one paragraph, explain why the $Z \mapsto$ map can be viewed as a representation problem on the gauged Hopf lattice.

7.B (hand). Why might Magic Islands near noble-gas Z be especially stable in the Model? Connect to closed-shell ideas and Chapter 6 reduced configurations.

7.C (code). Complete Labs 7.A–7.B. Tabulate `stability_score` and `is_noble_gas` for $Z = 1$ to 30. Identify the strongest islands.

7.D (code). Complete Lab 7.E for $Z = 2$ and $Z = 26$. What differences appear in extended fields?

7.E (visual). Render or inspect flux-derived clouds for He and Fe. Describe how Model geometry differs between a closed-shell case and a mid-table case.

7.F (Hatcher bridge). In Hatcher Chapter 6, representation by quadratic forms uses genus, characters, and quadratic reciprocity. Sketch how the $Z \mapsto$ map might eventually use "genus" of flux configurations or topological characters on the lattice.

7.G (forward). Why will class-group ideas (Chapter 8) be relevant to which Z land in Magic Islands?

7.H (software honesty). Distinguish in one paragraph: (i) the coded $Z \mapsto$ map as software, (ii) its use as a periodic-table proxy (**Model**), (iii) any claim that it derives real chemistry from topology (**Hypothesis**).

7.I (Open Problem 4 teaser). Fix a small set of gauge sequences from Chapter 4. For two nearby Z, inspect whether flywheel metrics could plausibly share an orbit. What would count as non-uniqueness of the map?

7.7 Code and asset pointers

```
kingdom.core.flux_flywheel.map_z_to_flywheel[_extended]
qga/lib/flux_topograph.stability_landscape_z
kingdom.viz.magic_island.build_magic_island_heatmap
kingdom.viz.electron_cloud.build_electron_cloud_figure
kingdom/z_knowns/frame_XXXX.png   # flywheel stills
```

Figures: `scripts/generate_ch7_figures.py` **Related portal:** Flux Flywheel tab (interactive Z slider). **Open problems:** OP4 (uniqueness); OP3 (island invariants); OP5 ($350/\pi$, Ch. 10).

7.8 Looking ahead

The $Z \mapsto$ flux map links the classified arithmetic objects of Chapters 5–6 to concrete atomic numbers. In **Chapter 8** we lift the class group itself to the quaternionic / Hopf setting, giving a deeper arithmetic home for Magic Island counts and equivalence classes. **Chapter 10** returns to observational validation of the whole construction—including statistical protocols for the $350/\pi$ signature and Z-map correlations.

With the representation bridge in place, we are ready for the ideal-class-group lift.

Part IV

Arithmetic Depth

Chapter 8

Composition and Class Groups in the Quaternionic Setting

This chapter lifts Gauss composition of binary quadratic forms and the ideal class group (Hatcher Chapters 7–8) to the gauged Hopf lattice and quaternion orders. We define composition of flux configurations / flywheels, construct class-group analogues, and examine how these structures organize Magic Islands and equivalence classes. The theory remains a **Model** construction, with **Open Problem 6** as its central open edge.

Learning goals

1. Recall Gauss composition of quadratic forms and its arithmetic meaning.
2. Lift composition to flux configurations and flywheels on the gauged Hopf lattice.
3. Construct class-group analogues for flux topographs (and, later, quaternion orders).
4. Connect class-group structure to Magic Island counts and equivalence classes.
5. Prepare the ground for quaternion algebras and ideal theory (Chapter 9).

Figures in this chapter

Tag	File	Role
Fig. 8.1	`figures/fig8_1_gauss_composition.png`	Classical Gauss composition diagram
Fig. 8.2	`figures/fig8_2_flywheel_composition.png`	Composition of two flux topographs
Fig. 8.3	`figures/fig8_3_class_group_analogue.png`	Class-group analogue on the lattice
Fig. 8.4	`figures/fig8_4_island_from_class.png`	Magic Island from class-group data
Aux A8.1	`figures/aux8_1_composition_table.png`	Small composition table

Claim discipline

Claim	Type
Classical Gauss composition and ideal class groups (Hatcher Ch. 7–8)	**Theorem** (cited)
Composition of flux configurations / flywheels; class-group analogues	**Model** (core of OP6)
`qga/lib/composition.py` helpers	**Software fact**

8.1 Gauss composition — classical reminder

In Hatcher Chapter 7, Gauss defined a composition law on (primitive) binary quadratic forms of the same discriminant. Up to equivalence, the law is associative and turns the set of classes into a finite abelian group—the **class group** of the discriminant. The same group appears as the ideal class group of the corresponding quadratic order (Hatcher Chapter 8).

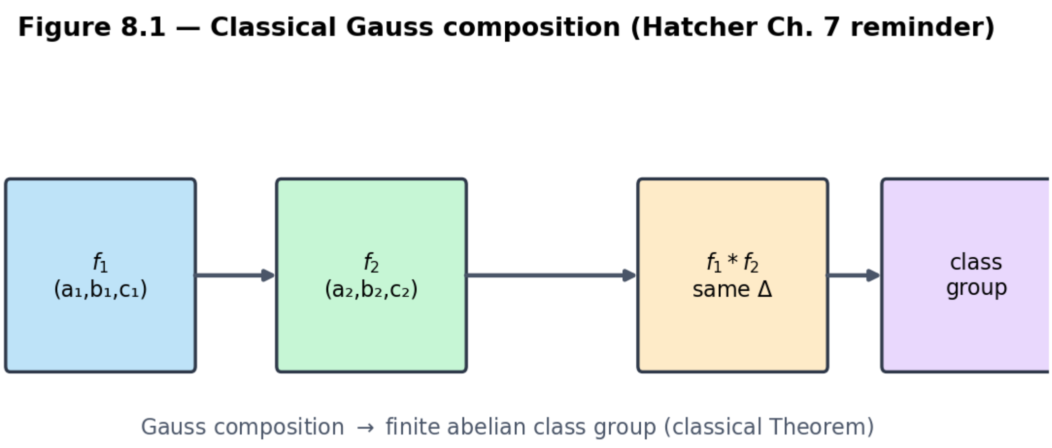


Figure 8.1: Forms f_1, f_2 of fixed discriminant compose to a third form; classes form a finite abelian group. This is classical **Theorem** territory; we cite Hatcher / Gauss rather than re-prove.

Why it matters here: class groups encode failure of unique factorization and control representation theory—exactly the arithmetic depth we want to lift for Magic Islands and the $Z \mapsto$ map.

8.2 Lifting composition to flux configurations and flywheels

On the gauged Hopf lattice we define a **composition** of two flux topographs (or flywheels) by combining values and/or points, then reducing with the Chapter 6 normalization.

8.2.1 Candidate methods (Model)

Method (code)	Rule
`value_sum`	Pointwise average of values; averaged then renormed points
`value_product`	Pointwise product (scaled)
`value_bilinear`	Soft bilinear blend $(V_a + V_b + V_a V_b)/3$
`point_mult`	Quaternion multiply corresponding lattice points; recompute functional

```
compose_flywheels(topo_a, topo_b, method="value_sum")
reduce_composition(composed) -> topograph, classification, magic_island_score
```

Honest status. On current candidate adjacency and modest angle lattices, composition tables often show **poor closure** and **low associativity scores** (**Software fact** from the sandbox). That is not a failure of the chapter—it is the experimental content of **Open Problem 6**.

Open Problem 6 (core of this chapter). Does a well-defined, associative composition law exist on equivalence classes of flux configurations / flywheels that:

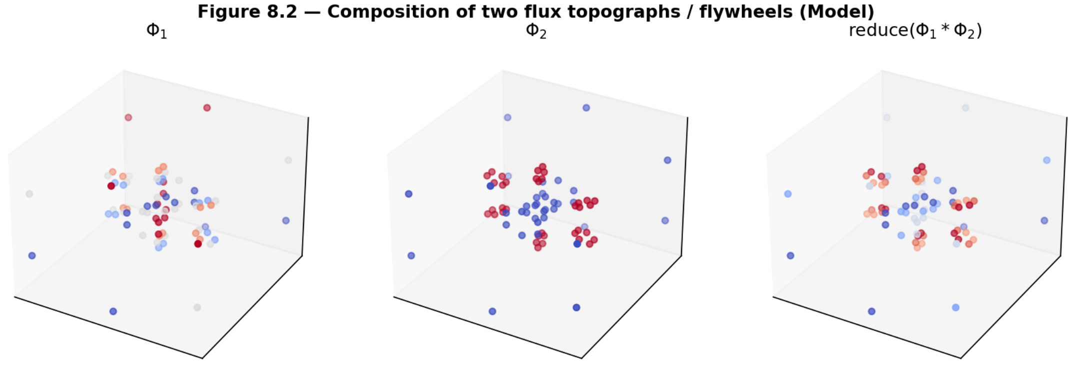


Figure 8.2: Φ_1, Φ_2, and reduce($\Phi_1 * \Phi_2$) for the pedagogical `value_sum` method.

1. Reduces to classical Gauss composition under a suitable projection or restriction?
2. Is compatible with the gauge actions of Chapter 4?
3. Produces a finite (or finitely generated) abelian group whose order/structure predicts Magic Island counts?
   Sandbox: `qga/lib/composition.py`. Depends on OP1–OP3.

8.3 Class-group analogues

The set of equivalence classes of reduced flux topographs, equipped with a composition law, forms a **class-group analogue**. Its order is a **class-number analogue**—related to Chapter 6's `class_number_analogue`, now enriched by composition tables and associativity diagnostics.

```
class_group_analogue(topo_or_list, method=..., dedup_tol=...)
  -> order, structure, table, closure_fraction, associativity, representatives

composition_table(reduced_set)
is_associative_up_to_equivalence(reduced_set, samples=...)
```

Magic Islands can be viewed as **distinguished classes** (or clusters of classes) where composition yields high `magic_island_score` or high portal `stability_score`. Predicting islands from group structure alone is OP3 $\cap$ OP6.

8.4 First computational labs

Helpers: `lib/composition.py` · **Appendix C §C.4**.

- **8.A** `compose_flywheels` + `reduce_composition`.
- **8.B** `composition_table` / `closure_fraction`.
- **8.C** `class_group_analogue` order and structure.
- **8.D** `is_associative_up_to_equivalence` (expect low scores — OP6 data).
- **8.E** Optional: compare composition methods.

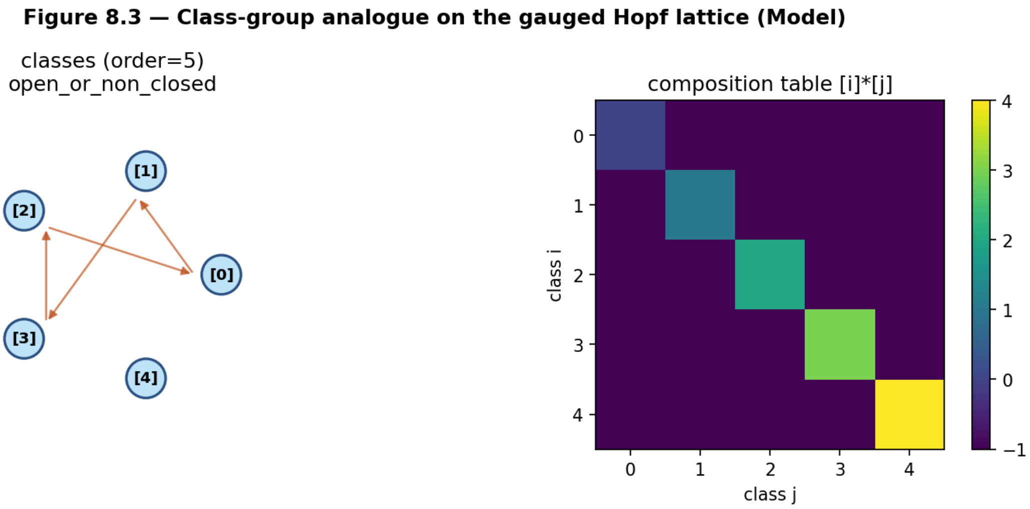


Figure 8.3: Equivalence classes with composition arrows, plus a numerical composition table. `structure` strings such as `partial_magma` or `open_or_non_closed` flag OP6 incompleteness.

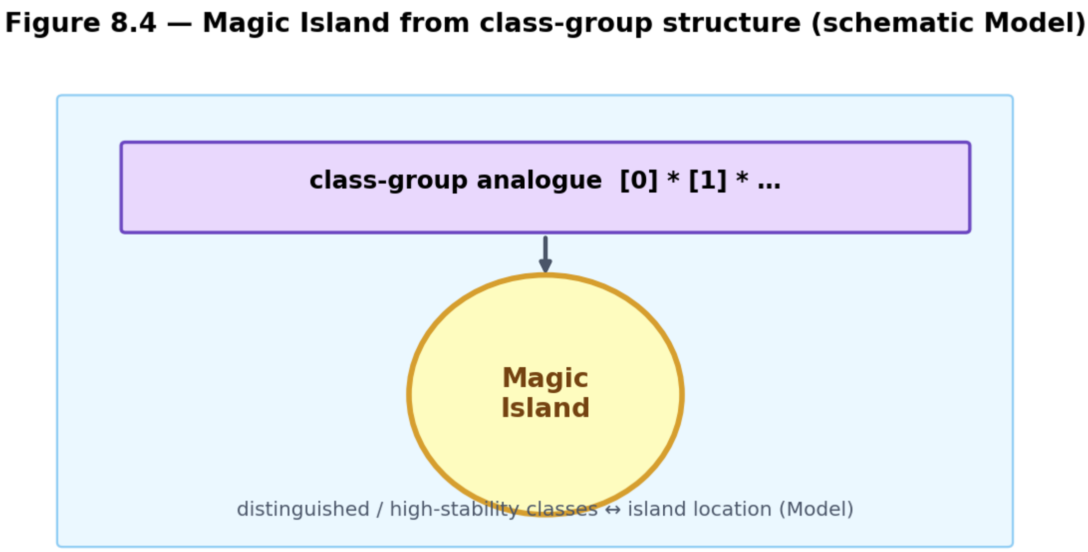


Figure 8.4: Schematic: class-group analogue → distinguished high-stability region. **Model** narrative, not a derived theorem.

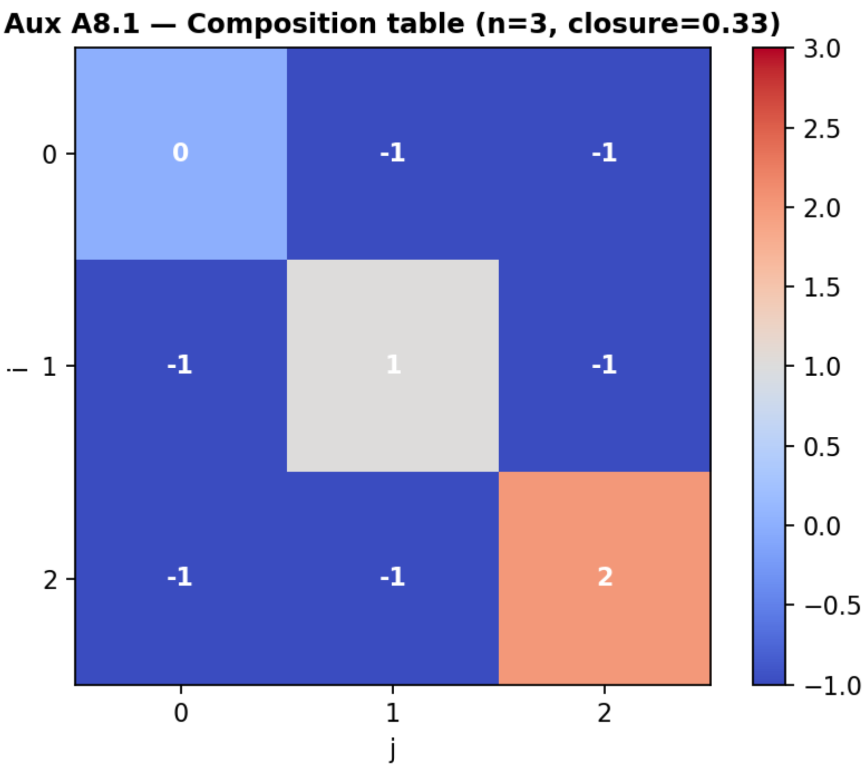


Figure 8.5: Small $n \times n$ table of reduced class indices under composition. Entries -1 mean "not closed in the set within tolerance."

```
from lib.composition import compose_flywheels, reduce_composition
# build t1, t2 as in Appendix C SC.4
# print(reduce_composition(compose_flywheels(t1, t2))["classification"]["type"])
```

8.5 Exercises

8.A (hand). State Gauss composition for binary quadratic forms in one sentence and explain why it produces a group law on classes.

8.B (hand). Why is associativity (up to equivalence) the key property needed for a class-group analogue?

8.C (code). Complete Labs 8.A–8.B. Report the type of the composed configuration and the composition table's `closure_fraction`.

8.D (code). Complete Lab 8.C. What is the size of the class-group analogue on your seed topograph?

8.E (code). Run Lab 8.D. What associativity score do you obtain? Record failures.

8.F (Hatcher bridge). In Hatcher, the class group encodes the failure of unique factorization in quadratic fields. Sketch how a class-group analogue on the Hopf lattice might encode "failure of unique representation" by stable flywheels / Z-map preimages.

8.G (Open Problem 6 teaser). Using `compose_flywheels` and `is_associative_up_to_equivalence`, test associativity on the current candidate adjacency. Document numerical failures—they *are* OP6 progress.

8.H (forward). Why will the quaternion algebra theory of Chapter 9 be the natural home for a rigorous version of these class groups?

8.I (software honesty). Distinguish: (i) classical class groups (**Theorem**), (ii) `class_group_analogue` outputs (**Model** + software), (iii) any claim that Magic Islands *are* class groups of a quaternion order (**Hypothesis** until proved).

8.6 Code and asset pointers

```
qga/lib/composition.py
qga/lib/flux_topograph.py
qga/lib/hopf_lattice.py
kingdom.core.flux_flywheel   # optional cross-check with Z stability
```

Figures: `scripts/generate_ch8_figures.py` **Open problems:** OP6 (this chapter); depends on OP1–OP3. **Related portal:** no dedicated composition tab yet—Lattice Simulator / Flux Flywheel remain dynamical counterparts.

8.7 Looking ahead

We now have a working **Model** lift of Gauss composition and the class group to flux configurations on the gauged Hopf lattice—with explicit experimental gaps (closure, associativity). In **Chapter 9** we move to quaternion algebras and their ideal theory, the rigorous arithmetic home for class groups of orders (Hurwitz, Eichler, ...). **Chapter 10** returns to observational validation, including whether class-group structure predicts $350/\pi$ or Z-map patterns.

With composition and class-group language in place, we are ready for the ideal-theoretic foundation.

Chapter 9

Quaternion Algebras and Ideal Theory

This chapter supplies the rigorous algebraic foundation for the class-group analogues and composition laws of Chapter 8. We introduce quaternion algebras over $\mathbb{Q}$, their orders (especially the Hurwitz order), and the ideal theory that turns equivalence classes of ideals into a group. This framework is the natural home for making the Model constructions of previous chapters more arithmetic and for attacking Open Problem 6 with algebraic tools.

Learning goals

1. Define quaternion algebras over $\mathbb{Q}$ and understand ramification.
2. Work with orders (Lipschitz, Hurwitz, maximal) and their ideal theory.
3. See how ideal classes form a group and how this relates to the class-group analogues of Chapter 8.
4. Connect ideal theory back to composition of forms / flywheels and to Magic Islands (**Model** dictionary).
5. Prepare algebraic language for rigorous uniqueness and composition statements (OP6 and beyond).
6. Separate **algebraic**, **sequential-label**, and **angle-lock** invariants when discrete moduli label continuous geometric orbits (supporting stack `vortex_math`).

Figures in this chapter

Tag	File	Role
Fig. 9.1	`figures/fig9_1_quaternion_algebra.png`	Quaternion algebra ramification diagram
Fig. 9.2	`figures/fig9_2_hurwitz_order.png`	Hurwitz order units / comparison
Fig. 9.3	`figures/fig9_3_ideal_class_group.png`	Ideal class group schematic
Fig. 9.4	`figures/fig9_4_ideal_to_flywheel.png`	Ideal class → flywheel (Model bridge)
Fig. 9.5	`figures/fig9_5_modulus_invariants.png`	Three-layer modulus invariants (algebra · flow · lock control)
Aux A9.1	`figures/aux9_1_ramification_table.png`	Hilbert-symbol ramification table

Claim discipline

Claim	Type
Quaternion algebras, Hilbert symbols, Hurwitz Euclidean property, Hurwitz class number 1	**Theorem** (classical)
Dictionary between ideal classes and flux topographs / Magic Islands	**Model**
Separation of modulus invariants on irrational rotations (algebra vs progression vs angle lock)	**Model** (experimentally controlled; not a theorem of TN)
`qga/lib/quaternion_algebra.py` helpers; external `vortex_math` metrics	**Software fact**

9.1 Quaternion algebras over $\mathbb{Q}$

A **quaternion algebra** over $\mathbb{Q}$ is a 4-dimensional central simple algebra over $\mathbb{Q}$. Up to isomorphism, each is given by square-free integers $a, b \neq 0$ via

$$\left(\frac{a,b}{\mathbb{Q}}\right) = \mathbb{Q} + \mathbb{Q}i + \mathbb{Q}j + \mathbb{Q}ij, \qquad i^2 = a,\ j^2 = b,\ ij = -ji.$$

The Hamilton algebra corresponds (over $\mathbb{R}$) to $a = b = -1$; as an algebra over $\mathbb{Q}$ one studies $\left(\frac{-1,-1}{\mathbb{Q}}\right)$.

9.1.1 Ramification

At each place p (finite prime or ∞), the local Hilbert symbol $(a, b)_p \in \{\pm 1\}$ detects whether the algebra is a division algebra (-1, **ramified**) or isomorphic to M_2 (**split**). The set of ramified places is finite and of even cardinality; it classifies the algebra.

```
QuaternionAlgebra(a, b)
  .ramified_places()
  .is_definite()          # True iff ramified at inf
  .hilbert_at(p)
```

Figure 9.1 — Ramification of a quaternion algebra (example $(-1, -1/\mathbb{Q})$)

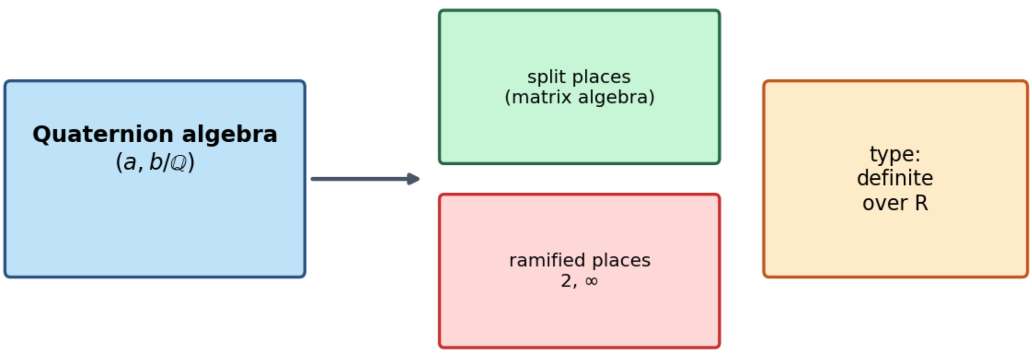


Figure 9.1: Example $\left(\frac{-1,-1}{\mathbb{Q}}\right)$: ramified at 2 and ∞ (definite over $\mathbb{R}$).

Claim type. Classification by ramification sets and Hilbert symbols: **Theorem** (classical). Implementation in the book helper: **Software fact** (pedagogical, not a full number-theory library).

Aux A9.1 — Hilbert symbols (a,b)_p (−1 = ramified)

(a,b)	2	3	5	7	11	13	∞
(-1,-1)	-1	1	1	1	1	1	-1
(2,3)	-1	-1	1	1	1	1	1
(-1,3)	-1	-1	1	1	1	1	1
(5,13)	-1	1	-1	1	1	-1	1

Figure 9.2: Hilbert symbols for several (a, b) at small places; red -1 marks ramification.

9.2 Orders in quaternion algebras

An **order** is a subring that is a full-rank $\mathbb{Z}$-lattice in the algebra. In the Hamilton setting:

Order	Definition	Units	Euclidean?
Lipschitz	$\mathbb{Z}[i, j, k]$	8	No
Hurwitz	all-integer or all-half-integer coords	24	Yes (maximal)

The Hurwitz order is preferred for arithmetic (Chapter 1 **Model** preference for lattice work; Euclidean algorithm is classical **Theorem**). Its unit group is the binary tetrahedral group—the set Λ_0 of Chapter 3.

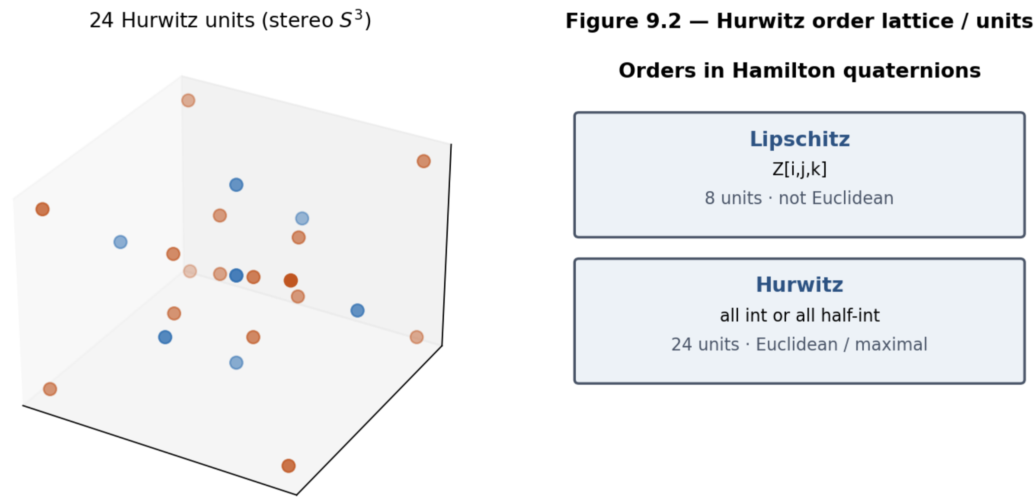


Figure 9.3: 24 Hurwitz units in stereographic S^3, and Lipschitz vs Hurwitz comparison.

```
HurwitzOrder().units          # 24
HurwitzOrder().is_euclidean() # True (classical fact)
HurwitzOrder().is_maximal()   # True (classical fact)
LipschitzOrder().n_units()    # 8
```

In general quaternion algebras, **maximal orders** play the role of rings of integers in number fields; different maximal orders need not be conjugate when the class number is nontrivial.

9.3 Ideal theory and class groups

One studies left ideals, right ideals, and two-sided ideals of an order. For **definite** quaternion algebras (ramified at ∞), the **left ideal class set** of a maximal order is finite — a *set*, not a group in general. Ideal multiplication induces a **group** structure on **two-sided** ideal classes (the Picard / two-sided class group). Related class sets control representation of integers by norms of ideals.

Theorem (Hurwitz, cited). The Hurwitz order is a left PID: every left ideal is principal, so the left class set has cardinality 1 (class number 1). The two-sided class group is likewise trivial. These are distinct objects that happen to be singletons here.

9.3.1 Hurwitz class number (classical)

The left ideal class number of the Hurwitz order is **1**: every left ideal is principal. The book helper reports this as a cited classical fact with a finite sample check of small-norm elements—not a full ideal enumeration.

```
left_ideal_class_set(HurwitzOrder())
  -> LeftIdealClassSetResult(cardinality=1,
    method='classical_hurwitz_left_pid_class_number_one', ...)
two_sided_class_group(HurwitzOrder())
  -> TwoSidedClassGroupResult(order=1, method='classical_hurwitz_two_sided_trivial', ...)
```

Claim type. Finiteness for definite maximal orders; Hurwitz class number 1: **Theorem** (classical). Full algorithmic enumeration of ideals for general (a, b): out of scope for this sandbox.

9.4 From ideals to flux topographs and flywheels (Model bridge)

Chapter 8 approximated class groups with composition of flux topographs. The natural dictionary is still a **Model**:

Algebraic object	Geometric / Model object
Ideal in a quaternion order	Flux configuration / supporting cycle on the gauged Hopf lattice
Ideal class	Equivalence class of reduced flux topographs
Ideal class group	Class-group analogue of Chapter 8
Norm of an ideal	Stability score or `magic_island_score`
Ideal multiplication	Composition of flywheels (when well-defined — OP6)

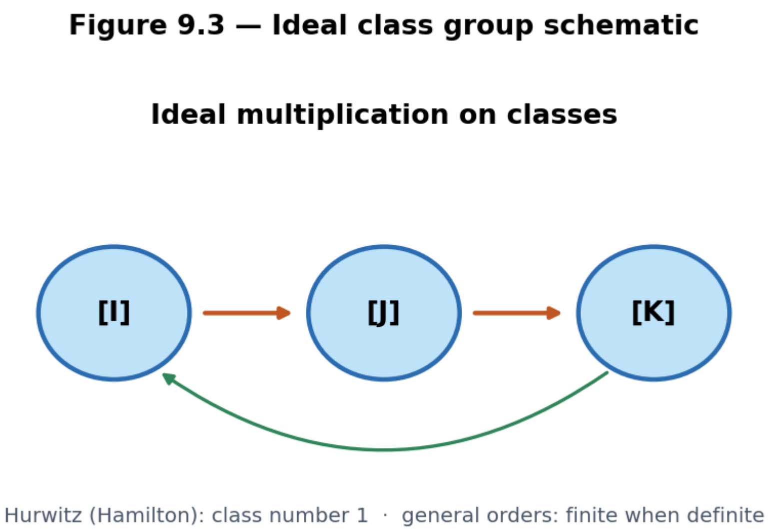


Figure 9.4: Ideal classes with multiplication. For Hurwitz, a single class; general definite orders may have larger finite class numbers.

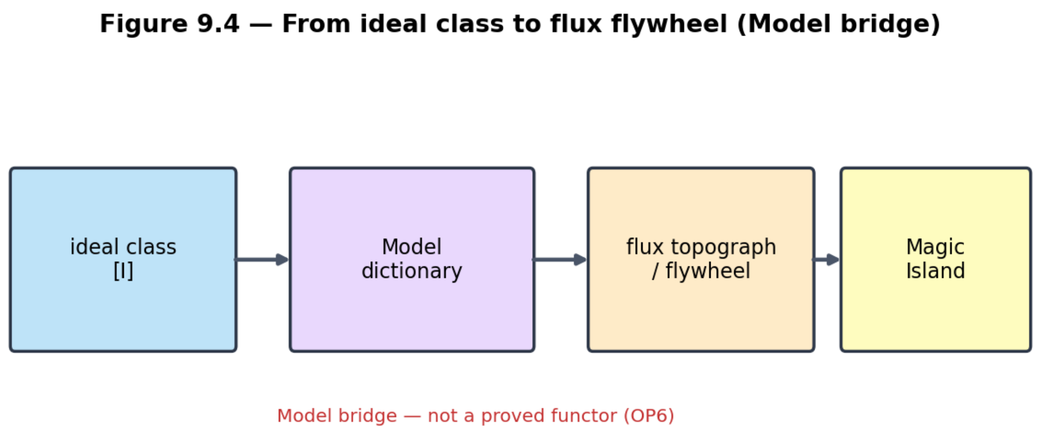


Figure 9.5: Schematic Model bridge. Not a proved functor.

Open Problem 6 (continued). Use ideal theory of quaternion algebras to define a rigorous composition law on classes of flux configurations that is associative, gauge-compatible, and reduces to Gauss composition in suitable limits. Determine whether the resulting class group predicts Magic Island location and size.

```
form_ideal_dictionary_entry()   # static dictionary for labs
class_group_analogue(...)       # Ch. 8 Model side
left_ideal_class_set(...)       # left class *set* (Hurwitz: cardinality 1)
two_sided_class_group(...)      # two-sided class *group* (Hurwitz: order 1)
```

9.5 Modulus, invariants, and angle–label locking

Supporting stack: `vortex_math` · *research note:* `notes/RESEARCH_NOTE_moduli.md`.

Chapters 3–8 build discrete geometric objects (gauged Hopf lattice, flux topographs) whose **labels** and **equivalence classes** carry arithmetic meaning. Before those dictionaries become too ambitious, it is worth isolating a cleaner, lower-dimensional lesson: when symbolic patterns of Vortex Math—digital roots, the doubling circuit, and the privileged status of 3-6-9—are placed onto points generated by *continuous geometry* rather than discrete integer sequences alone, the choice of labeling modulus does **not** act as a single uniform operator. It modifies *different invariants* depending on how it is attached to the system.

9.5.1 Geometric stage

Consider an irrational rotation on the unit circle whose step size is a fixed multiple of the circle constant:

$$\theta_k = k \cdot \frac{9}{\pi} \pmod{2\pi}.$$

Because $9/\pi$ is an irrational multiple of 2π, the orbit is dense: successive points come arbitrarily close to every location on the circle without closing into a finite regular polygon. The factor **9** echoes the digital-root base of Vortex Math; π is the circle's own constant. Geometry and numerology share the stage, but they are no longer forced to play the same role. (A coupled step $\Delta\theta = m/\pi$ will appear below when the modulus is allowed to set the winding rate.)

9.5.2 Three layers of invariant

Algebraic structure of the doubling map. Changing the modulus alters the cycle decomposition of repeated doubling on $\mathbb{Z}/m\mathbb{Z}$. The value $m = 37$ produces the cleanest long orbit: length 36 from 1, with 2 acting as a primitive root modulo 37. Composites built from 37—notably $111 = 3 \times 37$ and $333 = 9 \times 37$—inherit this long cycle length but fragment it according to the Chinese Remainder Theorem. Algebraic "cleanness" is therefore strongest at the repunit prime **37**, the prime factor of the length-3 repunit $R_3 = 111$.

Orderly progression of labels along the geometric orbit. When the rotational step itself is *coupled* to the modulus ($\Delta\theta = m/\pi$), labels advance with greater sequential coherence for certain composites. The value **111** frequently yields the highest scores for label progression and overall symmetry under this coupled regime. Here the modulus influences the *flow of labels relative to the accumulating points*, rather than the intrinsic algebraic cycles alone. Composite structure and winding rate interact; the prime-mover number 111 is not merely decorative in this layer.

Positional angle–label locking. Only one labeling method creates a genuine *statistical* dependence between label and actual geometric angle: explicit angular binning (`angle_bin`). Under the default sequential method (`step_index`)—labels from the step count and its digital root or modular residue—normalized mutual information between labels and angles remains at or below chance once a proper null model (label shuffling with the same marginals) is applied.

Apparent visual symmetry in sequential plots therefore reflects **orderly sequential progression**, not a hidden locking of labels to preferred angles on the circle. That outcome is precisely what one should expect from an irrational rotation labeled by step index: the orbit has no preferred angles unless the labeling method itself introduces angular bins by construction.

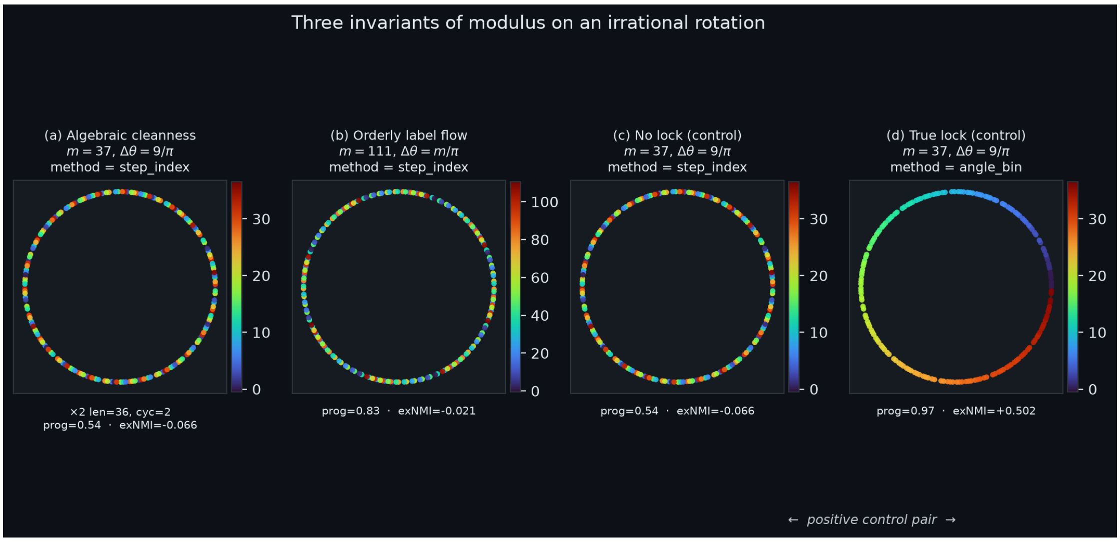


Figure 9.6: **(a)** Algebraic cleanness: $m = 37$, fixed step $9/\pi$, sequential labels—long clean $\times 2$ structure.

(b) Orderly label flow: $m = 111$, coupled step m/π, sequential labels—higher label-progression along the orbit. **(c)–(d)** Positive control pair at $m = 37$: sequential labeling yields excess NMI at or below the shuffle null; angular binning yields strongly positive excess NMI. Visual order under step-index is progression, not positional lock. *(Supporting generation: `vortex_math` · `assets/book_figure_three_layers.png`.)*

9.5.3 Controls and claim type

These layers were separated by controlled comparisons of mapping methods, step modes, and quantitative metrics: excess normalized mutual information (**exNMI**) against permutation baselines, label-progression scores, and angular uniformity. A positive control seals the analysis: when labels *are* angle bins, exNMI is strongly positive and many standard deviations above the null; when labels are step indices, exNMI is not.

Claim type. The three-layer separation and control are a **Model** for how arithmetic moduli interact with geometric orbits in this book's ecosystem—experimentally reproducible, not a classical theorem of Hatcher *Topology of Numbers*. Implementation and full tables: **Software fact** (`vortex_math` repo; note in `notes/RESEARCH_NOTE_moduli.md`).

9.5.4 Takeaway for the QGA spine

The practical consequence is concise:

> **Modulus changes the algebraic and sequential-label invariants; it does not create angle–label locking on an irrational rotation unless you label by angle itself.**

This distinguishes synchronicity that arises from composite number structure—notably **111** as prime-mover composite 3×37—from stronger geometric claims that would require explicit angular construction. Algebra crowns 37; coupled progression frequently crowns 111 under the m/π regime; true positional locking belongs only to labeling methods that bin explicitly by angle.

For the rest of the book: when Chapters 5–8 speak of "labels," "classes," and "Magic Islands," the same discipline applies—**name which invariant you are measuring**. Ideal class groups (sections 9.3–9.4) are algebraic invariants; flux scores and topograph diagrams are geometric or Model-level; statistical lock to a continuous base space is a third claim, and must be tested against a null, not inferred from visual order alone. Chapter 10 returns to observation with the same rule: multi-domain recurrence and visual order are not substitutes for a named invariant plus a null.

```
# External supporting stack (not vendored in lib/)
~/Projects/vortex_math
  python src/main.py --family-37 --num-steps 200
  python src/main.py --resonance-scan --method step_index --num-steps 600
  python src/main.py --resonance-scan --method angle_bin --num-steps 600
```

9.6 First computational labs

Helpers: `lib/quaternion_algebra.py` · **Appendix C §C.4**.

- **9.A** `QuaternionAlgebra(-1,-1).ramified_places()`.
- **9.B** Hurwitz: 24 units, Euclidean, maximal.
- **9.C** `left_ideal_class_set` → cardinality 1 (cited: Hurwitz left PID).
- **9.D** `two_sided_class_group` vs Model `class_group_analogue`.
- **9.E** Compare algebraic class number 1 with the Model analogue (one 9.E).
- **9.F** Hilbert symbols table (own number).
- **9.G (optional, external)** `vortex_math` resonance control.

```python
from lib.quaternion_algebra import (
    QuaternionAlgebra, HurwitzOrder, left_ideal_class_set, two_sided_class_group,
)
print(QuaternionAlgebra(-1, -1).ramified_places())
print(HurwitzOrder().n_units(), left_ideal_class_set().cardinality)
print(two_sided_class_group().order)
```

9.7 Exercises

9.A (hand). Define a quaternion algebra over $\mathbb{Q}$ and state what ramification means.

9.B (hand). Why is the Hurwitz order preferred over the Lipschitz order for most arithmetic work?

9.C (code). Complete Labs 9.A–9.B. Confirm 24 Hurwitz units and Euclidean / maximal flags.

9.D (code). Run Lab 9.C. What class number do you obtain for Hurwitz, and what does the `method` string say?

9.E (code). Complete Lab 9.D. Compare algebraic class number 1 with the Model `class_group_analogue` order. Why can they differ?

9.F (Hatcher bridge). In Hatcher Chapter 8, forms correspond to ideals. Sketch how section 9.4's dictionary might be made precise for a quaternion order (even if only conjecturally).

9.G (Open Problem 6). Using Hurwitz class number 1, what would a *successful* OP6 composition law look like when restricted to configurations "coming from" principal ideals? What obstructions remain for non-principal classes in other algebras?

9.H (forward). Why will Chapter 10's observational validation benefit from a rigorous ideal-theoretic home for class-group analogues?

9.I (software honesty). Distinguish: (i) two-sided class group vs left class set (**Theorem**), (ii) `left_ideal_class_set` / `two_sided_class_group` toy reports (**software** citing theorem), (iii) Ch. 8 `class_group_analogue` (**Model**).

9.J (modulus invariants, hand). Using section 9.5:

1. Name the three layers (algebra / progression / angle lock) in one sentence each.
2. Which modulus wins algebraic cleanness of $\times 2$ on $\mathbb{Z}/m\mathbb{Z}$, and what cycle length from 1 does it give?
3. Which modulus often wins label progression under coupled step m/π?
4. Why is strongly positive exNMI under `angle_bin` a *positive control* for the null baseline, rather than a discovery that "37 locks labels to angles"?
5. State the section's one-line claim in your own words, then apply it to one Ch. 5–8 phrase ("Magic Island," "class," or "stability score"): which invariant class does that phrase live in?

9.K (code, external). If `~/Projects/vortex_math` is available:

```
cd ~/Projects/vortex_math && source .venv/bin/activate
python src/main.py --resonance-scan --method step_index --num-steps 400
python src/main.py --resonance-scan --method angle_bin --num-steps 400
```

For $m \in \{9, 37, 111\}$ under m/π, copy **exNMI** for both methods into a small table. Confirm: `step_index` exNMI $\lesssim 0$; `angle_bin` exNMI $\gg 0$. One paragraph: what does this say about visual symmetry under sequential labeling?

9.8 Code and asset pointers

```
qga/lib/quaternion_algebra.py
  QuaternionAlgebra, hilbert_symbol,
  HurwitzOrder, LipschitzOrder,
  left_ideal_class_set, two_sided_class_group, form_ideal_dictionary_entry

qga/lib/composition.py
qga/lib/hopf_lattice.py   # HURWITZ_UNITS shared with Ch. 3

# Supporting stack (external)
~/Projects/vortex_math/          # active development
qga/lib/vortex_math/README.md    # pointer into this book's tree
qga/notes/RESEARCH_NOTE_moduli.md
```

Figures: `scripts/generate_ch9_figures.py` · Fig. 9.5 from `vortex_math` book figure **Open problems:** OP6 (composition via ideal theory); OP1–OP3 for the geometric side of the dictionary. **Research note:** `notes/RESEARCH_NOTE_moduli.md`

9.9 Looking ahead

We now have the algebraic backbone—quaternion algebras, orders, and ideal class groups—that can host rigorous versions of the Model constructions in Chapters 5–8, together with a controlled lesson on **which invariant a modulus is measuring** when discrete labels meet continuous orbits. In **Chapter 10** we return to the observational layer: statistical protocols for $350/\pi$, the $Z \mapsto$ map, Magic Islands, and any class-group predictions that become precise enough to test—always distinguishing algebraic claims, Model dictionaries, and empirical lock statistics.

With the ideal-theoretic foundation in place, the full construction is ready for observational scrutiny.

Part V

Observations and Models

Chapter 10

Observations, Hypotheses, and Validation

This chapter collects the observational layer of the Kingdom Come Model. It presents the recurring numerical signature $W_g = 350/\pi$, the $Z \mapsto$ flux map correlations, Magic Island alignments, and other multi-domain patterns. All claims here are labeled explicitly as **Model** or **Hypothesis**, and the chapter supplies concrete validation protocols (Table **T4**) so that the construction can be tested, refined, or falsified.

Learning goals

1. Review the major observational signatures that motivated the project.
2. Distinguish Model statements from testable Hypotheses.
3. Apply the validation protocols to the $350/\pi$ signature and the $Z \mapsto$ map.
4. Summarize the status of all Open Problems.
5. Provide a clear roadmap for future empirical work.

Figures in this chapter

Tag	File	Role
Fig. 10.1	`figures/fig10_1_350_over_pi_domains.png`	Multi-domain recurrence of $350/\pi$
Fig. 10.2	`figures/fig10_2_z_map_correlation.png`	Stability score vs chemical proxy
Fig. 10.3	`figures/fig10_3_magic_island_validation.png`	Islands vs observed specialness
Fig. 10.4	`figures/fig10_4_validation_flowchart.png`	Validation protocol (Table T4)
Aux A10.1	`figures/aux10_1_pulsar_bitcoin_overlay.png`	Pulsar / Bitcoin observation assets

Claim discipline (throughout the book)

Label	Meaning
Theorem	Proved here or classical
Model	Consistent mathematical or physical construction, not claimed as nature's unique law
Hypothesis	Observational claim needing validation or falsification
Software fact	Reproducible output of a named implementation

Invariant discipline (from Chapter 9 section 9.5). When reading or testing observations, also name *which invariant* is at stake:

Layer	Question	Example in this chapter
Algebraic	Does a discrete arithmetic object have a clean structure?	Ideal class number; composition law (OP6)
Sequential / Model flow	Do labels or scores progress coherently along a construction?	Stability scores along Z; topograph evolution
Statistical lock	Is there dependence on a continuous base beyond chance?	Multi-domain recurrence of $350/\pi$ vs null (Table T4)

Visual order and multi-domain co-occurrence are *not* automatic geometric lock. Chapter 9's modulus lesson (and its `angle_bin` control) is the template: measure excess association against a null, do not infer locking from appearance alone. Research note: `notes/RESEARCH_NOTE_moduli.md`.

10.1 The major observational signatures

The Kingdom Come project was motivated by several recurring numerical and structural patterns:

- The constant

$$W_g = \frac{350}{\pi} \approx 111.408$$

(`WG_FROM_350_OVER_PI` in Kingdom Come / book helper `WG_350_OVER_PI`) appearing across pulsar-timing narratives, Bitcoin Pi Cycle notes, TLS tree analysis, cuprate sketches, and structural constants.

- Alignment of high `stability_score` regions (Magic Islands) with noble-gas Z and selected mid-table elements.
- Topological protection (linked Hopf fibers) as a geometric language for stability that classical elementary number theory only hints at.

These patterns live in the portal **Observations** tab and in `kingdom/observations/`.

Figure 10.1 — Multi-domain recurrence of $350/\pi$ (schematic Hypothesis map)

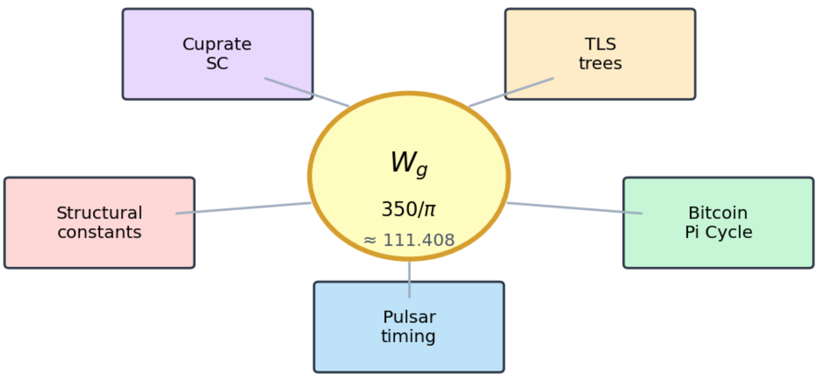


Figure 10.1: Domains associated with the $W_g = 350/\pi$ signature in the observation program (schematic).

Claim type. Raw portal constants and plotted observation assets: **software / project facts**. Per-domain clustering near $W_g = 350/\pi$: **Hypothesis H1a–H1e** (OP5), not one bundled clock.

Figure 10.2: Portal observation assets (pulsar / Bitcoin notes associated with $W_g = 350/\pi$, not 350π). Visual motivation only—not a completed statistical test.

10.2 The $Z \mapsto$ flux map — current status

Chapter 7 defined

$$Z \mapsto (\text{flywheel state}, \texttt{stability\_score}, \texttt{stability\_class}, \ldots)$$

via `map_z_to_flywheel` / `map_z_to_flywheel_extended`. Outputs show peaks near noble-gas Z and partial alignment with ionization-energy proxies in extended fields.

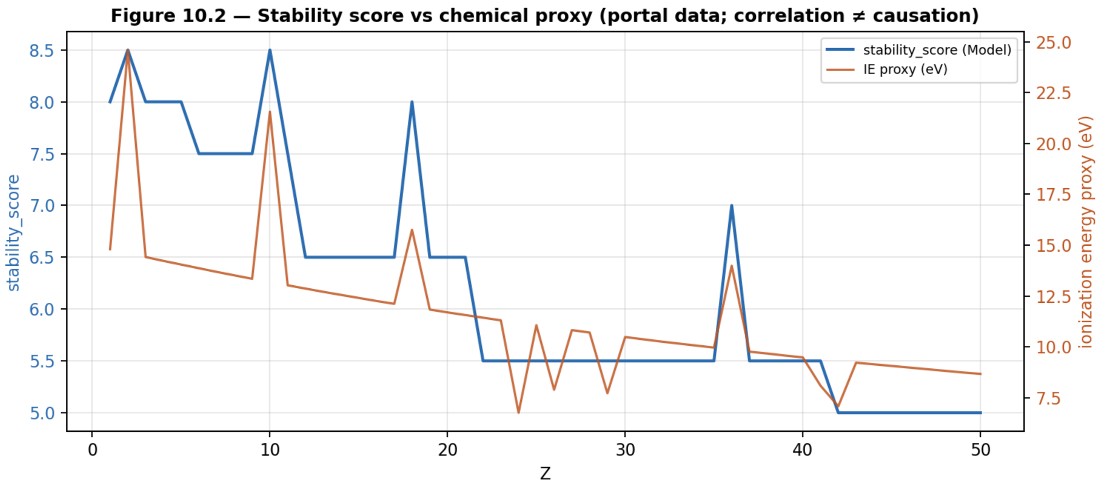


Figure 10.3: Portal `stability_score` vs an IE proxy across Z. **Correlation is not causation**; ablation and out-of-sample tests are required (Table T4).

Claim type. Reproducible curves from the current implementation: **Software fact**. Physical emergence of the periodic table: **Model** / **Hypothesis**.

Open Problem 4 (recap). Up to gauge, is the $Z \mapsto$ flywheel map unique under stated axioms? See Chapter 7 and `notes/open_problems.md`.

10.3 Magic Islands and chemical/nuclear specialness

Magic Islands (Chapters 5–6) appear as regions of high periodicity, controlled variance, and high `stability_score` / `magic_island_score`. Some coincide with elements that are chemically or nuclearly special.

Figure 10.3 — Magic Island predictions vs observed specialness (schematic)

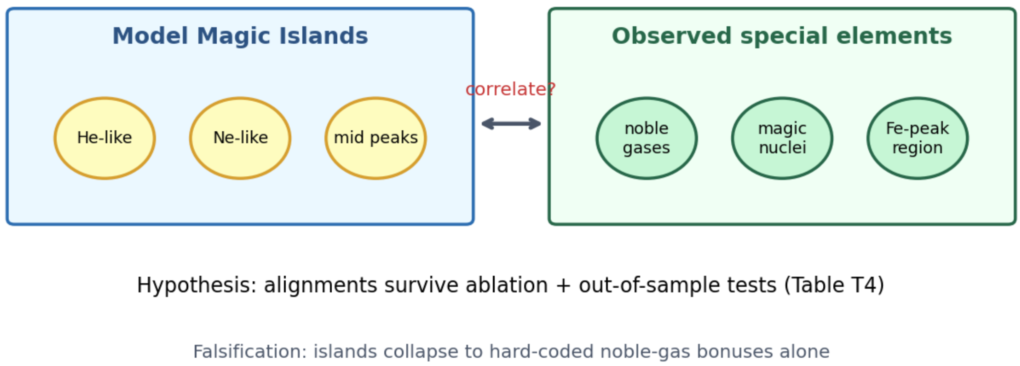


Figure 10.4: Model islands vs observed special classes (schematic). Encouraging correlation inside the working Model—not yet derived from first principles.

Falsification sketch. If ablating explicit noble-gas bonuses removes all predictive alignment with held-out properties, the "emergence" reading of islands fails the Table T4 protocol for that claim.

10.4 Summary of Open Problems

#	Problem	Home	Status (one line)
OP1	Canonical quaternionic Farey	Ch. 3	Open — `candidate_adjacency`
OP2	Flux topograph axioms	Ch. 5	Open — `flux_topograph`
OP3	Class number $\leftrightarrow$ Magic Island	Ch. 6	Open — heuristic
OP4	$Z \to$ flywheel uniqueness	Ch. 7 / 10	Open
OP5	$350/\pi$ first principles / falsify	Ch. 10	Open — Table T4
OP6	Flywheel composition (Gauss lift)	Ch. 8–9	Open — low sandbox closure

Full statements, sandboxes, success criteria, and dependency sketch: Appendix B. Helpers: `lib.validation.open_problems_status_table()`.

10.5 Validation protocols (Table T4)

A pre-registered validation checklist for major hypotheses. **Full protocol, hypothesis catalog, and statistical helpers: Appendix D**.

Eight elements (names only): null definition · data sources · test statistic · multiple testing · power · falsification · pre-registration · reproducibility.

```
from lib.validation import table_t4_checklist, default_hypotheses, run_table_t4_demo
print(len(table_t4_checklist()), [h.name for h in default_hypotheses()])
demo = run_table_t4_demo(seed=1, alpha=0.01)  # toy null -- not evidence for 350/pi
print(demo["decision"], demo["disclaimer"])
```

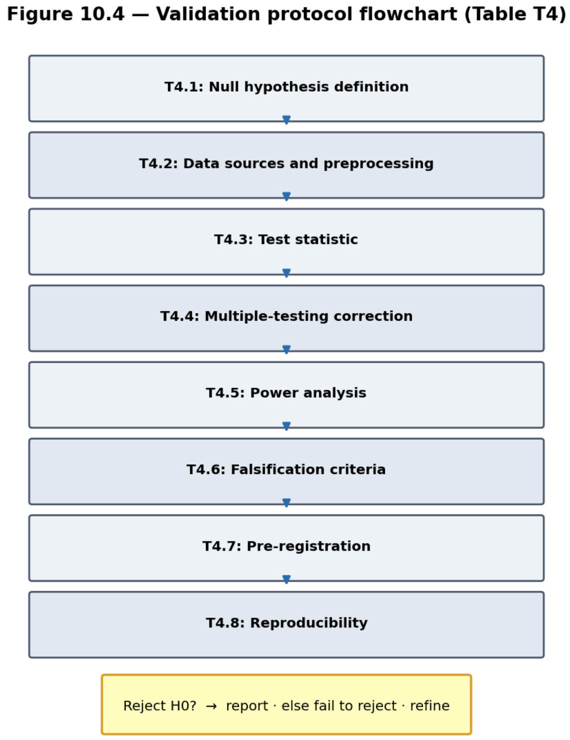


Figure 10.5: High-level Table T4 decision tree (details in Appendix D).

Claim type. Existence of the protocol and helpers: **Software fact**. "Hypothesis X passed T4": **Hypothesis** until the full checklist is executed and reviewed.

Open Problem 5. Derive $W_g = 350/\pi$ from lattice geometry, **or** falsify multi-domain recurrence via pre-registered tests (Appendix D / Table T4).

10.6 Summary of the entire construction

Parts I–II (Chapters 1–4) built the geometric foundation: quaternions, the Hopf fibration, the gauged Hopf lattice, and its symmetries—the higher-dimensional lift of Hatcher's Farey diagram.

Part III (Chapters 5–7) developed the visual and representation-theoretic core: flux topographs, classification and Magic Islands, and the $Z \mapsto$ flux map.

Part IV (Chapters 8–9) supplied arithmetic depth: composition, class-group analogues, and quaternion algebra / ideal theory as a rigorous home for those analogues. Section 9.5 added a controlled lesson on **modulus invariants**—algebraic cleanness (e.g.\ $m = 37$), sequential label progression (often $m = 111$ under coupled step), and true angle–label lock only under explicit angular binning—so that arithmetic labels are never confused with geometric locking.

Part V (this chapter) returns to observation and validation, closing the loop between geometry, arithmetic, and empirical patterns.

Every major claim is labeled so readers can separate theorems, models, and hypotheses; every observational pattern should also name the **invariant layer** it claims (section 9.5; Table T4).

10.7 Outlook and invitation

The six Open Problems are the main research edges. Advances on **OP1** (canonical adjacency), **OP2** (topograph axioms), or **OP6** (associative composition via ideal theory) especially strengthen the whole framework.

Validation can proceed incrementally: a documented failure to reject the null for $350/\pi$ in one domain, or a refined class-number invariant with better island prediction, both count as progress. Apply the same null-first discipline as Chapter 9 section 9.5: if a claim is *lock* or *recurrence*, define the null; if it is *algebraic structure*, report the discrete object; if it is *Model flow*, say so and do not upgrade it to geometry by rhetoric.

The qga repository and Kingdom Come portal remain open for extension. **Book Mode** is live in the portal (chapter map → Hopf / Lattice / Flux / Observations mini demos and tab jumps; see `HOW_TO_USE.md` §6). Future work includes fuller composition / validation dashboards.

10.8 Exercises (final chapter)

10.A (hand). Using **Appendix B**, choose OP1, OP2, or OP6 and write a one-page research proposal: sandbox, diagnostic, and success criterion.

10.B (code). Using **Appendix D** / `run_table_t4_demo`, run under the null toy generator and with hand-chosen p-values. Report Bonferroni threshold, Fisher combined p, and decision. Toy data are **not** evidence for $350/\pi$.

10.C (code). Using `proximity_to_wg` (Appendix D), measure closeness of a few constants to $350/\pi$. Diagnostic only, not a significance test.

10.D (forward). If OP6 were solved with an associative law from ideal multiplication, what new prediction about Magic Islands or the $Z \mapsto$ map could be tested under Table T4?

10.E (philosophical). In one paragraph, explain why Theorem / Model / Hypothesis labeling is essential for a project at the intersection of pure mathematics and speculative physics.

10.F (project). Open `default_hypotheses()` and add one new `HypothesisSpec` for a domain you care about, with null, falsification, and data sources.

10.G (invariant discipline). Using the three-layer table in this chapter's claim discipline (from Chapter 9 section 9.5), classify each of the following as algebraic / sequential-Model / statistical-lock (or mixed), and name one null or control you would demand: (i) Hurwitz class number 1; (ii) Magic Island high `stability_score` band; (iii) multi-domain recurrence of $350/\pi$; (iv) visual "symmetry" of a step-index vortex plot.

10.9 Code and asset pointers

```
qga/lib/validation.py
Appendix B (Open Problems) . Appendix D (Table T4 full)
Appendix C SC.4 (lab listings)
kingdom/observations/ . kingdom.core.flux_flywheel
```

Figures: `scripts/generate_ch10_figures.py` · **Appendix E** (atlas) **Related portal:** Observations tab; Flux Flywheel.

10.10 Closing

This book has attempted to lift Allen Hatcher's visual, geometric approach to number theory into the richer world of quaternions and the Hopf fibration, while labeling every claim carefully. The result is a coherent framework connecting classical number theory, topology, and a speculative model of emergent physics.

Whether the Model survives empirical scrutiny is now a question for systematic validation. The tools in Chapters 1–9 and the protocols in this chapter are designed to make that question answerable.

Appendix A

Terminology and Notation

Nonstandard terms and the minimal notation sheet used throughout the book. (Expanded from the Preface and `HOW_TO_USE.md`.)

A.1 Claim labels

Label	Meaning
Theorem	Proved here or classical
Model	Consistent construction; not claimed as nature's unique law
Hypothesis	Observational claim needing validation or falsification
Software fact	True of the current code or portal implementation

---

A.2 Nonstandard terminology

Term	Working meaning
Gauged Hopf lattice	Discrete point set in S^3 (or dynamics on site quaternions) with Hopf projection, adjacency, and left/right gauge actions
Flux flywheel	Closed, topologically protected circulating flux configuration on the lattice
Flux topograph	Value landscape of a flux functional on lattice sites/edges, with separators (lift of Conway topographs)
Magic Island	Region of enhanced stability / periodicity in parameter or Z space (Model)
Porous vacuum	Informal name for incomplete / gauge-flexible ambient lattice substrate
Separator structure	Locus where a flux functional changes sign or crosses a threshold
Class-group analogue	Equivalence classes of reduced topographs/flywheels with a candidate composition law (Model; OP6)
Class-number analogue	Count of inequivalent reduced representatives (`class_number_analogue`)
$Z \mapsto$ **map**	Portal map from atomic number to flywheel metrics (`map_z_to_flywheel`)
Table T4	Pre-registered validation checklist for Hypotheses (Ch. 10, Appendix D)
QVPIC	Quaternion vortex persistent identity (related Kingdom Come / QVPIC work; optional cross-reference)
W_g	Working topological clock $350/\pi \approx 111.408$ (Hypothesis layer)

A.3 Notation sheet

Symbol	Meaning
$\mathbb{H}$	Hamilton quaternions
S^3	Unit quaternions
$h : S^3 \to S^2$	Hopf map
$N(q) = q\bar{q}$	Multiplicative (squared) norm
$\mathcal{H}$	Hurwitz order
$L = \mathbb{Z}[i, j, k]$	Lipschitz order
Λ_0	$\mathcal{H} \cap S^3$ (24 units)
Λ_{ang}	Angle-sampled lattice
Λ_{dyn}	Two-gyro dynamical sites
$E_{\parallel}, E_{\perp}$	Along-fiber / inter-fiber edges
$\left(\frac{a,b}{\mathbb{Q}}\right)$	Quaternion algebra over $\mathbb{Q}$
OP1–OP6	Open problems (Appendix B)

A.4 Software path cheat sheet

Path	Role
`lib/hopf_lattice.py`	Lattice, adjacency (OP1), gauge sequences
`lib/flux_topograph.py`	Topographs, classification (OP2–OP3)
`lib/composition.py`	Composition, class-group analogue (OP6)
`lib/quaternion_algebra.py`	Algebras, Hurwitz, toy class number
`lib/validation.py`	Table T4, OP5 diagnostics
`kingdom.core.*`	Live portal modules
`kingdom.simulations.lattice`	`TwoGyroLattice`

Appendix B

Open Problems (Extended)

Living research edges of the project. The main text (especially Chapters 3, 5–10) states each problem briefly; this appendix keeps the full statements, sandboxes, and status tracking.

Status vocabulary: `Open` · `In progress` · `Partial result` · `Resolved` · `Deferred`

B.1 Status table

#	Problem	Home	Status	Sandbox
OP1	Canonical quaternionic Farey structure	Ch. 3	Open	`lib.hopf_lattice.candidate_adjacency`
OP2	Flux topograph axioms	Ch. 5	Open	`lib.flux_topograph`
OP3	Class number ↔ Magic Island	Ch. 6	Open	`classify_topograph_type`, `class_number_analogue`
OP4	$Z \to$ flywheel uniqueness	Ch. 7 / 10	Open	`map_z_to_flywheel`, `stability_landscape_z`
OP5	$350/\pi$ first principles or falsification	Ch. 10	Open	`lib.validation` · Table T4 (Appendix D)
OP6	Composition of flywheels (Gauss lift)	Ch. 8–9	Open	`lib.composition` · `lib.quaternion_algebra`

B.2 Problem statements

B.2.1 OP1 — Canonical quaternionic Farey structure

Home: Chapter 3

Prove uniqueness (or classify) discrete adjacency / mediant rules on the gauged Hopf lattice that reduce to classical Farey under a fixed embedding $\mathbb{Q} \cup \{\infty\} \hookrightarrow$ lattice/base. Without a chosen primary rule, "quaternionic Farey" remains a family of metaphors rather than a single theory.

Sandbox: `candidate_adjacency` (along-fiber phase neighbors + base angular threshold). Not claimed canonical. **Diagnostics:** `adjacency_equivariance_score` (Ch. 4, Exercise 4.H). **Success criteria:** equivariant rule + documented Farey reduction + comparable mediant algorithm.

B.2.2 OP2 — Flux topograph axioms

Home: Chapter 5

Which properties of Conway topographs (separator structure, periodicity, river, face/edge values) survive for (a) quaternion norm forms and (b) lattice flux functionals? State a minimal axiom system such that Hatcher's binary case is a specialization.

Sandbox: `build_flux_topograph`, `detect_separators`, `periodicity_score`, `separator_equivariance_score`. Depends on OP1. **Success criteria:** axioms + proof or counterexample of reduction to TN Ch. 4.

B.2.3 OP3 — Class number ↔ Magic Island

Home: Chapter 6

Is there a precise arithmetic invariant (class number, type number, discriminant, topological charge, ...) whose magnitude or type predicts Magic Island stability scores? Correlation is not enough; seek a structural map or a clear negative result.

Sandbox: `classify_topograph_type`, `enumerate_reduced`, `class_number_analogue`, `magic_island_score`. Depends on OP1–OP2. **Success criteria:** invariant with out-of-sample predictive power or rigorous non-existence under stated axioms.

B.2.4 OP4 — $Z \to$ flywheel uniqueness

Home: Chapters 7 and 10

Up to gauge equivalence, is the map from atomic number Z to flywheel configuration unique under stated axioms? If not, classify the ambiguity and its chemical consequences.

Sandbox: `map_z_to_flywheel[_extended]`, `stability_landscape_z`. **Success criteria:** uniqueness theorem up to gauge, or explicit moduli of ambiguity.

B.2.5 OP5 — $350/\pi$ first principles or falsification

Home: Chapter 10

Derive $W_g = 350/\pi$ from lattice geometry / topological clock axioms, **or** falsify multi-domain recurrence as coincidence via pre-registered statistical tests (Table T4, Appendix D).

Sandbox: `table_t4_checklist`, `default_hypotheses`, `run_table_t4_demo`, `combine_p_values_fisher`, `proximity_to_wg`. **Success criteria:** derivation paper, or registered negative result with full T4 compliance.

B.2.6 OP6 — Composition of flywheels (Gauss lift)

Home: Chapters 8–9

Does Gauss-style composition admit a dynamical or ideal-theoretic realization on flywheels / flux classes that is associative up to gauge equivalence, compatible with Ch. 4 actions, and reduces to classical composition under restriction?

Sandbox: `compose_flywheels`, `composition_table`, `is_associative_up_to_equivalence`, `class_group_analogue`; algebraic side `QuaternionAlgebra`, `HurwitzOrder`, `left_ideal_class_set`, `two_sided_class_group`. **Current note:** flux-composition experiments often show **low closure and low associativity** — document failures as progress. **Success criteria:** associative group law on classes + comparison with Hurwitz / other orders.

B.3 How to update

1. Change **Status** when work starts or a result lands.
2. Put initials or issue links in an Owner column (optional).
3. When resolved, add a one-line pointer to the theorem / section / notebook and keep the row for history.
4. Keep `notes/open_problems.md` in sync with this appendix (or treat this appendix as the print form of that file).

B.4 Dependency sketch

```
OP1 (adjacency)
  +-> OP2 (topographs)
        +-> OP3 (class number / islands)
              +-> OP4 (Z-map uniqueness)
              +-> OP6 (composition) <-- Ch. 9 ideal theory
OP5 (350/pi) -- independent empirical track, informed by geometry
```

Appendix C

Laboratory Code Reference

Full lab listings moved out of the main chapter flow. Chapters keep short "call" versions and exercises; use this appendix when typing or copying longer scripts.

Setup (all labs) — one install path. From the clone root:

```
python3 -m pip install -e .
python3 -m pip install -e ".[portal]"   # flux-hopf-lib (PyPI) + kingdom-come (git pin)
```

```
from lib.hopf_lattice import hopf_map, HURWITZ_UNITS
```

Do not hard-code `~/Projects/...` and do not use a commented `PYTHONPATH` as the official path.

C.1 Chapters 1–2 (foundations)

C.1.1 Lab 1.A — Multiplication table

```
from kingdom.core.quaternion import Quaternion

i = Quaternion(0, 1, 0, 0)
j = Quaternion(0, 0, 1, 0)
k = Quaternion(0, 0, 0, 1)
print(i.multiply(j))  # ~ k
print(j.multiply(i))  # ~ -k
print(i.multiply(i))  # ~ -1
```

C.1.2 Lab 1.B — Norm multiplicativity

```
import numpy as np
from kingdom.core.quaternion import Quaternion

rng = np.random.default_rng(0)

def rand_q():
    a = rng.normal(size=4)
    return Quaternion(*a)

q1, q2 = rand_q(), rand_q()
```

```
prod = q1.multiply(q2)
n1, n2 = q1.norm() ** 2, q2.norm() ** 2
print(abs(prod.norm() ** 2 - n1 * n2))
```

C.1.3 Lab 1.C — Small Lipschitz norms

```
from itertools import product

def lipschitz_norms(nmax=3):
    found = {}
    for a, b, c, d in product(range(-nmax, nmax + 1), repeat=4):
        n = a * a + b * b + c * c + d * d
        if 0 < n <= nmax:
            found.setdefault(n, []).append((a, b, c, d))
    return {k: found[k][:8] for k in sorted(found)}

print(lipschitz_norms(3))
```

C.1.4 Lab 1.D — Double cover

```
import numpy as np
from kingdom.core.quaternion import Quaternion

q = Quaternion.from_axis_angle(np.array([0.0, 0.0, 1.0]), np.pi / 3)
mq = Quaternion(-q.w, -q.x, -q.y, -q.z)
v = Quaternion(0, 1, 0, 0)
rq = q.multiply(v).multiply(q.inverse())
rm = mq.multiply(v).multiply(mq.inverse())
print(rq, rm)  # same rotation
```

C.1.5 Lab 2.A–B — Hopf image and structure-group fiber

```
from lib.hopf_lattice import hopf_map, common_phase, sample_structure_group_fiber
import numpy as np

q = np.array([0.5, 0.5, 0.5, 0.5]); q = q / np.linalg.norm(q)
print(hopf_map(q), np.linalg.norm(hopf_map(q)))
print("h(0,0,1,0) =", hopf_map(np.array([0.0, 0.0, 1.0, 0.0])))

fiber = sample_structure_group_fiber(q, n_points=64)
base = np.stack([fiber["y1"], fiber["y2"], fiber["y3"]], axis=1)
print(np.linalg.norm(np.stack([fiber["x1"], fiber["x2"], fiber["x3"], fiber["x4"]], axis=1),
    axis=1)[:5])
print("base constant:", np.allclose(base, base[0]))
```

C.1.6 Lab 2.C — Fiber constancy vs xi2-circle

```
from lib.hopf_lattice import hopf_map, common_phase, hopf_coordinates
import numpy as np

q = np.array([0.6, 0.3, 0.4, 0.6]); q = q / np.linalg.norm(q)
y0 = hopf_map(q)
print(max(np.linalg.norm(hopf_map(common_phase(q, p)) - y0) for p in np.linspace(0, 2 * np.pi,
    12, endpoint=False)))
```

```
# xi2-circle at fixed (eta, xi1) is NOT the structure-group fiber
xi2_pts = np.stack([hopf_coordinates(0.4, 0.3, t) for t in np.linspace(0, 2 * np.pi, 12,
    endpoint=False)])
xi2_base = np.stack([hopf_map(p) for p in xi2_pts])
print("xi2-circle base spread:", np.max(np.linalg.norm(xi2_base - xi2_base[0], axis=1)))
```

C.1.7 Lab 2.D — Random unit images

```
from lib.hopf_lattice import hopf_map
import numpy as np
rng = np.random.default_rng(0)
for _ in range(3):
    q = rng.normal(size=4); q = q / np.linalg.norm(q)
    y = hopf_map(q)
    print(y, np.linalg.norm(y))
```

C.1.8 Lab 2.E — Structure-group fiber family

```
from lib.hopf_lattice import sample_structure_group_fiber_family
family = sample_structure_group_fiber_family(n_fibers=8, n_points=64)
print(len(family), family[0]["base_y1"])
```

C.2 Chapters 3–4 (lattice and symmetries)

C.2.1 Lab 3.A–C — Hurwitz, adjacency, gauge

```
from lib.hopf_lattice import (
    HURWITZ_UNITS, hopf_project_points, sample_angle_lattice,
    candidate_adjacency, left_multiply, right_multiply, rounded_point_set,
)
import numpy as np

print(len(HURWITZ_UNITS), np.allclose(np.sum(HURWITZ_UNITS**2, axis=1), 1.0))
base = hopf_project_points(HURWITZ_UNITS)
print("distinct base ~", len(np.unique(np.round(base, 5), axis=0)))

pts = sample_angle_lattice(n_eta=3, n_xi1=8, n_xi2=8)
along, inter = candidate_adjacency(pts, base_angle_thresh=0.5)
print(len(pts), len(along), len(inter))

i = np.array([0.0, 1.0, 0.0, 0.0])
moved = left_multiply(HURWITZ_UNITS, i)
base0, baseL = hopf_project_points(HURWITZ_UNITS), hopf_project_points(moved)
same_set = rounded_point_set(base0) == rounded_point_set(baseL)
print("base set invariant:", same_set)
print("total-space moved:", not np.allclose(moved, HURWITZ_UNITS))
```

C.2.2 Lab 3.D — Two-gyro dynamics

```
from kingdom.core.lattice import LatticeConfig
from kingdom.simulations.lattice import TwoGyroLattice, run_lattice_comparison

stable, chaotic = run_lattice_comparison(frames=80, n_sites=48)
print(stable.stability_score, stable.total_bursts, chaotic.total_bursts)
```

C.2.3 Lab 4.A–B — Gauge sequences and orbits

```
from lib.hopf_lattice import (
    HURWITZ_UNITS, permutes_hurwitz_units, left_multiply,
    hopf_project_points, orbit_of_point, phase_unit,
)
import numpy as np

i = np.array([0.0, 1.0, 0.0, 0.0])
print(permutes_hurwitz_units(i, side="L"))
seq = [("R", phase_unit(np.pi / 3)), ("L", i)]
orbit = orbit_of_point(HURWITZ_UNITS[3], seq, max_periods=24, tol=1e-6)
print(len(orbit), np.linalg.norm(orbit[-1] - orbit[0]))
```

C.2.4 Lab 4.C — Identity preservation (portal extra)

```
from kingdom.core.lattice import LatticeConfig
from kingdom.simulations.lattice import run_lattice_comparison
stable, chaotic = run_lattice_comparison(frames=40, n_sites=32)
print("stable", stable.identity_preservation if hasattr(stable, "identity_preservation") else
    stable.stability_score)
print("chaotic bursts", chaotic.total_bursts)
```

C.2.5 Lab 4.D — Flux push-forward from the left action of i (OP1 diagnostic)

```
from lib.hopf_lattice import (
    sample_angle_lattice, candidate_adjacency, discrete_flux_cycle,
    left_multiply, transform_flux, adjacency_equivariance_score,
    nearest_index_map, HURWITZ_UNITS,
)
import numpy as np

pts = HURWITZ_UNITS
along, inter = candidate_adjacency(pts, base_angle_thresh=0.55, fiber_phase_bins=8)
edges = along[:8] if along else inter[:8]
Phi = discrete_flux_cycle(edges, value=1)
i = np.array([0.0, 1.0, 0.0, 0.0])
moved = left_multiply(pts, i)
imap = nearest_index_map(moved, pts)  # permutation induced by left action of i
print("imap unique", len(set(imap.values())), "of", len(pts))
print(len(transform_flux(Phi, imap)) // 2)
eq = adjacency_equivariance_score(pts, i, side="L", base_angle_thresh=0.55, fiber_phase_bins=8)
print(eq)  # OP1 diagnostic -- not a theorem of equivariance
```

C.3 Chapters 5–7 (topographs, classification, Z-map)

C.3.1 Lab 5.A–D — Topographs

```
from lib.hopf_lattice import sample_angle_lattice, candidate_adjacency, phase_unit
from lib.flux_topograph import (
    build_flux_topograph, detect_separators, arithmetic_progression_residuals,
    periodicity_score, separator_equivariance_score,
)
import numpy as np

pts = sample_angle_lattice(4, 12, 12)
along, inter = candidate_adjacency(pts, base_angle_thresh=0.5, fiber_phase_bins=12)
topo = build_flux_topograph(pts, edges=along + inter, functional="hopf_height")
seps = detect_separators(topo, mode="sign")
print(len(seps), sum(len(c) for c in seps), arithmetic_progression_residuals(topo))

i = np.array([0.0, 1.0, 0.0, 0.0])
print(periodicity_score(topo, [("L", i), ("R", phase_unit(np.pi / 2))], max_periods=8))
print(separator_equivariance_score(topo, [("L", i)]))
```

C.3.2 Lab 6.A–D — Classification

```
from lib.flux_topograph import (
    build_flux_topograph, classify_topograph_type, enumerate_reduced,
    class_number_analogue, equivalence_distance, apply_gauge_to_topograph,
)
from lib.hopf_lattice import sample_angle_lattice, candidate_adjacency
import numpy as np

pts = sample_angle_lattice(3, 10, 10)
along, inter = candidate_adjacency(pts, base_angle_thresh=0.5, fiber_phase_bins=10)
topo = build_flux_topograph(pts, edges=along + inter, functional="hopf_height")
print(classify_topograph_type(topo))
print("n reduced", len(enumerate_reduced(topo, dedup_tol=0.05)))
print(class_number_analogue(topo, dedup_tol=0.05)["class_number_analogue"])
i = np.array([0.0, 1.0, 0.0, 0.0])
print(equivalence_distance(topo, apply_gauge_to_topograph(topo, [("L", i)])))
```

C.3.3 Lab 7.A–E — Z-map with H2 ablation (not findings)

```
from kingdom.core.flux_flywheel import map_z_to_flywheel
from lib.validation import h2_ablation_table

rows = [map_z_to_flywheel(z) for z in range(1, 51)]
report = h2_ablation_table(rows, high_threshold=7.5, seed=0)
print(report["disclaimer"])
print("alignment with bonus", report["alignment_with_bonus"])
print("alignment ablated   ", report["alignment_ablated"])
print("null (shuffled labels)", report["null_alignment_ablated"])
# Do not print noble-gas bonuses as findings. Compare ablated alignment to the null.
```

C.4 Chapters 8–10 (composition, algebras, validation)

C.4.1 Lab 8.A–D — Composition

```
from lib.hopf_lattice import sample_angle_lattice, candidate_adjacency
from lib.flux_topograph import build_flux_topograph, class_number_analogue
from lib.composition import (
    compose_flywheels, reduce_composition, composition_table,
    class_group_analogue, is_associative_up_to_equivalence,
)

pts = sample_angle_lattice(2, 6, 6)
along, inter = candidate_adjacency(pts, base_angle_thresh=0.55, fiber_phase_bins=6)
edges = along + inter
t1 = build_flux_topograph(pts, edges=edges, functional="hopf_height")
t2 = build_flux_topograph(pts, edges=edges, functional="hopf_y1")
print(reduce_composition(compose_flywheels(t1, t2,
    method="value_sum"))["classification"]["type"])

cg = class_group_analogue(t1, dedup_tol=0.1, samples=10)
print(cg["order"], cg["structure"], cg["associativity"])
reps = cg["representatives"]
print(composition_table(reps, dedup_tol=0.1)["closure_fraction"])
print(is_associative_up_to_equivalence(reps, samples=20, tol=0.1))
```

C.4.2 Lab 9.A–E — Quaternion algebras

```
from lib.quaternion_algebra import (
    QuaternionAlgebra, HurwitzOrder, LipschitzOrder,
    left_ideal_class_set, two_sided_class_group, hilbert_symbol,
)

A = QuaternionAlgebra(-1, -1)
print(A.presentation(), A.ramified_places(), A.is_definite())
O = HurwitzOrder()
print(O.n_units(), O.is_euclidean(), O.is_maximal())
print(left_ideal_class_set(O).cardinality, left_ideal_class_set(O).method)
print(two_sided_class_group(O).order, two_sided_class_group(O).method)
```

C.4.3 Lab 9.F — Hilbert symbols (own number)

```
from lib.quaternion_algebra import hilbert_symbol
print({p: hilbert_symbol(-1, -1, p) for p in [2, 3, 5, "inf"]})
```

C.4.4 Lab 10.B-style — Table T4 demo

```
from lib.validation import (
    run_table_t4_demo, combine_p_values_fisher, bonferroni_threshold,
    proximity_to_wg, WG_350_OVER_PI, default_hypotheses, table_t4_checklist,
)

print("W_g", WG_350_OVER_PI)
print(len(table_t4_checklist()), [h.name for h in default_hypotheses()])
demo = run_table_t4_demo(seed=1, alpha=0.01)
print(demo["bonferroni_threshold"], demo["decision"], demo["disclaimer"])
print(proximity_to_wg([111.4, 111.5, 110.0, 111.408]))
```

C.4.5 Lab 10.A — OP research proposal (hand)

Choose OP1, OP2, or OP6 from Appendix B. Write sandbox, diagnostic, and success criterion (one page). This listing is the prompt, not a script.

C.5 Common pitfalls

Symptom	Fix
`ModuleNotFoundError: lib`	`python3 -m pip install -e .` from the clone root
`ModuleNotFoundError: kingdom`	`python3 -m pip install -e ".[portal]"` (not a `PYTHONPATH` to `~/Projects`)
`magic_flag` KeyError	Use `stability_score` / `is_noble_gas`
`step` AttributeError	Use `step_frame()`
Slow `enumerate_reduced`	Smaller lattices (`n_eta<=3`, `n_xi<=10`)

Appendix D

Table T4 Validation Protocols (Full)

Pre-registered validation machinery for Part V Hypotheses. Chapter 10 keeps a short summary; this appendix is the full checklist, hypothesis catalog, and demo usage.

Helpers: `lib/validation.py`.

D.1 Core checklist (Table T4)

ID	Element	Description
T4.1	Null hypothesis definition	State H_0 with significance level α (default 0.01).
T4.2	Data sources and preprocessing	List datasets, windows, and cleaning rules.
T4.3	Test statistic	Define the scalar or vector statistic used for decision.
T4.4	Multiple-testing correction	Bonferroni or FDR control across domains/tests.
T4.5	Pre-registered sample size	Was the pre-registered (n) achieved? Not post-hoc power after T4.4.
T4.6	Falsification criteria	What result counts as strong evidence against the alternative.
T4.7	Pre-registration	Timestamped commit or external registry **before** looking at new data.
T4.8	Reproducibility	Full code, seeds, and environment for every figure and table.

```python
from lib.validation import table_t4_checklist
for row in table_t4_checklist():
    print(row["id"], row["element"])
```

D.2 Catalog of major hypotheses

These match `default_hypotheses()` in code.

D.2.1 H1a–H1e — $W_g = 350/\pi$ per domain (not one bundled clock)

Domain list locked 2026-08-28 **before** p-values. One symbol: $W_g = 350/\pi$ (not 350π). Bonferroni $n = 5$. Each row has its own T4 checklist, MDE, and pre-registered n (T4.5).

ID	Domain	Claim (Hypothesis)
H1a	pulsar_timing	Clustering near $W_g = 350/\pi$ in pulsar-timing narratives
H1b	bitcoin_pi_cycle	Clustering near $W_g = 350/\pi$ in Bitcoin Pi Cycle notes
H1c	tls_trees	Clustering near $W_g = 350/\pi$ in TLS tree analysis
H1d	cuprate_superconductors	Clustering near $W_g = 350/\pi$ in cuprate sketches
H1e	structural_constants	Clustering near $W_g = 350/\pi$ in structural constants

Field	Content (each H1$*$)
Type	Hypothesis
Null	Recurrence in *that* domain is consistent with random coincidence at $\alpha = 0.01$.
Correction	Bonferroni $n = 5$ (locked with the domain list)
T4.5	Was the pre-registered (n) for that domain achieved?
Falsification	Fail to reject H_0 in that domain after pre-registration; or pre-registered n not achieved

Do not recombine H1a–H1e into a single “shared clock” claim. No new observational domains on H1 in this revision.

D.2.2 H2 — $Z \mapsto$ map as periodic-table proxy

Field	Content
Claim	`map_z_to_flywheel` stability peaks reflect genuine chemical/nuclear specialness beyond model tuning.
Type	Hypothesis
Null	High scores near noble gases are explained by explicit model bonuses alone (no extra predictive content).
Domains	periodic table; ionization energy; nuclear magic numbers
Statistic	Out-of-sample correlation or ablation study
Falsification	Ablating noble-gas bonuses removes all predictive alignment; held-out properties not above chance

D.2.3 H3 — Magic Island $\leftrightarrow$ class-number association

Field	Content
Claim	Magic Island structure is predicted by class-number-like invariants.
Type	Hypothesis
Null	Island locations independent of `class_number_analogue`.
Statistic	Association / rank correlation
Correction	FDR
Falsification	No significant association after pre-registration

```python
from lib.validation import default_hypotheses
for h in default_hypotheses():
    print(h.name, h.alpha, h.null_hypothesis[:60], "...")
```

D.3 Statistical helpers

```python
from lib.validation import (
    WG_350_OVER_PI,
    bonferroni_threshold,
    combine_p_values_fisher,
    proximity_to_wg,
    run_table_t4_demo,
    toy_multidomain_pvalues,
)

print("W_g =", WG_350_OVER_PI)
print("Bonferroni thr (alpha=0.01, n=5):", bonferroni_threshold(0.01, 5))

# Toy null demo -- NOT evidence for 350/pi
demo = run_table_t4_demo(seed=1, alpha=0.01)
print(demo["decision"], demo["fisher"])

# Diagnostic closeness of constants to W_g
print(proximity_to_wg([111.4, 111.5, 110.0, 111.408]))
```

Under the **null toy** generator, expect frequent `fail_to_reject_H0`. Real domain p-values without pre-registration do **not** count as T4 success.

D.4 Decision flowchart (narrative)

1. Write H_0, α, statistic, and falsification (T4.1, T4.3, T4.6).
2. Lock data sources and preprocessing (T4.2).
3. Pre-register (T4.7) — e.g. git tag / commit hash.
4. Compute statistics; apply multiple-testing correction (T4.4).
5. Check that the pre-registered (n) was achieved (T4.5) — not post-hoc power.
6. Decide: reject H_0 / fail to reject / refine design.
7. Publish code and seeds (T4.8).

Figure 10.4 in Chapter 10 is the visual summary of this flow.

D.5 Claim discipline for validation results

Statement	Label
"Table T4 exists in the repo"	**Software fact**
"Hypothesis X passed T4"	**Hypothesis** until full checklist executed and reviewed
"W_g is a law of nature"	**Hypothesis** (not default)

Appendix E

Figure Atlas

Index of main and auxiliary figures. Paths are relative to `book/figures/`. Full attribution: `figures/ATTRIBUTION.md`. Regenerate with `scripts/generate_chN_figures.py`.

E.1 By chapter

E.1.1 Chapter 0

ID	File
Fig. 0.1–0.5	`fig0_1_...` ... `fig0_5_...`
Aux A0.1–A0.2	`aux_z_map_he.png`, `aux_z_map_fe.png`

E.1.2 Chapters 1–2

ID	File
Fig. 1.1–1.4, Aux A1.1	`fig1_*`, `aux1_1_*`
Fig. 2.1–2.4, Aux A2.1–A2.2	`fig2_*`, `aux2_*`

E.1.3 Chapters 3–4

ID	File
Fig. 3.1–3.5, Aux A3.1	`fig3_*`, `aux3_1_*`
Fig. 4.1–4.4, Aux A4.1	`fig4_*`, `aux4_1_*`

E.1.4 Chapters 5–7

ID	File
Fig. 5.1–5.4, Aux A5.1	`fig5_*`, `aux5_1_*`
Fig. 6.1–6.4, Aux A6.1	`fig6_*`, `aux6_1_*`
Fig. 7.1–7.4, Aux A7.1	`fig7_*`, `aux7_1_*`, `aux_z_map_au.png`

E.1.5 Chapters 8–10

ID	File
Fig. 8.1–8.4, Aux A8.1	`fig8_*`, `aux8_1_*`
Fig. 9.1–9.5, Aux A9.1	`fig9_*`, `aux9_1_*`
Fig. 10.1–10.4, Aux A10.1	`fig10_*`, `aux10_1_*`

E.2 Naming conventions

- Main figures: `fig{N}_{k}_descriptive_name.png`
- Auxiliary: `aux{N}_{k}_...` or `aux_z_map_*.png`
- In LaTeX: `\includegraphics{...}` with `graphicspath` → `figures/`
- Labels: `\label{fig:filename-stem}`

E.3 Portal sources (selected)

Book figure	Kingdom Come source (if vendored)
Fig. 0.1	`app/assets/home/hopf_linked_fibers.png`
Fig. 0.2	`app/assets/hopf_preview.png`
Fig. 0.3	`app/assets/home/hopf_fibration_bundle.png`
Fig. 0.4–0.5	`app/assets/bitcoin_pi/...`
Aux A10.1	`app/assets/pulsars/...`, `app/assets/bitcoin_pi/...`
Aux A7.1	`z_knowns/frame_*.png`

Generated diagrams (most of Ch. 1–10) are produced by `scripts/generate_chN_figures.py`.

Appendix F

Hatcher Parallel Dictionary

Condensed from root `HATCHER_MAP.md` for print. Use the free TN PDF: https://pi.math.cornell.edu/~hatcher/TN/TNbook.pdf

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F.1 Chapter map

Hatcher *Topology of Numbers*	This book (QGA)	Lift
Ch. 0 Preview	Ch. 0	Pythagorean → four squares → Hopf glimpse
(none)	Ch. 1–2	Quaternions; Hopf fibration
Ch. 1 Farey diagram	Ch. 3	Gauged Hopf lattice; OP1
Ch. 2 Continued fractions	Ch. 3–4	Phase walks; lattice paths
Ch. 3 Symmetries of Farey	Ch. 4	Left/right multiplications; LFT comparison
Ch. 4 Quadratic forms / topographs	Ch. 5	Flux topographs; OP2
Ch. 5 Classification of forms	Ch. 6	Types, reduced configs, Magic Islands; OP3
Ch. 6 Representations	Ch. 7	$Z \mapsto$ map; OP4
Ch. 7 Class group	Ch. 8	Composition; class-group analogue; OP6
Ch. 8 Quadratic fields	Ch. 9	Quaternion algebras; ideal theory
(applications)	Ch. 10	Observations; Table T4; OP5

F.2 Concept dictionary

Hatcher	QGA
Farey edge / mediant	Lattice adjacency / candidate mediant (Model)
Continued-fraction zigzag	Fiber phase walk; gauge sequence
Linear fractional transformation ($\mathrm{PGL}(2,\mathbb{Z})$; det ± 1)	Unit quaternion left/right multiplication
Conway topograph	Flux topograph
Separator line	Separator structure / surface
Reduced form	Reduced flux configuration
Class number	`class_number_analogue` / island counts (Model)
Gauss composition	`compose_flywheels` (Model; OP6)
Ideal class group	Two-sided class group (Theorem) vs left class set; Model bridge
Representation by forms	Representation by norms + $Z \mapsto$ flux configs

---

F.3 What is deliberately not 1:1

- No reuse of Hatcher's prose, figures, or exercises.
- "Quaternionic Farey" is **defined** for the gauged Hopf lattice (OP1), not assumed unique.
- Physics / observation chapters are extra floors, not TN theorems.

Bibliography (seed)

- Allen Hatcher, *Topology of Numbers*, AMS, 2022. Free PDF: `https://pi.math.cornell.edu/~hatcher/TN/TNbook.pdf`.
- Classical quaternion arithmetic: Hurwitz; Conway–Smith, *On Quaternions and Octonions*.
- Quaternion algebras: Vignéras; Voight, *Quaternion Algebras*.
- Hopf fibration surveys; topological phases / monopoles literature.
- Kingdom Come repository and observation program: `https://github.com/kinaar8340/kingdom_come`.
- This book repository: `https://github.com/kinaar8340/qga`.